

Virtuoso® Layout Pro: T9 Virtuoso Design Planner

Course Version IC 25.1

Lecture Manual

Revision 1.0

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Module 1

About This Course

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Course Objectives

In this course, you:

- Set up and analyze the design planning environment
- Use the virtual hierarchy in design planning
- Analyze early design planning
- Use the Design Planner in the Team environment



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Course Agenda

- Setting Up the Design Planning Environment
 - Lab: Setting Up the Working Directory
 - Lab: Analyzing the Design Planning Environment
- Design Planning
 - Submodule: Generating the Virtual Hierarchy
 - Submodule: Setting up the Soft Block P-cells Using CSV in Design Planning
 - Submodule: Accessing the Design Planning Data Sets in the Navigator Assistant
 - Submodule: Viewing the Virtual Hierarchy
 - Submodule: Analyzing the Net Connectivity
 - Lab: Generating the Virtual Hierarchy
 - Lab: Setting Up the Soft Block Pcells Using CSV in Design Planning
 - Lab: Accessing the Design Planning Data Sets in the Navigator Assistant
 - Lab: Viewing the Virtual Hierarchy
 - Lab: Analyzing the Net Connectivity
- Early Design Planning
 - Lab: Manipulating the Virtual Hierarchy
- Using the Design Planner in the Team Environment
 - Submodule: Creating Rows and Running Automatic Placement
 - Submodule: Running Congestion Analysis
 - Submodule: Creating Real Layout Views for the Virtual Hierarchies
 - Submodule: Creating Virtual Hierarchies for the Layout Views
 - Lab: Creating Rows and Running Automatic Placement
 - Lab: Running Congestion Analysis
 - Lab: Creating Real Layout Views for the Virtual Hierarchies
 - Lab: Creating Virtual Hierarchies for the Layout Views

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Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/86265.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.



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How do I register to take the exam?

- Log in to our [Learning Management System](#), click on the course in your transcript, and go to the **Content** tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

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Icons Used in This Class



Best Practice



Language/
Command Syntax



Concept/
Glossary



Frequently Asked
Questions/
Quiz



Error Message

Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.



Problem & Solution



Quick Reference



GUI and Command



How To



Lab List

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Module 2

Setting Up the Design Planning Environment

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Module Objectives

At the end of the module, you will be able to:

- Set up the Virtuoso® Design Planning environment in Layout EXL
- Identify the prerequisites required to access the Virtuoso Design Planner
- Navigate the Design Planning workspace, menu, toolbar, and associated assistants
- Analyze the Virtuoso Design Planner workflow and its key planning stages
- Differentiate the Virtuoso Design Planner approach from related planning flows and tools



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Useful Acronyms and Terms



Generate All From
Source (GFS)

Generates the layout from the schematic



Configure Physical
Hierarchy (CPH)

Defines how the layout is generated based on the schematic



Check Against
Source (CAS)

Compares the layout with the schematic and reports mismatches



Update
Components and
Nets (UCN)

Updates the layout based on the schematic modifications



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What Is the Virtuoso Design Planner?

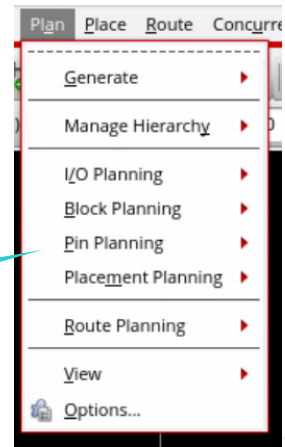


The Virtuoso® Design Planner (VDP) is a connectivity-based tool in Layout EXL. VDP optimizes and supports seamless layout planning, placement and routing.

The Design Planner supports built-in automatic placement and congestion analysis capabilities.

- It allows you to optimize layout planning at the top level, block level, and cell level.
- It offers a new method for efficient physical layout planning, providing a smooth transition between placement and routing.

In the Layout EXL menu bar, choose the **Plan** menu to access all the Design Planner commands.



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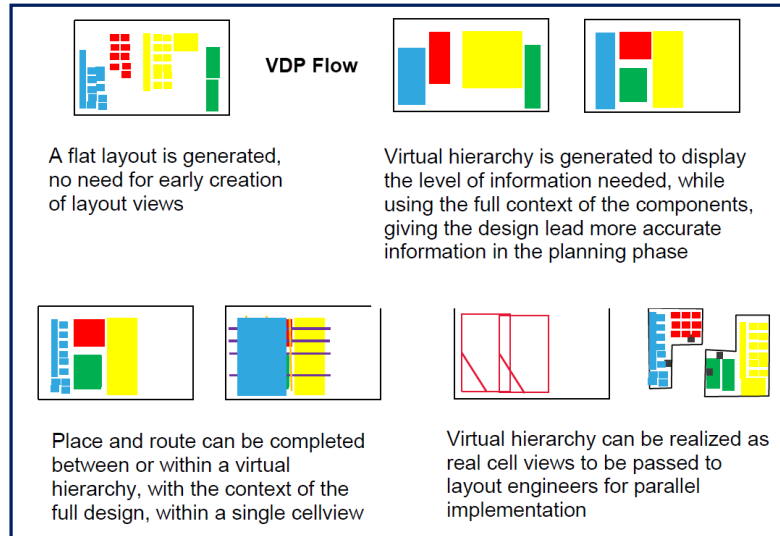
The Virtuoso Design Planner is a connectivity-based tool in the Layout EXL that optimizes layout planning and supports seamless layout planning, placement, and routing. The Design Planner supports in-built automatic placement and congestion analysis capabilities that allow optimizing layout planning at top level, block level, and cell level. The Design Planner offers a new method for efficient physical layout planning, providing a smooth transition between placement and routing.

The Virtuoso Design Planner provides an innovative layout-place-route solution for advanced node and mature node designs.

Advantages of the Virtuoso Design Planner Flow



The VDP flow provides more information in the planning phase, leading to less rework, a more productive flow, and a shorter design cycle.



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Virtuoso Design Planner Flow – Overview

The Virtuoso Design Planner offers a new method for efficient layout planning that can be applied at the top, block, or cell level. It introduces virtual layout hierarchy. This is a new concept that allows the design team to benefit from the advantages of both the flat and hierarchical design flows, while providing a smooth transition to layout, place, and route.

Advantages of the VDP Flow

- A flat layout is generated, and there is no need for early creation of layout views.
- Virtual hierarchy is generated to display the level of information needed, while using the full context of the components, giving the design lead more accurate information in the planning phase.
- Place and route can be completed between or within a virtual hierarchy, with the context of the full design, within a single cellview.
- Virtual hierarchy can be realized as real cell views to be passed to layout engineers for parallel implementation.

Features of the Virtuoso Design Planner



The Virtuoso Design Planner supports:

- **Hierarchical schematic-driven layouts** that combine the best of bottom-up and top-down design methodologies while avoiding the shortcomings of each.
- **Hierarchical visualization** that lets you easily view or hide details on any level, anywhere in your design, so you can see only what you need when you need it.
- **Hierarchical and congestion-aware design planning and placement** that provides automated and assisted generation.
- **Hierarchical routing and congestion analysis** that makes real routing and congestion analysis information available upfront.

Virtuoso Design Planner is **NOT** a replacement for the Virtuoso Floorplanner. The two approaches are complementary.



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How to Use the Virtuoso Design Planner Flow



In this procedure, you will learn how to use the Virtuoso Design Planner flow to create and refine a layout. You will choose the design source, generate the layout, place instances, and route the design. Apply this workflow when planning and implementing a top-level layout in Virtuoso Design Planner.



In this procedure, you will learn how to use the Virtuoso Design Planner flow to create and refine a top-level layout. The workflow consists of four key steps: choosing the design source, generating the layout, placing instances, and routing the design. Each step builds on the previous one to help you create an optimized and design-aware layout.

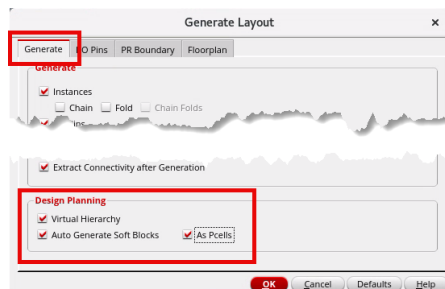
First, you will select the appropriate source information, such as the schematic hierarchy, hard blocks, soft blocks, and constraints. Next, you will generate the layout and create the virtual hierarchy needed for design planning. You will then place the instances and organize the layout structure. Finally, you will route the design and analyze connectivity to prepare the layout for further implementation and optimization.

Let's begin by learning how to choose the source for layout generation in Virtuoso Design Planner.

1. Choosing the Source

Schematic hierarchy, hard blocks, soft blocks, and constraints.

- If no hard or soft block exists, the Design Planner uses the schematic hierarchy to create a Virtual Hierarchy (VH) that entirely matches the schematic.
- If a hard block exists, the Design Planner recognizes it and uses it as a source for generating the layout view.
- If a soft block exists, the Design Planner leverages the soft block pin information to plan the top-cell layout.
- For schematic instances that have constraints added, the Design Planner transfers the constraints to the generated virtual hierarchy. Any constraints that exist within a VH are restricted to the VH.



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In this step, we will learn how to choose the source for layout generation in Virtuoso Design Planner. The source information can come from the schematic hierarchy, hard blocks, soft blocks, or design constraints.

If hard blocks or soft blocks are not available, Design Planner uses the schematic hierarchy to create a Virtual Hierarchy that matches the schematic. If hard blocks are available, the tool recognizes them and uses them as a source for generating the layout view.

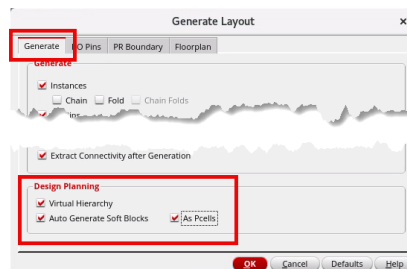
When soft blocks are present, Design Planner uses the soft block pin information to help plan the top-level layout. If constraints are added to schematic instances, those constraints are transferred to the generated Virtual Hierarchy and remain restricted within that Virtual Hierarchy.

This step is important because selecting the correct source ensures that the layout generation starts with the right design information.

2. Generating the Layout

You can use *Generate All From Source (GFS)* to generate a VH for schematic instances that have no corresponding layout views.

- You can access the generated VH easily using the Navigator assistant, where a new data subset has been added to allow this.
- If multiple instances of a VH need to be generated, the Design Planner creates clones of the hierarchy.
 - You can access these clones using the new data subset created in the Navigator assistant.
- You can realize a block from the top-cell layout at any time by using the *Make Cell* command.
 - You can apply the congestion-aware pin optimization.
- You can virtualize a block by using the *Make Virtual Hierarchy* command.



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In this step, you will generate the layout using the **Generate All From Source**, or **GFS**, capability in Virtuoso Design Planner. This command creates a Virtual Hierarchy for schematic instances that do not have corresponding layout views.

The generated Virtual Hierarchy can be accessed through the Navigator assistant, where a dedicated data subset is created for easy navigation. If multiple instances require virtualization, Design Planner automatically creates hierarchy clones, which can also be accessed from the Navigator assistant.

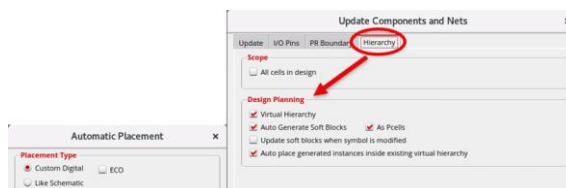
As the design evolves, you can convert a virtual block into a physical layout block by using the **Make Cell** command. During this process, congestion-aware pin optimization can be applied to improve layout planning. Conversely, you can virtualize an existing block at any time by using the **Make Virtual Hierarchy** command.

Generating the layout provides the foundation for the next stage of the design planning flow, where you will place and organize the generated instances.

3. Placing the Instances

The Design Planner creates an area boundary around the top-level VH.

- It is driven by the size of the layout views present inside the hierarchy.
- It allows for more realistic placement of the blocks at the top level.
- You can use the built-in *Analyze Connectivity* tool to make more informed decisions when placing the top-level objects manually.
- You can stretch, resize, or move the area boundary, if needed, to manually place the generated blocks at the top level.
- You can choose to run one of the built-in automatic placers to perform a “like-schematic, analog, or custom placement”.
- You can run the *Update Components and Nets (UCN)* command for any schematic changes to bring the layout back in sync with the schematic.



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In this step, you will place the instances within the generated Virtual Hierarchy. Design Planner automatically creates an area boundary around the top-level Virtual Hierarchy based on the size of the layout views contained within the hierarchy.

This boundary enables a more realistic representation of the design and provides a framework for organizing blocks at the top level. You can use the built-in **Analyze Connectivity** tool to better understand design connections and make informed placement decisions.

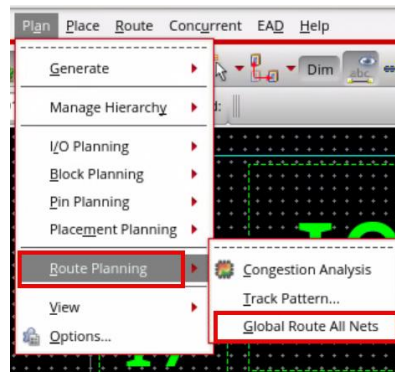
If necessary, you can manually adjust the area boundary by stretching, resizing, or moving it to accommodate the generated blocks. Design Planner also provides automatic placement options, allowing you to perform a like-schematic, analog, or custom placement based on your design requirements.

As schematic changes occur, you can use the **Update Components and Nets**, or **UCN**, command to synchronize the layout with the latest schematic information. Once the placement is complete, you are ready to proceed to the final step of the workflow—routing the design.

4. Routing the Design

The Design Planner supports a built-in global router to allow routing for all nets.

- You can route the critical nets manually, if needed.
- To perform a detailed analysis of the generated routes, you can call the built-in *Congestion Analysis* tool.



In this final step, you will route the design using the routing capabilities available in Virtuoso Design Planner. The tool includes a built-in global router that can automatically generate routes for all nets in the design.

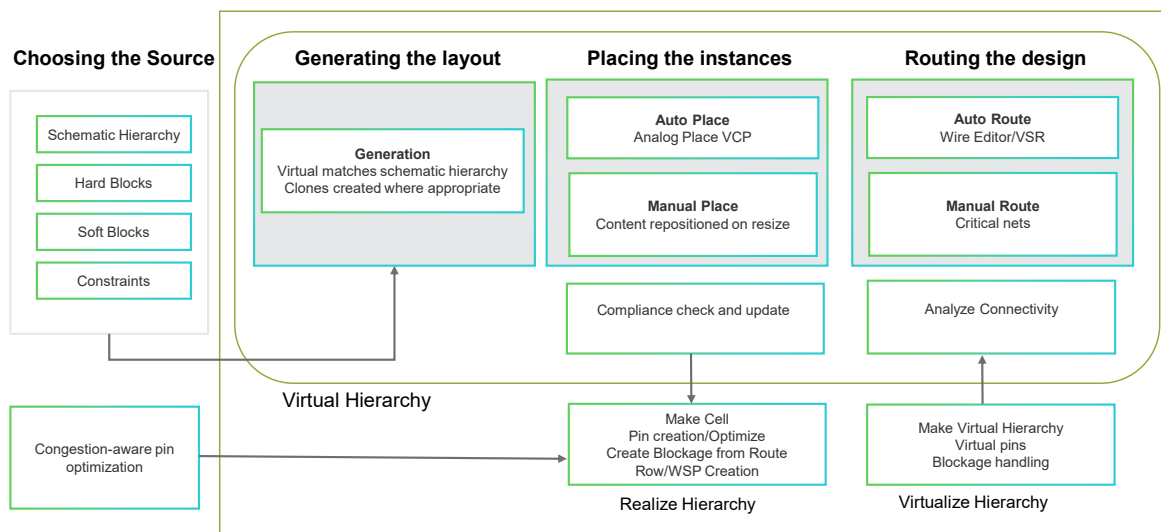
For critical nets that require special attention, you can manually create or adjust the routing to meet specific design requirements. This provides greater control over key signal paths while still benefiting from automated routing for the rest of the design.

After routing is complete, you can use the **Congestion Analysis** tool to evaluate routing quality and identify potential congestion areas within the layout. This analysis helps you optimize routing resources and improve overall design implementation.

By completing the routing process, you finalize the primary stages of the Virtuoso Design Planner flow, transforming the planned hierarchy into a connected and layout-ready design.

The Virtuoso Design Planner Flow

The Virtuoso Design Planner supports seamless transitions across the generate-place-route phases until an optimal layout plan is created.



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The Virtuoso Design Planner Flow

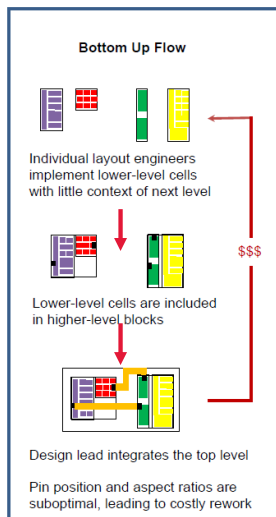
For designs that have unimplemented layout views existing at the top level and the levels below, the layout designers need to work across multiple cellviews. Also, the placement at the top level is oblivious to the actual lower level instances, so the top-level placement is based on estimation. After the design is routed, there might be changes needed to the top level that are often expensive to accommodate. With the Design Planner, layout engineers can seamlessly lay out, place, and route at both advanced and mature nodes, backed by real routing data.

The Virtuoso Design Planner flow is a schematic-driven flow that offers an innovative solution for generating, placing, and routing advanced and mature node designs seamlessly. The Design Planner flow is expected to be useful for designs that have no layout views existing at the top level and the levels below – the premise being that the Design Planner will use the existing schematic hierarchy, where available, to create a Virtual Hierarchy that can be realized, when appropriate. Blocks that have an existing IP in place and soft blocks from an existing Virtuoso Floorplanner flow are both recognized by the Design Planner as valid inputs for design planning.

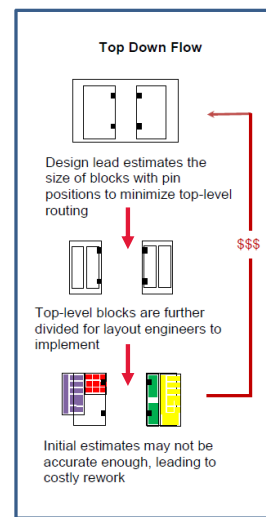
The Virtuoso Design Planner also recognizes any constraints that may exist on schematic instances and generates the Virtual Hierarchy with constraints in place, where needed. However, the constraints cannot be transferred outside a Virtual Hierarchy.

Drawbacks of the Bottom-Up and Top-Down Flows

Bottom-Up Flow: Pin positions and aspect ratios are suboptimal, leading to costly rework.



Top-Down Flow: Initial estimates may not be accurate enough, leading to costly rework.



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Drawbacks of the bottom-up and down-down flows – Overview

Creation of a top- or block-level layout is a complex process. Physical design planning is mandatory to increase the silicon usage and design performance. Although physical planning is a time-consuming task, it is an important stage of the layout design process. Working with a flat hierarchy can be impractical due to the large number of components in a design. However, introducing hierarchy to allow the work to be divided amongst a design team also has its challenges. Whether the team works in a bottom-up or a top-down flow, it will face challenges such as having to create the layout cells early in the design and to go through multiple iterations for Placement and Routing.

Drawback of the Bottom-Up Flow

- Individual layout engineers implement lower-level cells with little context of next level.
- Lower-level cells are included in higher-level blocks.
- Design lead integrates the top level.
- Result: Pin position and aspect ratios are suboptimal, leading to costly rework.

Drawback of the Top-Down Flow

- Design lead estimates the size of blocks with pin positions to minimize top-level routing.
- Top-level blocks are further divided for layout engineers to implement.
- Layout engineers implement the divided blocks.
- Result: Initial estimates may not be accurate enough, leading to costly rework.

Licensing Requirements to Access the Virtuoso Design Planner



To use the Virtuoso Design Planner (VDP), you must have access to the following design environment capabilities:

- **Virtuoso Release:** *IC 25.1*
- **Virtuoso Layout Suite Application:** *Virtuoso Layout Suite EXL*

Design Planning functionality is available in **Layout EXL** from **ICADVM18.1** onwards.



Accessing the Design Planner

To use the Virtuoso Design Planner, you must have access to the following design environment capabilities:

- **Virtuoso Release:** *ICADVM20.1*
 - For information about licensing in the Virtuoso design environment, see the *Virtuoso Software Licensing and Configuration User Guide, Product Version ICADVM20.1*.
- **Virtuoso Layout Suite Application:** *Virtuoso Layout Suite EXL*
 - For information about the Layout Suite EXL editor, see the *Virtuoso Layout Suite EXL Reference, Product Version ICADVM20.1*.

Software Product License Management for VDP

License Requirement for VDP

Software License Management			
Status Token Usage Ordering Diagnostics Performance Tests			
Checked out Licenses			Relevant Licenses
Number	Product Name	In Use	Number
111	Cadence(R) Design Framework II	1	12141
95011	Virtuoso Advanced Node Framework	1	206
95115	Virtuoso(R) Schematic Editor XL	1	21060
95321	Virtuoso(R) Layout Suite - GXL	12	21400
95800	Virtuoso Layout Suite EXL	1	276
			32100
			32101
			32500
			32760
			33500
			365
			38500
			70000
			71110
			71120
			71130
			900
			90006
			95015
			95022
			95100
			95127
			95130
			95200
			95230
			95250
			95252
			067456



Software Product License Management for VDP is given here.

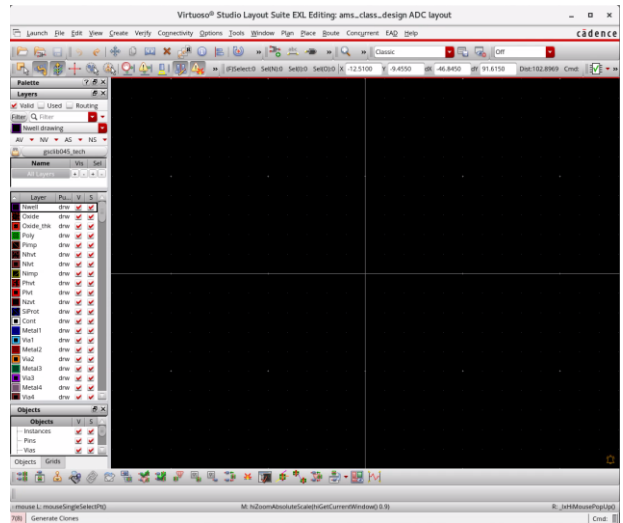
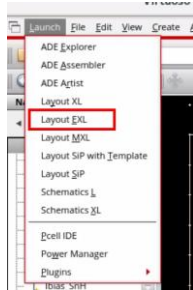
The table lists the product number and the product name, and the number of licenses required to use VDP.

Opening the Virtuoso Layout EXL Canvas



In the schematic, choose **Launch – Layout EXL** to open the new/existing layout in EXL mode.

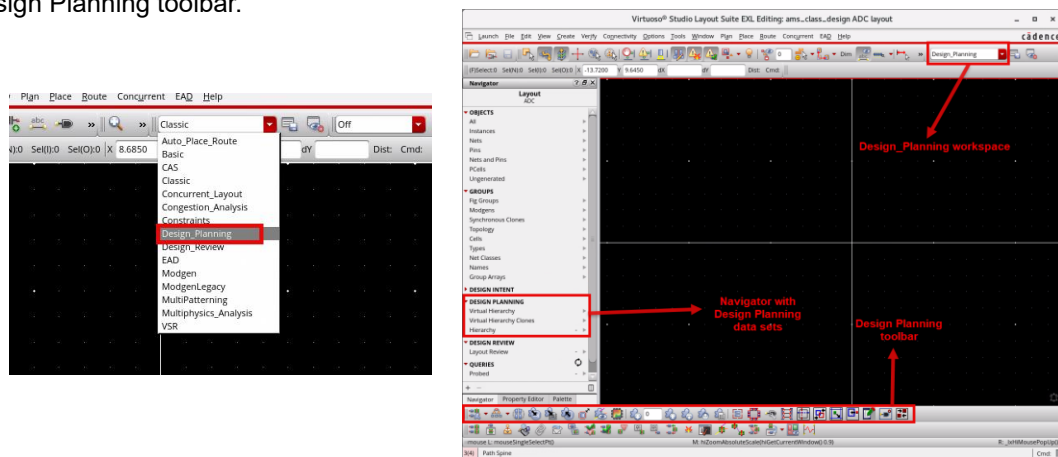
- Use the Virtuoso Layout EXL window to start or access the Virtuoso Design Planner.



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Selecting the Design_Planning Workspace in the Layout Canvas

- From the Workspace Configuration pull-down, select **Design_Planning** to change the workspace.
 - Opens the layout canvas in the Design_Planning workspace.
 - Enables Navigator with DESIGN PLANNING Data Sets, Palette, and Property Editor assistants.
 - Enables Design Planning toolbar.



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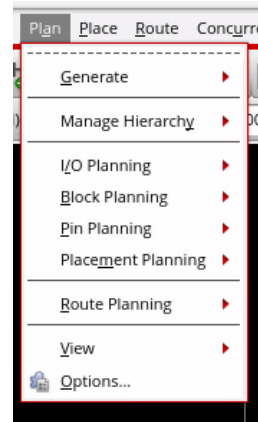
Starting the Virtuoso Design Planner: Design Planning Menu



In the Layout EXL menu bar, choose **Plan** to display the Design Planning menu.

- The Design Planning menu gives you access to the various commands and related forms supported by the Design Planner.

The commands in the Plan menu are also available on the Design Planner toolbar.



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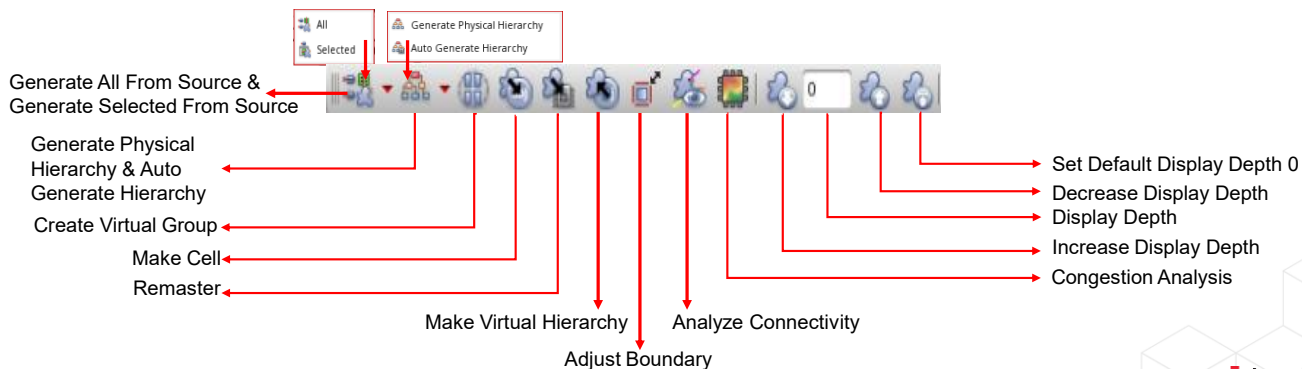
Starting the Virtuoso Design Planner: Design Planner Toolbar



In the Layout EXL window, choose **Window – Toolbars – Design Planning** to display the Design Planner toolbar.

Most commands available on the Design Planner toolbar can also be accessed using the Design Planning menu.

- The Design Planner toolbar lets you access the various commands and related forms supported by the Design Planner.
- As with all Virtuoso toolbars, you can move the Design Planner toolbar anywhere within the layout window and dock it at an appropriate position.



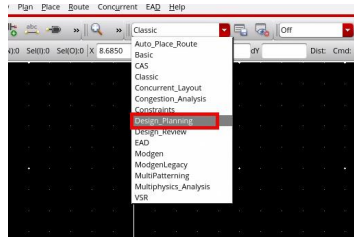
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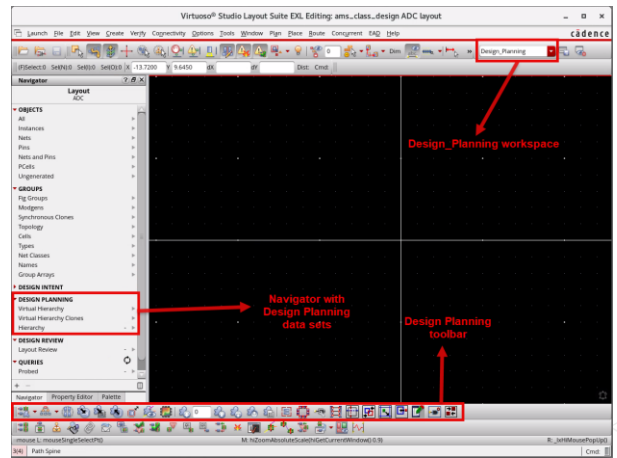
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Starting the Virtuoso Design Planner: Design_Planning Workspace

- From the Workspace Configuration pull-down, select the **Design_Planning** workspace to access the Design Planning Environment.
- It consists of three main components:
 - Design_Planning workspace
 - Navigator with DESIGN PLANNING Data Sets
 - Design Planning toolbar



The default workspace for Design Planner is called **Design_Planning**.



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Quiz: Virtuoso Release Support for Design Planning Functionality



Design Planning functionality is available in which of the following Virtuoso releases?
Select all that apply.

- A. IC6.1.8
- B. IC 25.1
- C. ICADVM20.1
- D. ICADV12.3
- E. All of the Above

Answer: B and C

Design Planning functionality is available in the Virtuoso ICADVM20.1 and IC 25.1 releases.



Answer: B and C

Design Planning functionality is available in the Virtuoso ICADVM18.1 and ICADVM20.1 releases.

Quiz: VLS Tier Support in Virtuoso Design Planner



The Virtuoso Design Planner is available in which of the Virtuoso Layout Suite Applications?
Select all that apply.

- A. Virtuoso Layout Suite L
- B. Virtuoso Layout Suite XL
- C. Virtuoso Layout Suite GXL
- D. Virtuoso Layout Suite EXL
- E. None of the Above

Answer: D

Virtuoso Design Planner is available in the Virtuoso Layout Suite EXL Application.



Answer: D

Virtuoso Design Planner is available in the Virtuoso Layout Suite EXL Application.

Quiz: Virtuoso Design Planner and Virtuoso Floorplanner



The Virtuoso Design Planner is a replacement for Virtuoso Floorplanner.

- A. True
- B. False

Answer: B

False. Virtuoso Design Planner is NOT a replacement for Virtuoso Floorplanner. The two approaches are complementary.



Answer: B

False. Virtuoso Design Planner is NOT a replacement for Virtuoso Floorplanner. The two approaches are complementary.

Quiz: Default Workspace for Virtuoso Design Planner



The default workspace for the Virtuoso Design Planner is called _____.
Enter the answer in the blank field.

Answer: Design_Planning

The default workspace for Virtuoso Design Planner is called Design_Planning.



Answer: Design_Planning

The default workspace for Virtuoso Design Planner is called Design_Planning.

Quiz: Connectivity in Virtuoso Design Planner



The Virtuoso Design Planner is a non-connectivity-based tool.

- A. True
- B. False

Answer: B

False. Virtuoso Design Planner is a connectivity-based tool.



Answer: B

False. Virtuoso Design Planner is a connectivity-based tool.

Quiz: Activating the Design Planner Toolbar



What are the ways you can use to activate the Design Planner toolbar in the layout canvas?
Select all that apply.

- A. From the **Window – Toolbars** menu
- B. **Right-click** the menu bar and from the Toolbars menu
- C. From the **Window – Assistants** menu
- D. **Right-click** the menu bar and from the Assistants menu
- E. All of the above

Answers: A and B

You can activate the Design Planner toolbar in the layout canvas from the **Window – Toolbars** menu and **right-click** the menu bar and from the **Toolbars** menu.



Answers: A and B

You can activate the Design Planner toolbar in the layout canvas from the Window – Toolbars menu and right-click the menu bar and from the Toolbars menu.

Quiz: Design Planning Menu and Design Planner Toolbar



The commands in the Design Planning menu are also available on the Design Planner toolbar and vice-versa.

- A. True
- B. False

Answer: B

False. Most commands available on the Design Planner toolbar can also be accessed using the Design Planning menu.



Answer: B

False. Most commands available on the Design Planner toolbar can also be accessed using the Design Planning menu.

Module Summary

In this module, you learned to:

- Set up the working directory correctly to ensure a structured, efficient design planning workflow.
- Verify required libraries, files, and paths to create a reliable design environment baseline.
- Understand the design planning environment layout to navigate tools and features effectively.
- Analyze available options and settings to control design behavior and outputs.
- Use the environment insights to make informed decisions during design planning.



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Labs



Lab 2-1 Setting Up the Working Directory

Lab 2-2 Analyzing the Design Planning Environment



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Module 3

Design Planning

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Module Objectives

At the end of this module, you will be able to:

- Generate the virtual hierarchy
- Soft block P-cells using CSV in Design Planning
- Access the Design Planning data sets in the Navigator assistant
- View the virtual hierarchy
- Analyze the net connectivity



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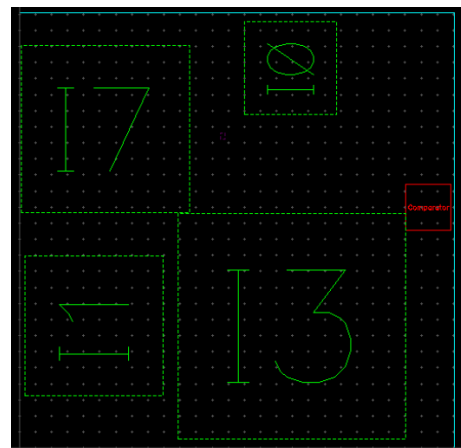
What Is a Virtual Layout Hierarchy?



A Virtual Layout Hierarchy allows you to access and edit all the objects in the hierarchy at the top level.

It eliminates the need to manage and maintain multiple hierarchical cellviews.

- You can realize a block from the top-cell layout at any time using the *Make Cell* command.
 - You can virtualize the block using the *Make Virtual Hierarchy* command.
 - This back-and-forth can be performed any number of times during a design cycle without losing any pin locations and block sizes.
- Having access to the hierarchical layout information at the top level allows for more accurate design planning.
 - The soft block sizes and the pin information for each block are based on real data.
- You can view the net connections through the entire virtual hierarchy using Navigator.
 - You can position the lower-level objects more accurately from the top level, driven by context.



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The Virtual Layout Hierarchy

For designs that have unimplemented layout views existing at the top level and the levels below, the layout designers need to work across multiple cellview. Also, the placement at the top level is oblivious to the actual lower level instances, so the top-level placement is based on estimation. After the design is routed, there might be changes needed to the top level that are often expensive to accommodate. With the Design Planner, layout engineers can seamlessly lay out, place, and route at both advanced and mature nodes, backed by real routing data.

Virtual Layout Hierarchy could be defined as the “yet-to-be-realized” layout hierarchy, which is entirely based on the schematic hierarchy.

The Virtual Layout Hierarchy is a “virtual” representation of not only the top-cell objects but also the objects at the lower levels, generating a completely flat hierarchy at the top-cell level.

Enhancements in the Virtuoso Design Planner

- The *Generate All From Source (GFS)* and *Generate Selected From Source (GSFS)* commands can be invoked from the Design Planner toolbar.
 - You can generate the layout as usual.
 - You can also choose to generate a layout hierarchy using the new *Virtual Hierarchy* option on the *Generate* tab.
- The generated virtual hierarchy is accessible as a data set in the Navigator assistant.
- Repetitive blocks of virtual hierarchy can be generated as clones.
 - These clones, too, are accessible through the Navigator assistant.
- The Design Planner enables you to adjust the boundary of a virtual hierarchy block.
 - Make Cell from the top-cell layout to be realized outside of the cellview.
 - Virtualize it back into the top cellview.
- The integrated *Congestion Analysis* tool enables you to plan your design more accurately, with actual routing data available to guide you for pin placement and positioning.

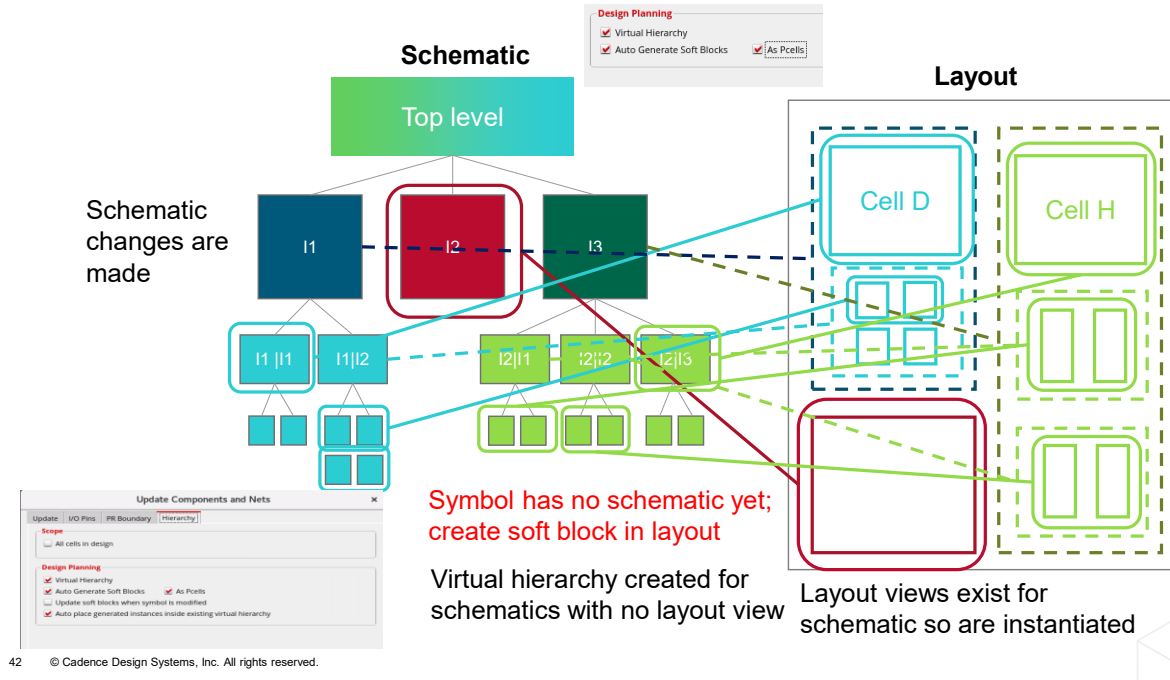
The Design Planner supports several enhancements to enable layout planning using a virtual hierarchy.



How to Generate a Virtual Layout Hierarchy

To generate a Virtual Layout Hierarchy, you can use the Virtuoso Design Planner, if you have opened your design in Layout EXL. The Design Planner supports several enhancements to enable layout planning using a virtual hierarchy.

The Design Planning: Generation Flow



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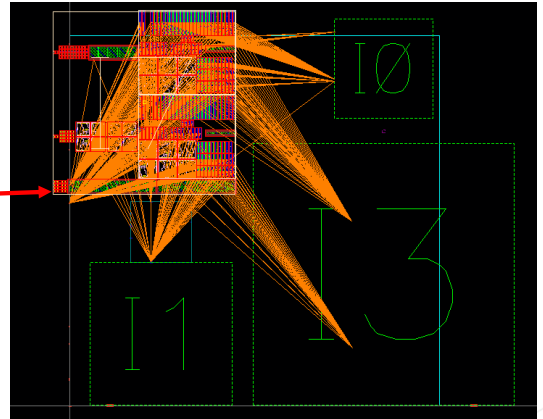
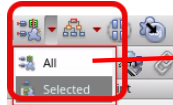
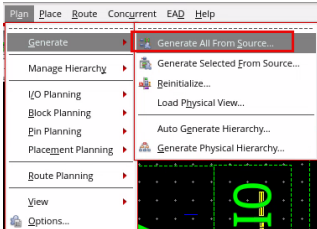
The Design Planning Toolbar: Generate Command



In the layout EXL canvas, you have **Design Planning > Generate Selected From Source (GSFS)** support in addition to **Generate All From Source (GFS)**.

These commands are available under the **Plan – Generate** menu as well in the layout EXL canvas.

- The virtual hierarchy can be created incrementally using GSFS.
- GFS is the expected use model.

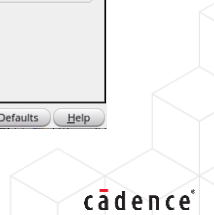


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The Generate Layout Form: PR Boundary Tab



- Controls the size of the virtual hierarchy and soft blocks
- Shape — Rectangle or Polygon, with Origin as the reference point
- Area Estimation controls the size calculation:
 - Utilization (%) and Aspect Ratio (W/H) define block area and shape
 - Estimator options: PR Boundary Based, User-defined, or Area Estimate CSV File (new)
 - Area Estimate CSV File imports exact dimensions for precise block sizing
- Virtual Hierarchy Area Boundary — Enclose by or Utilization (%), applied at Top Level or selected hierarchy
- Soft Block global control: Size = Area × Utilization. Acts as a placeholder unless overridden by the CSV file
- User-defined estimation functions can be registered via Register... for both virtual hierarchy and soft blocks



Let us discuss the PR Boundary tab in the Generate Layout form.

You can control the size of the virtual hierarchy and the soft blocks.

Virtual hierarchy has Enclose by or Utilization option.

Soft Block has a global control as shown here.

You can calculate the size by multiplying the Area and the Utilization values.

It is intended as a placeholder and will require further resizing.

User-defined estimation functions can be registered for both the virtual hierarchy and for the soft blocks.



Submodule 3-1

Generating the Virtual Hierarchy

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Submodule Objectives

In this submodule, you

- Generate the layout with virtual hierarchy
- Identify the virtual hierarchy
- View and control the levels of virtual hierarchies displayed
- Perform cross-selection of virtual hierarchies in the schematic and layout canvases



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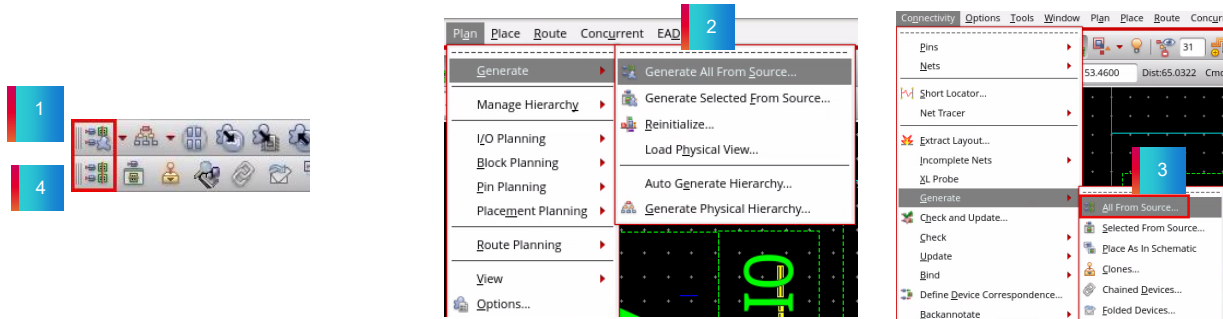
Generating the Layout with Virtual Hierarchy



To generate the layout with a virtual hierarchy, do one of the following:

1. Click the **Generate All From Source (GFS)** icon in the Design Planning toolbar.
2. Choose the **Plan – Generate – Generate All From Source** menu command.
3. Choose the **Connectivity – Generate – All From Source** menu command.
4. Click the **Generate All From Source (GFS)** icon in the Layout XL toolbar to launch the Generate Layout form.

Use the *GFS* command to generate the layout representations of the schematic design components.



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Generating the Layout with Virtual Hierarchy

Let us see how to generate the layout with Virtual Hierarchy.

To generate the layout with virtual hierarchy:

Click the Generate All From Source icon in the Design Planning toolbar.

Or choose Plan – Design Planning – Generate All From Source menu command.

Or choose Connectivity – Generate – All From Source menu command.

Or click the Generate All From Source icon in the Layout XL toolbar to launch the Generate Layout form.

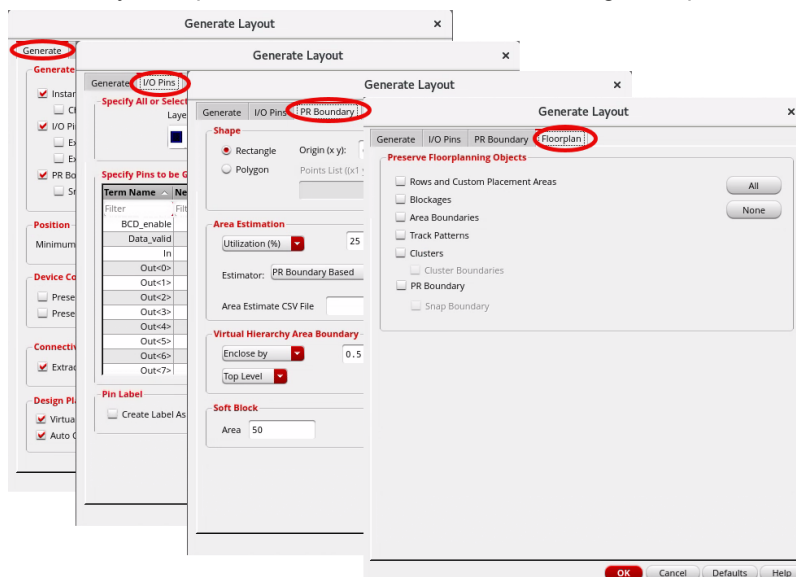
You can use the GFS command to generate the layout representations of the schematic design components.

Generate All From Source Interface

Use the Generate Layout form to generate the layout representations of the schematic design components.

The form is split into four tabs:

- Generate
- I/O Pins
- PR Boundary
- Floorplan



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Generate All From Source Interface

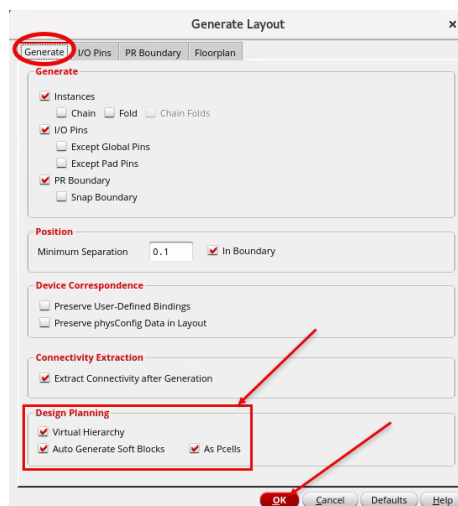
Generate All From Source Interface is shown here.

You can use the Generate Layout form to generate the layout representations of the schematic design components.

The form is split into four tabs namely, Generate, I/O Pins, PR Boundary, and Floorplan.

Generate Layout Form

- In the Generate Layout form, in the Design Planning group box:
 - Keep the *Virtual Hierarchy*, *Auto Generate Soft Blocks* and *As Pcells* options selected.
 - When the *Virtual Hierarchy* option is true, a virtual hierarchy is generated for schematics that have no corresponding layout.
 - When the *Auto Generate Soft Blocks* option is true, a soft block is created for symbols that do not have corresponding schematics.
 - When the *As P-cells* option is true, soft blocks are generated as parameterized cells (P-cells), allowing their dimensions to be driven by parameters or by the Area Estimate CSV File for precise block sizing.
 - Keep the other values as default in all the tabs.
 - Click **OK** in the form to generate the layout.



The Design Planning group box is available only when you have the Layout EXL license checked out.

In the Generate Layout form, in the Design Planning group box:

Keep Virtual Hierarchy, Auto Generate Soft Blocks, and As Pcells options selected.

When the Virtual Hierarchy option is true, a virtual hierarchy is generated for schematics that have no corresponding layout.

When the Auto Generate Soft Blocks option is true, a soft block is created for symbols that do not have corresponding schematics.

When the As Pcells option is true, soft blocks are generated as parameterized cells, allowing their dimensions to be driven by parameters or by the Area Estimate CSV File for precise block sizing.

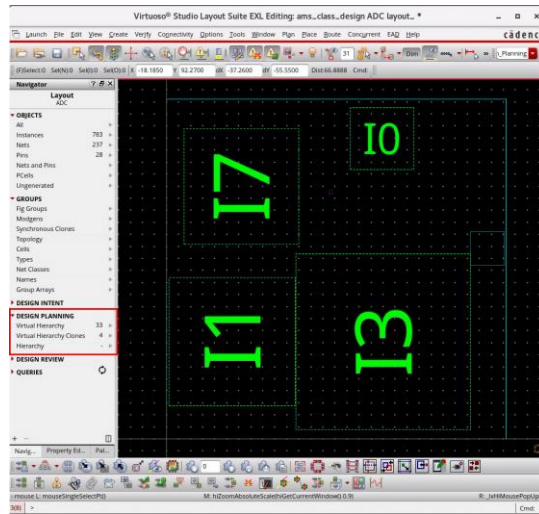
Keep the other values as default in all the tabs.

Click OK in the form to generate the layout.

Design Planning group box is available only when you have the Layout EXL license checked out.

Generated Layout

The layout is generated flat (opaque, Display level = 0), containing virtual hierarchies and a soft block.



I0, I1, I3, and I7 are virtual hierarchies, I11 is Virtual Hierarchy Clone, and I2 is a soft block.

Generated Layout

You see the generated layout here.

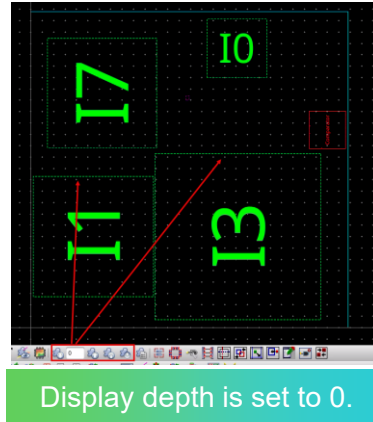
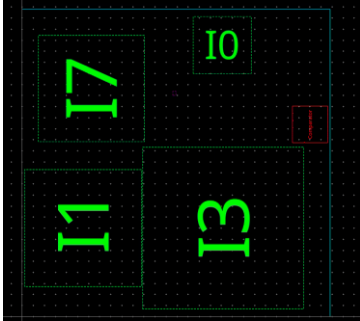
The layout is generated flat, that is opaque with display level 0, containing virtual hierarchies and a soft block.

In this layout, I0, I1, I3, and I7 are Virtual Hierarchies, I11 is Virtual Hierarchy Clone, and I2 is Soft Block.

Identifying the Virtual Hierarchy



- Virtual hierarchies are displayed with a green dashed line to make them stand out from real hierarchies.
- By default, virtual hierarchies are created opaque with their contents hidden.
- This can be seen in the Design Planning toolbar, where the display level is set to 0.



Let us see how to identify the Virtual Hierarchy.

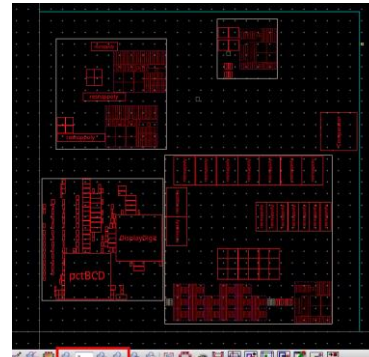
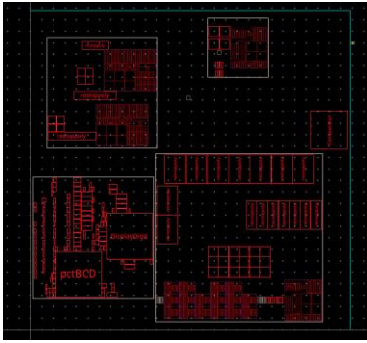
Virtual hierarchies are displayed with a green dashed line to make them stand out from other real hierarchies.

By default, the virtual hierarchies are created opaque with their contents hidden.

This can be seen in the Design Planning toolbar, where the display level is set to 0.

Identifying the Virtual Hierarchy (continued)

- You have the flexibility to display as much of the hierarchy as is needed.
- This can be seen in the Design Planning toolbar, where the display level is set to 3.
- As the virtual hierarchy is flat, you can hide levels that are not important for the current task.
- If you can see an object in a virtual hierarchy, you can interact with it.



Display depth is set to 3.

You have the flexibility to display as much of the hierarchy as is needed.

This can be seen in the Design Planning toolbar, where the display level is set to 3.

As the virtual hierarchy is flat, you can hide levels that are not important for the current task.

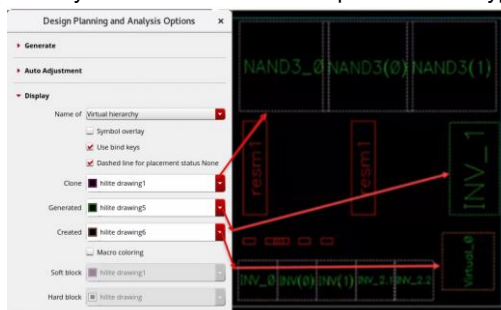
If you can see an object in a virtual hierarchy, you can interact with it.

Viewing the Virtual Hierarchy Types



To support ease of identification, the bounding boxes of the virtual hierarchies in the layout can be color-coded using the **Display** options in the **Design Planning and Analysis Options** form.

- Virtual hierarchies can be *created*, *cloned*, or *generated*.
- This means a layout in EXL can have three types of virtual hierarchies depending on how the virtual hierarchies were formed.
- You can display the virtual hierarchy bounding box using the default color for each type or select a color of your choice.
- The text labels on each virtual hierarchy use the same color irrespective of the type of virtual hierarchy they belong to.



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Viewing the Virtual Hierarchy Types

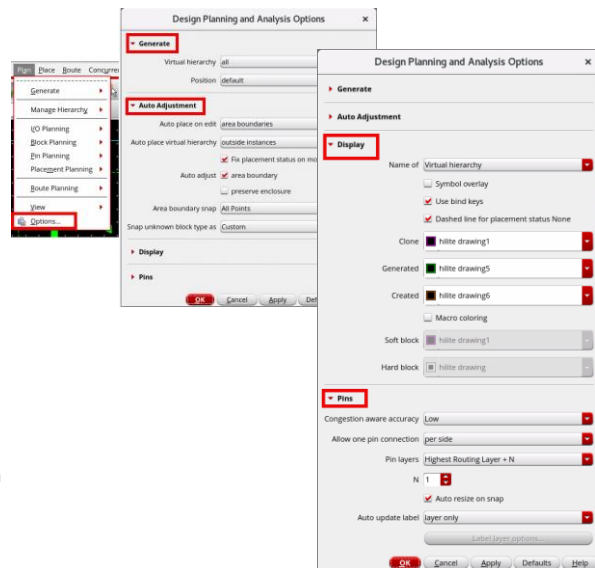
To support ease of identification, the bounding box of the virtual hierarchies in the layout can be color-coded differently using the **Display** options in the **Design Planning and Analysis Options** form.

- Virtual hierarchies can be *created*, *cloned*, or *generated*.
- This means a layout in EXL can have three different types of virtual hierarchies depending on how the virtual hierarchies were formed.
- You can display the virtual hierarchy bounding box using the default color for each type or select a color of your choice.
- The text labels on each virtual hierarchy use the same color irrespective of the virtual hierarchy type they belong to.

Configure the Design Planning and Analysis Options



Plan → Options → Design Planning and Analysis Options → configure, Generate, Auto Adjustment, Display, and Pins sections → OK



Open the Design Planning and Analysis Options form

- Configure the Generate section to define the virtual hierarchy and position settings.
- Adjust the Auto Adjustment section to control automatic placement behavior on edit, move, and area boundary settings.
- Set the Display section to manage naming conventions, symbol overlay, bind keys, and coloring options for blocks.
- Configure the Pins section to define congestion accuracy, pin connection rules, routing layers, and label update preferences.

The Design Planning and Analysis options control virtual hierarchy generation, block adjustment, display, and pin planning.

To open this form, select Plan → Options.

The form has four sections:

Generate — Define the virtual hierarchy and position settings.

Auto Adjustment — Control automatic placement behavior on edit, move, and area boundary settings.

Display — Manage naming conventions, symbol overlay, bind keys, and coloring options for blocks.

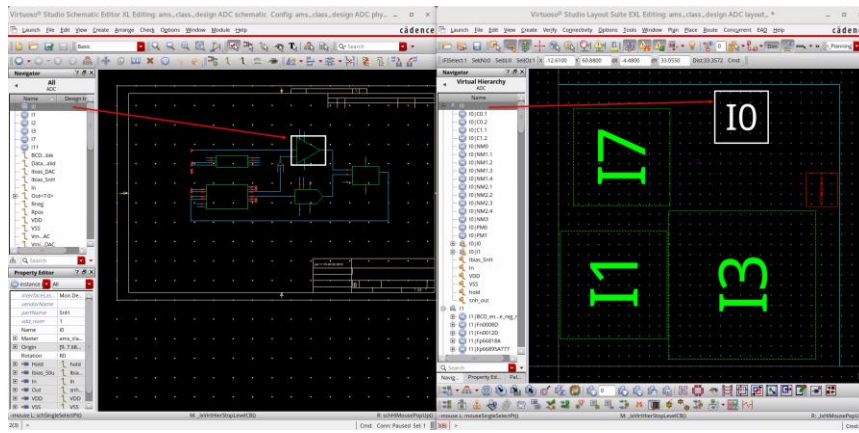
Pins — Set congestion accuracy, pin connection rules, routing layers, and label update preferences.

Once configured, click OK to apply.

Cross Selection of the Virtual Hierarchies in Schematic/Layout



- Selecting a Virtual Hierarchy in the layout canvas cross-selects its counterpart in the schematic.
- Virtual Hierarchies are connectivity-compliant and correct by construction.



Schematic Instance is bound to the corresponding Virtual Hierarchy Layout Instance as shown by the cross selection between Navigators and Canvases.



Cross Selection of the Virtual Hierarchies in Schematic/Layout

Cross Selection of the Virtual Hierarchies in the Schematic and Layout Canvases.

Selecting a Virtual Hierarchy in the layout canvas cross-selects its counterpart in the schematic.

Virtual Hierarchies are connectivity-compliant and correct by construction.

Schematic instance is bound to the corresponding Virtual Hierarchy Layout Instance as shown by the cross selection between Navigators and Canvases.



Submodule 3-2

Setting up the Soft Block P-cells Using CSV in Design Planning

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Module Objectives

In this module, you

- Generate Soft Blocks as P-cells
- Provide block dimensions using a CSV file
- Load the CSV file in the Generate All From Source (GFS) form
- Regenerate layout with updated parameters without deleting existing cell views



In this module, you will learn how to generate soft blocks as P-cells using the Design Planning tool. You will understand how to provide block dimensions through a CSV file, load that CSV file into the Generate All From Source (GFS) form, and regenerate the layout with updated parameters — without deleting existing cell views.

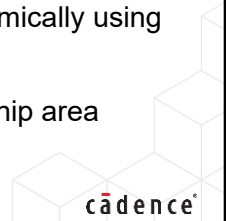
Soft Blocks as P-cells in Design Planning



In the Virtuoso Layout Suite, Design Planning enables engineers to plan the chip's physical layout at an early stage. When a block has a symbol, but no schematic or layout, a Soft Block is created as a placeholder. Soft Block P-cells are parameterized versions of these blocks, allowing designers to dynamically resize and regenerate them using external inputs such as CSV files, without manually deleting and recreating layouts.

Soft Blocks Overview

1. Design Planning functionality is available in VLS EXL and MXL tiers of the Virtuoso Layout Suite
2. A Soft Block is a placeholder block created when a symbol has no schematic or layout
3. A P-cell (Parameterized Cell) allows the block to be resized and regenerated dynamically using parameters
4. Soft Blocks are used during early floor planning to estimate block sizes and plan chip area



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How to Provide CSV File for Area Estimation?

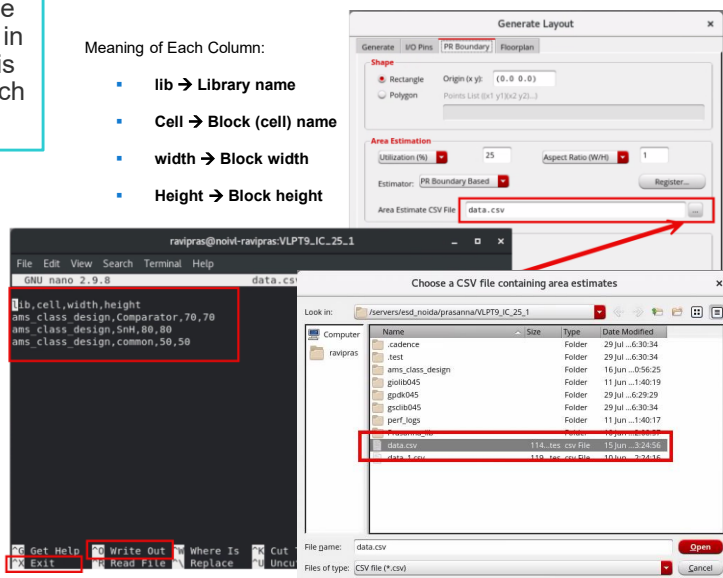


A CSV (Comma-Separated Values) file is a simple text file used to store data in a table format. In Design Planning, this file defines the width and height of each soft block.

Meaning of Each Column:

- lib → Library name
- Cell → Block (cell) name
- width → Block width
- Height → Block height

1. **Open** Terminal
2. **Create** a CSV file (data.csv) with block dimensions
3. In the PR Boundary → Area Estimation section, **click** the three-dot (...) button.
4. **Select** the CSV file that you created earlier (data.csv).
5. **Verify** that the file name is displayed in the field.



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A **CSV file**, or Comma-Separated Values file, is a simple text file used to store data in a table format. In Design Planning, this file defines the **width and height** of each soft block.

The CSV file has four columns — **lib** refers to the library name, **cell** refers to the block or cell name, **width** defines the block width, and **height** defines the block height. The values in the file depend on the designer's design requirements and block specifications.

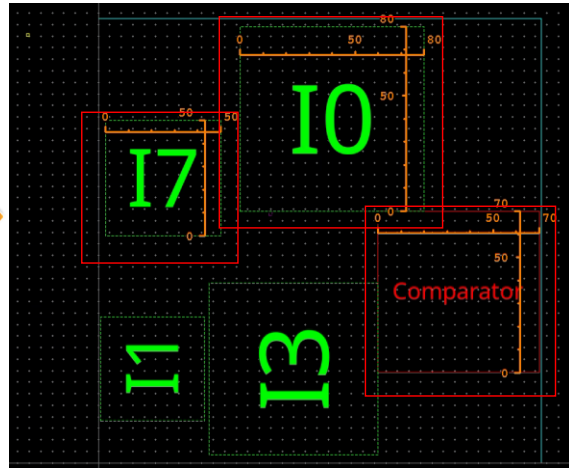
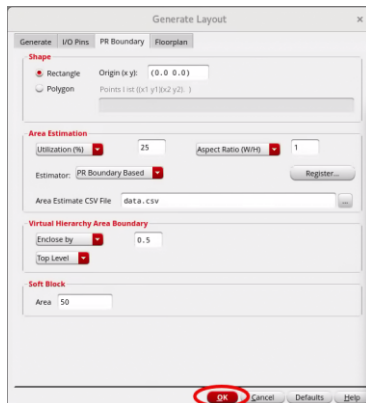
To create and load the CSV file:

Open the **Terminal** and create a CSV file named **data.csv** using the **nano** command with the required block dimensions based on your design needs. Alternatively, you can also use other Linux text editors like **vi** or **vim**, or use the **cat** command with output redirection to create the file directly from the terminal. Within Virtuoso, you can also generate a CSV file using **SKILL scripting** from the CIW console.

Then, go to the **PR Boundary** tab in the Generate Layout form. Under the **Area Estimation** section, click the **three-dot (...)** button to browse and select the CSV file you created. Verify that the file name is displayed in the field.

How to Provide CSV File for Area Estimation? (continued)

6. The Area Estimate CSV File is set to data.csv
7. Click OK to generate the layout.



Once the **Area Estimate CSV File** is set to **data.csv** in the PR Boundary tab, click **OK** to generate the layout.

The tool reads the block dimensions from the CSV file and generates the soft blocks on the layout canvas. As you can see, the block dimensions have changed based on the values provided in the CSV file.

The **I0** block is now **80 × 80**, the **I7** block is now **50 × 50**, and the **Comparator** block is now **70 × 70** — all matching the width and height values defined in the CSV file.

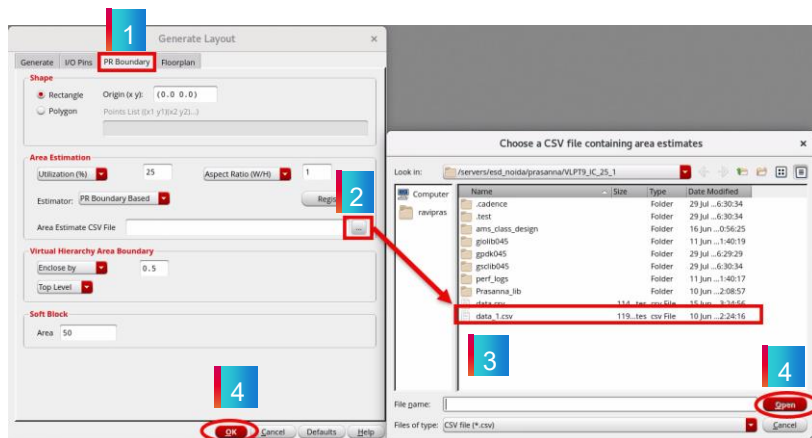
This confirms that the tool has successfully picked up the dimensions from the CSV and applied them to the corresponding soft blocks during layout generation.

How to Regenerate Layout Using a Revised CSV File



When block dimensions change, you can update the layout by simply loading a revised CSV file — no need to delete existing layouts. The tool automatically regenerates all soft block P-cells with the new dimensions from the updated file.

1. Navigate to the PR Boundary tab
2. Select the browse button next to Area Estimate CSV File
3. Select data_1.csv from the file browser
4. Select Open, then click OK



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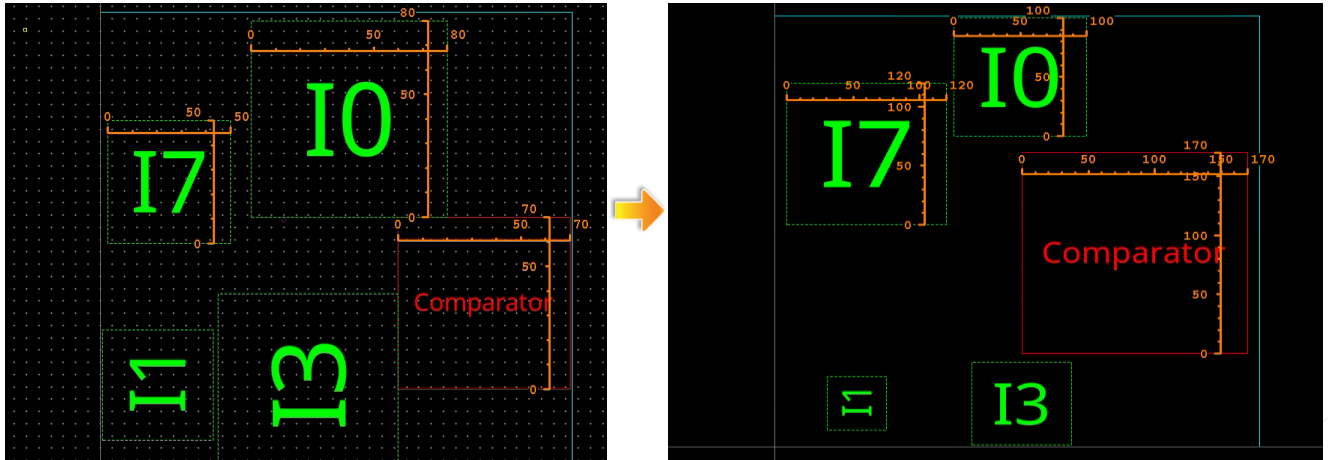
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When block dimensions change, you can update the layout by simply loading a revised CSV file — there is no need to delete existing layouts. The tool automatically regenerates all soft block P-cells with the new dimensions from the updated file.

To load the revised CSV file, navigate to the **PR Boundary** tab in the Generate Layout form. Under the **Area Estimation** section, click the **browse button (...)** next to the Area Estimate CSV File field. A file browser window opens — select the updated file, in this case **data_1.csv**, and click **Open**. Then click **OK** to regenerate the layout with the updated block dimensions.

How Revised CSV Dimensions Change the Layout: `data.csv` → `data_1.csv`

- After clicking OK, the blocks will regenerate with the revised dimensions from `data_1.csv`



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After clicking **OK**, the tool regenerates the layout with the revised dimensions from `data_1.csv` — without deleting the existing cell views.

As you can see in the comparison, the layout on the left shows the blocks generated using the original `data.csv` file, where **I0** was 80×80 , **I7** was 50×50 , and the **Comparator** block was 70×70 .

On the right, the layout has been regenerated using the updated `data_1.csv` file. The block dimensions have changed — **I0** is now 100×100 , **I7** is now 120×120 , and the **Comparator** block is now 170×170 — as per the new values defined in the revised CSV file.

This demonstrates the advantage of using soft blocks as P-cells — block dimensions can be updated dynamically by simply loading a revised CSV file, without the need to manually delete and recreate the layout.

Challenges in Traditional Soft Block Creation



In Design Planning, soft blocks are created when no schematic or layout exists. While the Adjust Boundary command allows resizing, it only provides temporary changes. To apply new dimensions, the existing layout must be deleted and regenerated, and only the deleted blocks get updated. This makes the process slow and inefficient, especially when handling multiple blocks.

Addressing the Issue with Soft Block P-cells

- Soft Block P-cells provide a parameterized approach to generate soft blocks
- Instances are automatically created in the design library
- Blocks can be resized by updating the area estimation
- No need to delete existing layouts during regeneration
- Ensures all blocks are updated consistently
- Improves efficiency and iteration speed in design planning



In Design Planning, soft blocks are created when no schematic or layout exists for a block. The **Adjust Boundary** command allows resizing, but the changes are only temporary. To apply new dimensions permanently, the existing layout must be deleted and regenerated, and only the deleted blocks get updated. This makes the process slow and inefficient, especially when handling multiple blocks.

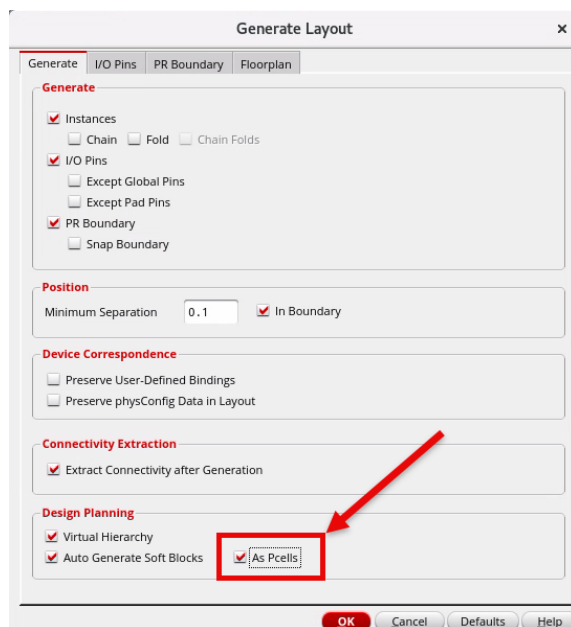
This challenge is addressed by **Soft Block P-cells**, which provide a parameterized approach to generate soft blocks. Instances are automatically created in the design library, and blocks can be resized simply by updating the area estimation, without the need to delete existing layouts during regeneration. This ensures all blocks are updated consistently and significantly improves efficiency and iteration speed in design planning.

Generating Soft Blocks as P-cells

By enabling this P-cell option in the Generate Layout command, you can generate soft blocks as parameterized P-cells and easily resize them by updating the area estimation without deleting existing layouts.

Soft Blocks Overview

1. Enable As P-cells in Generate Layout
2. Provide area estimation (CSV or default)
3. Click OK to generate layout
4. Regenerate with updated values



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By enabling the **As P-cells** option in the **Generate Layout** command, soft blocks can be generated as parameterized P-cells and easily resized by updating the area estimation, without deleting the existing layouts.

The process involves four simple steps. First, enable **As P-cells** in the **Generate Layout** form under the **Design Planning** section. Next, provide the area estimation either through a **CSV file** or by using the default values. Then, click **OK** to generate the layout with soft block P-cells. Finally, regenerate the layout with updated values whenever block dimensions need to change, without removing the existing layout.

This parameterized approach makes soft block creation faster, more consistent, and significantly improves iteration speed during design planning.



Submodule 3-3

Accessing the Design Planning Data Sets in the Navigator Assistant

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Submodule Objectives

In this submodule, you

- Identify the Design Planning data sets in the Navigator assistant
- Access the virtual hierarchies
- Access the virtual hierarchy clones
- Access the hierarchical objects



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What Is the DESIGN PLANNING Data Set?

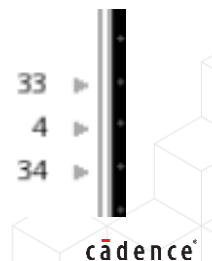
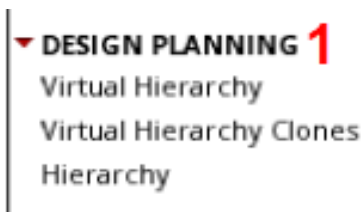


In Virtuoso Layout EXL, the DESIGN PLANNING data set is used for displaying the generated virtual hierarchies and related components.

The Navigator assistant supports three DESIGN PLANNING data sets: *Virtual Hierarchy*, *Virtual Hierarchy Clones*, and *Hierarchy*.

These predefined data sets are available under the DESIGN PLANNING category in the Navigator assistant.

- The counts under the Design Planning data subset show that virtual hierarchies and virtual hierarchy clones have been created, as shown in graphic 1.
- The *Hierarchy* subset does not display a count until you expand it for performance reasons, as shown in graphic 2.



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What Is DESIGN PLANNING Data Set?

In Virtuoso Layout EXL canvas, DESIGN PLANNING Data Set is used for displaying the generated virtual hierarchies and related components.

The Navigator assistant supports 3 DESIGN PLANNING Data Sets, namely: Virtual Hierarchy, Virtual Hierarchy Clones, and Hierarchy.

These predefined data sets are available under the DESIGN PLANNING category in the Navigator assistant.

The counts under the Design Planning data subset show that virtual hierarchies and virtual hierarchy clones have been created as shown in the graphic 1.

The Hierarchy subset does not display a count until you expand it for performance reasons as shown in the graphic 2.

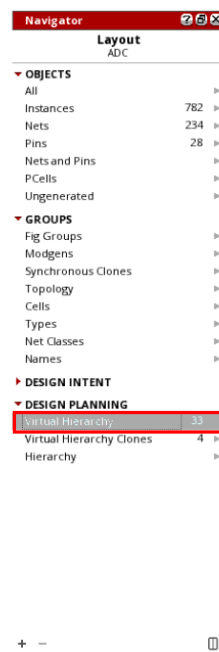
Accessing the Virtual Hierarchies



In the layout EXL canvas, select from the **Navigator** → **DESIGN PLANNING** → **Virtual Hierarchy** data set to access the generated virtual hierarchies and interface nets.

- The virtual hierarchies generated in the layout are categorized under the *Virtual Hierarchy* data set in the Navigator assistant.

Use the *Virtual Hierarchy* data set to access the generated virtual hierarchies and interface nets in the layout EXL canvas.



Accessing the Virtual Hierarchies

Let us discuss how to access the Virtual Hierarchies.

In the Layout EXL canvas, select from the **Navigator** – **Design Planning** – **Virtual Hierarchy** data set to access the generated virtual hierarchies and interface nets.

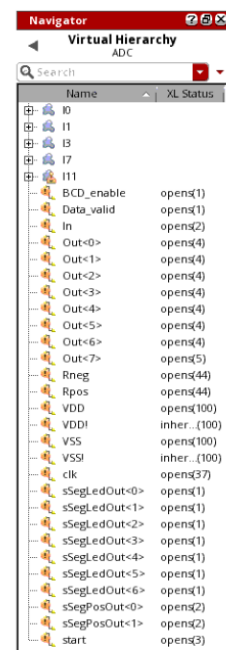
The virtual hierarchies generated in the layout are categorized under the Virtual Hierarchy data set in the Navigator assistant.

You can use the Virtual Hierarchy Data Set to access the generated virtual hierarchies and interface nets in the layout EXL canvas.

Displaying the Virtual Hierarchies

- The Navigator tree updates to display the available virtual hierarchies and the associated instances and nets.
- The *Virtual Hierarchy* is represented using an amoeba-shaped icon. 
- There are five top-level virtual hierarchies that have their own icon to help with identification.
- Top-level nets are also shown in the *Virtual Hierarchy* view.

If the *Virtual Hierarchy* is created as a group, it is represented using the group icon. 



Displaying the Virtual Hierarchies

The Navigator tree updates to display the available virtual hierarchies and the associated instances and nets.

The Virtual Hierarchy is represented using an amoeba-shaped icon.

There are five top-level virtual hierarchies that have their own icon to help with identification.

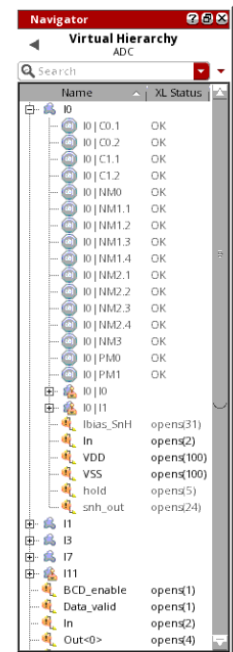
Top-level nets are also shown in the Virtual Hierarchy view.

If the Virtual Hierarchy is created as a group, it is represented using the group icon.

Expanding the Virtual Hierarchies

- Virtual hierarchies can be further expanded to show their contents.
- Here, **I0** is expanded.
 - The interface nets are shown at each level.
 - This allows you to easily find the connectivity between the virtual hierarchies.

The Navigator assistant displays the XL status of the virtual hierarchy components.



Expanding the Virtual Hierarchies

A virtual hierarchy can also contain an instance (or more) of another virtual hierarchy as one of the components. Moving such a virtual hierarchy around the layout canvas will display flight lines on the canvas. However, if the virtual hierarchy you move is opaque, which means it has no contents, no flight line will be generated to the virtual hierarchy.

Virtual hierarchies can be further expanded to show their contents.

Here, **I0** is expanded.

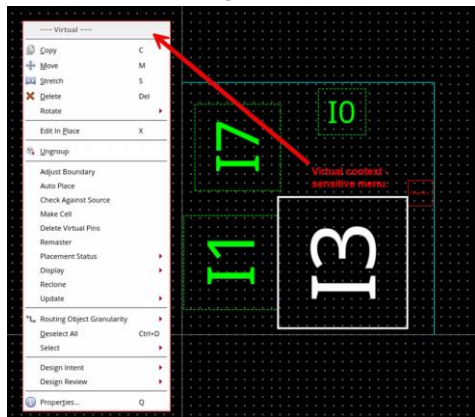
The interface nets are shown at each level.

This allows you to easily find the connectivity between the virtual hierarchies.

The Navigator assistant displays the XL Status of the virtual hierarchy components.

Using the Context-Sensitive Menu of the Virtual Hierarchies

Right-click a virtual hierarchy, say for example **I3**, in the Navigator tree or in the layout EXL canvas to perform any of the operations supported through the context menu.



When invoked from the Navigator assistant, the *Virtual* context-sensitive menu displays an additional command: *Show Content*.



Using the Context-Sensitive Menu of the Virtual Hierarchies

Right-click a virtual hierarchy, say for example **I3**, in the Navigator tree or in the layout EXL canvas to perform any of the operations supported through the context menu.

When invoked from the Navigator assistant, the virtual context-sensitive menu displays an additional command, namely, *Show Content* as seen here.

Accessing the Virtual Hierarchy Clones

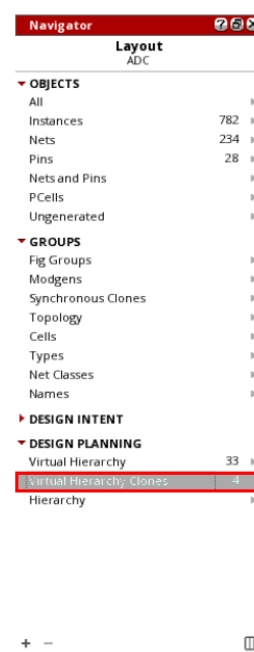


In the layout EXL canvas, select from the **Navigator** → **DESIGN PLANNING** → **Virtual Hierarchy Clones** data set to access the generated virtual hierarchy clones.

- The virtual hierarchy clones generated in the layout are categorized under the *Virtual Hierarchy Clones* data set in the Navigator assistant.
- If multiple instances of the selected schematic instance need to be created, the Design Planner clones the instances and adds them to the Design Planning Data Sets.

Use the *Virtual Hierarchy Clones* data set to access the generated virtual hierarchy clones in the layout EXL canvas.

Clones are created where virtual hierarchies have the same schematic master.



Accessing the Virtual Hierarchy Clones

Let us discuss how to access the Virtual Hierarchy Clones.

In the Layout EXL canvas, select from the Navigator – Design Planning – Virtual Hierarchy Clones data set to access the generated virtual hierarchy clones.

The virtual hierarchy clones generated in the layout are categorized under the Virtual Hierarchy Clones data set in the Navigator assistant.

If multiple instances of the selected schematic instance need to be created, the Design Planner clones the instances and adds them to the Design Planning Data Sets.

You can use Clones Data Set to access the generated virtual hierarchy clones in the layout EXL canvas.

Virtual Hierarchy Clones are automatically generated and are given the name of the first clone in the family.

Clones are created where virtual hierarchies have the same schematic master.

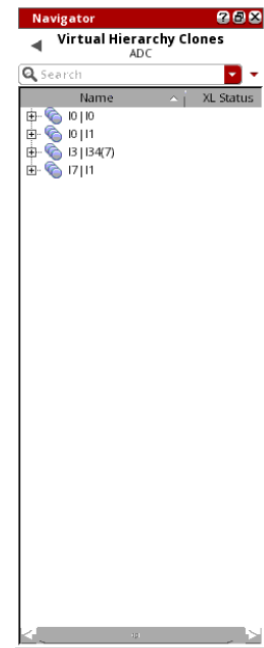
If you are creating layout views, these would be instances of the same layout master.

In the Design Planner, you use clones to keep the virtual hierarchies identical.

Displaying the Virtual Hierarchy Clone Families

- The Navigator tree updates to display the available virtual hierarchy clone families.
- The *Virtual Hierarchy Clone Family* is represented using the icon shown here in the Navigator assistant.
- There are four virtual hierarchy clone families.
- These are automatically generated and are given the name of the first clone in the family.

Virtual Hierarchy Clones are created where virtual hierarchies have the same schematic master.



Displaying the Virtual Hierarchy Clone Families

The Navigator tree updates to display the available virtual hierarchy clone families.

The Virtual Hierarchy Clone Family is represented using the icon shown here in the Navigator assistant.

There are four virtual hierarchy clone families as seen here.

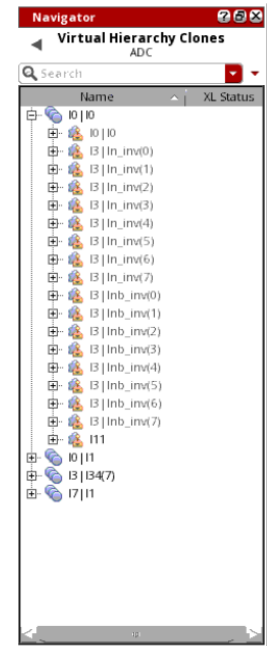
These are automatically generated and are given the name of the first clone in the family.

Virtual Hierarchy Clones are created where virtual hierarchies have the same schematic master.

Expanding the Virtual Hierarchy Clone Families

- Virtual hierarchy clone families can be further expanded to show their contents.
- Here, **I0|I0** in the Navigator tree is expanded.
 - The corresponding clone family members are shown here.

The *Virtual Hierarchy Clones* are represented using the icon shown here in the Navigator assistant.



Expanding the Virtual Hierarchy Clone Families

Virtual hierarchy clone families can be further expanded to show their contents.

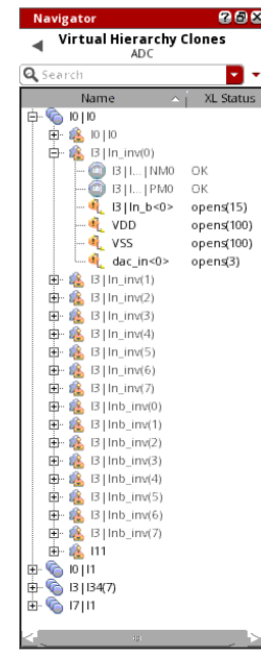
Here, **I0|I0** is expanded as seen in the Navigator tree.

The corresponding clone family members are shown here.

The Virtual Hierarchy Clones are represented using the icon shown here in the Navigator assistant.

Expanding the Virtual Hierarchy Clones

- Virtual hierarchy clones can be further expanded to show their contents.
- Here, **I3|In_inv(0)** in the Navigator tree is expanded.
 - The objects inside the virtual hierarchy clone are shown here.



Expanding the Virtual Hierarchy Clones

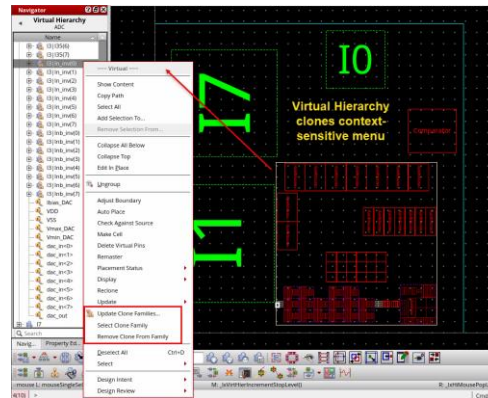
Virtual hierarchy clones can be further expanded to show their contents.

Here, **I3|In_inv(0)** is expanded as seen in the Navigator assistant.

The objects inside the virtual hierarchy clone are shown here.

Using the Context-Sensitive Menu of the Virtual Hierarchy Clones

Right-click a virtual hierarchy clone, say for example **I3|In_inv(0)**, in the Navigator tree or in the layout EXL canvas to perform any of the operations supported through the context menu.



A virtual hierarchy clone supports three additional commands in the context-sensitive menu: *Update Clone Families*, *Select Clone Family*, and *Remove Clone From Family*.



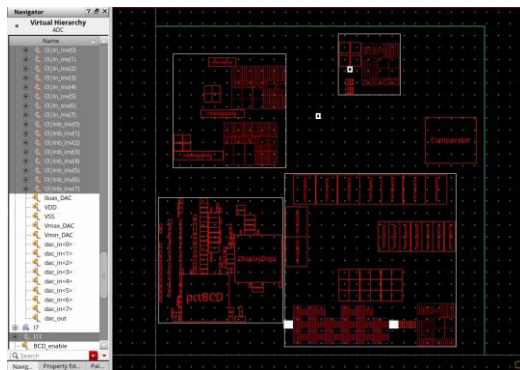
Using the Context-Sensitive Menu of the Virtual Hierarchy Clones

Right-click a virtual hierarchy clone, say for example **I3|In_inv(0)**, in the Navigator tree or in the layout EXL canvas, to perform any of the operations supported through the context menu.

A virtual hierarchy clone supports three additional commands in the context-sensitive menu, namely *Update Clone Families*, *Select Clone Family*, and *Remove Clone From Family*.

Cross Selection of the Contents in the Virtual Hierarchy Clone Families in Navigator/Layout

- In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to make all the virtual hierarchies opaque.
- In the layout canvas, in the Design Planning toolbar, set *Increase Display Depth* to **2** with no virtual hierarchies selected.
 - Click the (+) button next to **I0|I0** in the Navigator assistant to expand the clone family.
 - Select the contents of this clone family in the Navigator; they will cross-select in the canvas.



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Cross Selection of the Contents in the Virtual Hierarchy Clone Families in Navigator/Layout

In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to make all the virtual hierarchies opaque.

In the layout canvas, in the Design Planning toolbar, set *Increase Display Depth* to **2** with no virtual hierarchies selected.

Click the (+) button next to **I0|I0** in the Navigator assistant to expand the clone family.

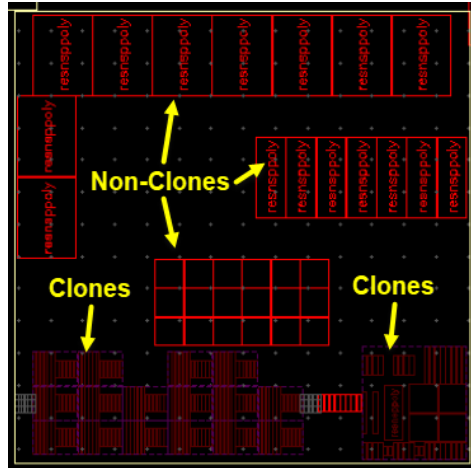
Select the contents of this clone family in the Navigator; they will cross-select in the canvas as well.

Clones do not need to be in the same virtual hierarchy.

The only condition that must be fulfilled for clones to be created is that they have the same schematic master.

Non-Clones and Clones Objects in the Virtuoso Design Planner

- Virtual hierarchy clones are controlled by the cloning infrastructure.
- You **cannot** select their contents without doing an **Edit In Place** first.
- Initially, the clones appear dimmed on the canvas which helps with their identification.



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Non-Clones and Clones Objects in the Virtuoso Design Planner

Virtual hierarchy clones are controlled by the cloning infrastructure.

You cannot select their contents without doing an Edit In Place first.

Initially, the clones appear dimmed on the canvas which helps with their identification.

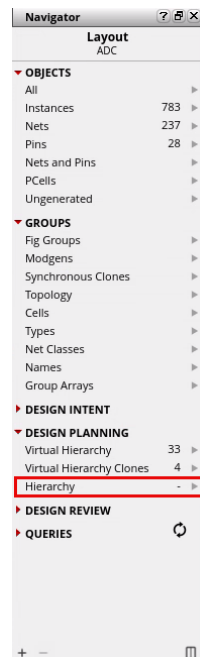
Accessing the Hierarchical Objects



In the layout EXL canvas, select from the **Navigator → DESIGN PLANNING → Hierarchy** data set to access the generated hierarchy in the layout canvas.

- The hierarchical objects associated with the generated virtual hierarchy are categorized under the *Hierarchy* data set in the Navigator assistant.

Use the *Hierarchy* data set to access the generated hierarchy in the layout EXL canvas.



Accessing the Hierarchical Objects

Let us see how to access the Hierarchical Objects.

In the layout EXL canvas, select from the **Navigator – DESIGN PLANNING – Hierarchy** data set to access the generated hierarchy in the layout canvas.

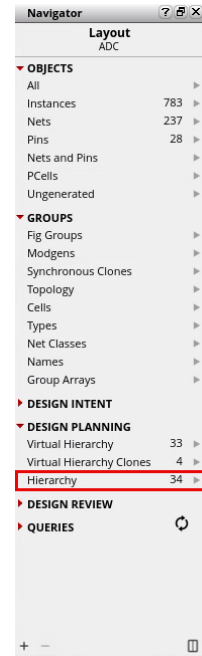
The hierarchical objects associated with the generated virtual hierarchy are categorized under the Hierarchy data set in the Navigator assistant.

You can use the Hierarchy Data Set to access the generated hierarchy in the layout EXL canvas.

Displaying the Counts in the Hierarchy Data Set

- Click the **Back**-arrow button to see that, in the Navigator assistant, the counts under the Design Planning data subset show that hierarchies have been created.

The *Hierarchy* subset does not display a count until you expand it for performance reasons.



Displaying the Counts in the Hierarchy Data Set

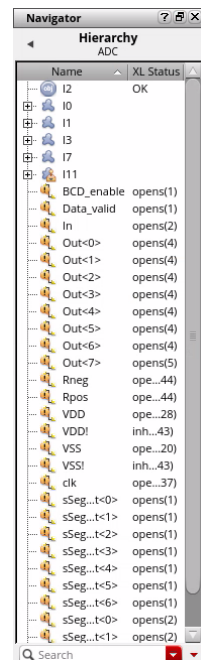
Click the back-arrow button to see that in the Navigator assistant, the counts under the Design Planning data subset show that hierarchies have been created.

The Hierarchy subset does not display a count until you expand it for performance reasons.

Displaying the Hierarchical Objects

- Click the **Hierarchy** data set again.
- The Navigator tree updates to display the available hierarchical instances.
- The *Hierarchy* data set shows both the virtual and real hierarchies.
 - This category has one more entry than the *Virtual Hierarchy*, a total of six.
 - This is because **I2** has been created as a soft block.

The Navigator assistant displays the *XL Status* of the hierarchical objects.



Displaying the Hierarchical Objects

Click the Hierarchy data set again.

The Navigator tree updates to display the available hierarchical instances.

The Hierarchy data set shows both the virtual and real hierarchies.

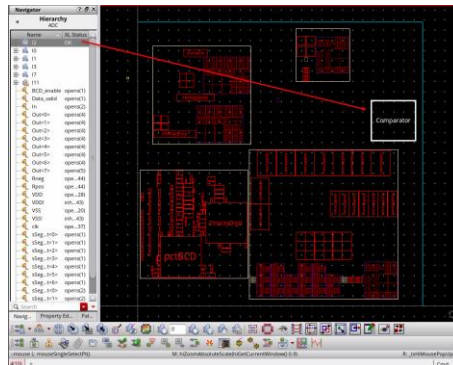
This category has one more entry than the virtual hierarchy, totally six.

This is because I2 has been created as a soft block.

The Navigator assistant displays the *XL Status* of the hierarchical objects.

Generated Layout with Depth Level Set as 2

- **I2** is created as a soft block.
 - There is no schematic for the **Comparator** symbol.
 - The *Auto Generate Soft Blocks* option was set to *true* during generation.



The soft block instance is represented using the icon  shown here in the Navigator assistant.



Generated Layout with Depth Level Set as 2

I2 is created as a soft block.

Because there is no schematic for the Comparator symbol.

The *Auto Generate Soft Blocks* option was set to true during generation.

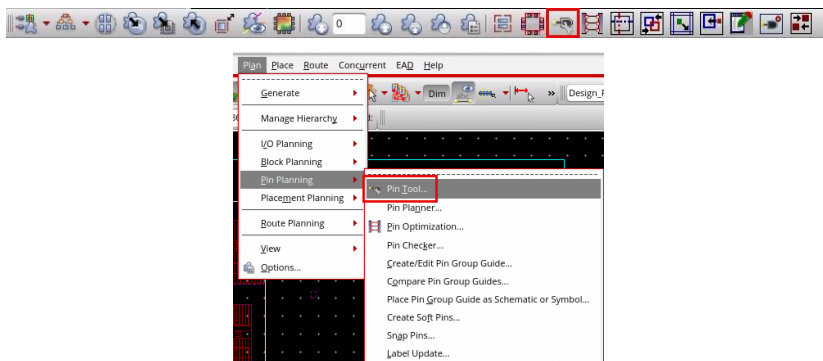
The soft block instance is represented using the icon shown here in the Navigator assistant.

Using the Pin Tool Form for Soft Blocks in the Design Planner



In the layout EXL canvas, choose the **Plan – Pin Planning – Pin Tool** menu command or click the **Pin Tool** icon in the Design Planning toolbar to open the Pin Tool form.

Use the *Pin Tool* form to review/update the pins in the created soft block.



Using the Pin Tool Form for Soft Blocks in the Design Planner

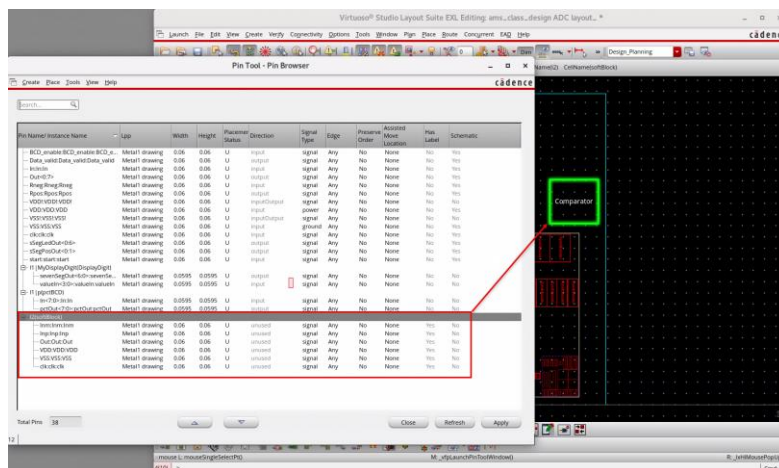
Virtuoso Floorplanner includes the Pin Tool, which is a unified interface for all pin-related tasks, such as creating pins, resizing pins, planning and optimizing pins, editing pin attributes, and setting their location constraints.

In the layout EXL canvas, choose the **Plan – Pin Planning – Pin Tool** menu command or click the **Pin Tool** icon in the Design Planning toolbar to open the Pin Tool form.

You can use the *Pin Tool* form to review and/or update the pins in the created soft block.

The Pin Tool Form

Use the *Pin Tool* form to select and review/update the created pins in the soft block.



The Pin Tool Form

Before launching the Pin Tool, ensure that the blocks in the design are placed.

You can then use the Pin Tool to define the pin location constraints, alignments, and pin group guides, and set the pin attributes.

You can use the Pin Tool form to select and review and/or update the created pins in the soft block.



Submodule 3-4

Viewing the Virtual Hierarchy

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Submodule Objectives

In this submodule, you

- Set display controls and adjust display depth in the Design Planning toolbar
- List display depth options
- Make virtual hierarchies opaque
- Control display levels of virtual hierarchies
- Add and remove display overrides



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What Is the Display Control Option?

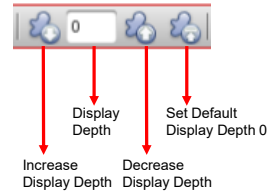


Use the **Display Control Options** in the Design Planning toolbar to set/control the display depth.

You can control how much of the virtual hierarchy is visible either on a global basis or individually by setting an override on selected virtual hierarchies.

The Design Planner toolbar provides the following Display Control options to:

- Increase Display Depth
- Display Depth
- Decrease Display Depth
- Set Default Display Depth 0



By default, when a virtual hierarchy is generated, the Design Planner sets its display depth to 0 to hide the contents within.



Setting the Display Controls

Since the layout is created flat, it is important to be able to control the display of the extra information contained in the layout view.

You can control how much of the virtual hierarchy is visible, either on a global basis or individually by setting an override on selected virtual hierarchies. You saw in the generation step that the default display depth is **0**. The display depth can be controlled from the toolbar for global and for the overrides or using the context menu for overrides only. Bindkeys are also supported.

What Is Display Controls?

You can use the Display Controls in the Design Planning toolbar to set and control the display depth.

You can control how much of the virtual hierarchy is visible, either on a global basis or individually by setting an override on selected virtual hierarchies.

The Design Planner toolbar provides the following display control options.

Increase Display Depth, Display Depth, Decrease Display Depth, and Set Default Display Depth 0.

By default, when a virtual hierarchy is generated, the Design Planner sets its display depth to 0 to hide the contents within.

Display Controls: Display Depth Options

Display Depth options in the Design Planner toolbar are listed in this table:

Increase Display Depth	Increases the display depth to display the objects at a level deeper in the hierarchy than the current display level. The maximum display depth level is 32 .
Display Depth	Enables you to type in the value of the required hierarchy level.
Decrease Display Depth	Decreases the display depth to display the objects at a level higher in the hierarchy than the current display level. The minimum display depth level is 0 .
Set Default Display Depth 0	Removes all stop level overrides and sets the default display depth to 0 .

You can use bindkeys to control the display depth.
Use **Ctrl, Shift +** to increment and **Ctrl -** to decrement the display depth.



Display Controls: Display Depth Options

By default, when a virtual hierarchy is generated, the Design Planner sets its display depth to **0** to hide the contents within. This helps reduce the clutter at the top level, which is otherwise possible, given that a virtual hierarchy is flat at the top level. You can, however, choose to display (or hide) the virtual hierarchy at different levels, depending on the task being completed. The Design Planner enables you to control the display level at a global level, although it also honors the display overrides that you may have set on a specific virtual hierarchy.

Display Controls: Display Depth Options

Display Depth Options in the Design Planner toolbar are:

- **Increase Display Depth:** Increases the display depth to display the objects at a level deeper in the hierarchy than the current display level
- **Display Depth:** Enables to type in the value of the required hierarchy
- **Decrease Display Depth:** Decreases the display depth to display the objects at a level higher in the hierarchy than the current display level
- **Set Default Display Depth 0:** Removes all stop-level overrides and sets the default display depth to **0**

You can use bindkeys to control the display depth.

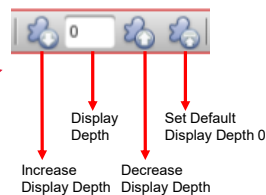
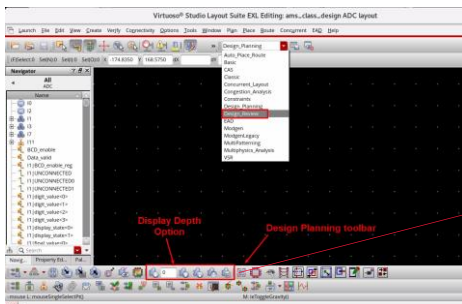
Use **Ctrl, Shift+** to increment, and **Ctrl-** to decrement the display depth.

Setting Display Depth



In the layout EXL canvas, select from **Design Planner → Design Planning Toolbar → Display Depth Options** to set/control the display depth.

- The Design Planner enables you to control the display depth globally for all the components in the design, or only for the selected components.
- You can choose to navigate and display the design.



Setting Display Depth

In the Layout EXL canvas, select from **Design Planner – Design Planning Toolbar – Display Depth Options** to set and control the display depth.

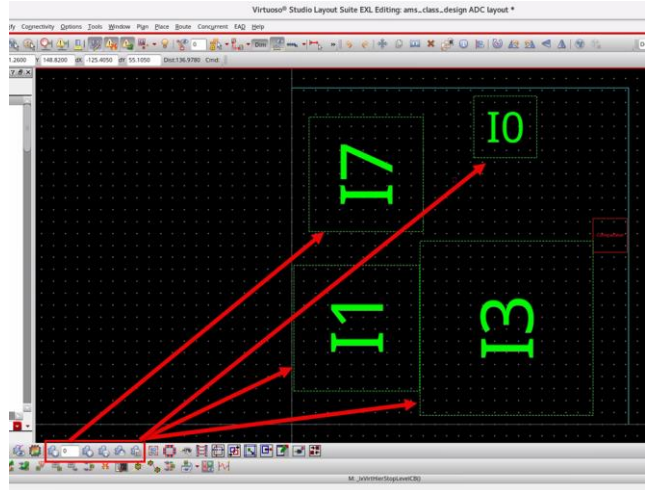
The Design Planner enables you to control the display depth globally for all the components in the design or only for the selected components.

You can choose to navigate and display the design.

Since the top cell layout of a virtual hierarchy is flat, setting the display control enables you to show connectivity to the lower-level instances or the virtual hierarchy at a lower level, which is not possible when working with real cellviews. This access to the connectivity information to the lower-level cells allows you to make more context-sensitive connections, and eventually helps with doing more efficient design plans.

Making All the Virtual Hierarchies Opaque

- In the layout EXL canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar.
- It makes all the virtual hierarchies opaque, Display Level = 0.



Making All the Virtual Hierarchies Opaque

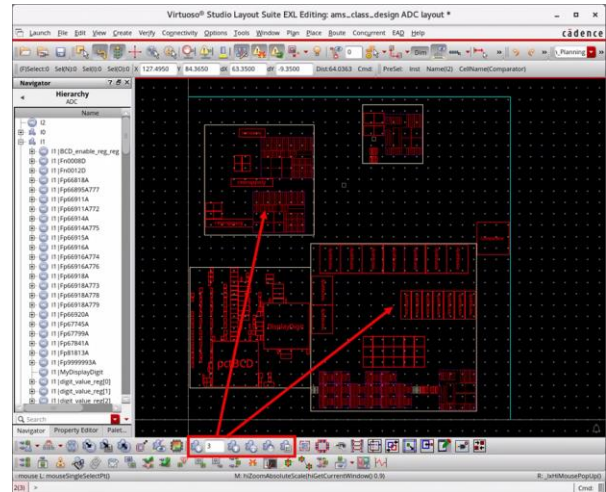
Let us see how to make all the Virtual Hierarchies opaque.

In the Layout EXL canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar.

It makes all the virtual hierarchies opaque, that is, Display level = 0.

Controlling the Levels of Virtual Hierarchies Displayed

- With no virtual hierarchy selected, click the **Increase Display Depth** button in the Design Planning toolbar to say, for example, **3**.
- This controls the number of levels of virtual hierarchies, in this example, **3** is displayed.
- The count in the *Global Display Depth* field will change to reflect the current level.
- You can also type in the required value in the *Display Depth* field if needed.
- You can use the *Decrease Display Depth* toolbar button to return to a lower level of hierarchy.



Controlling the Levels of Virtual Hierarchies Displayed

Let us see how to control the levels of Virtual Hierarchies displayed.

With no virtual hierarchy selected, click the **Increase Display Depth** button in the Design Planning toolbar, to say for example 3.

This controls the number of levels of virtual hierarchies displayed. In this case, the level is 3.

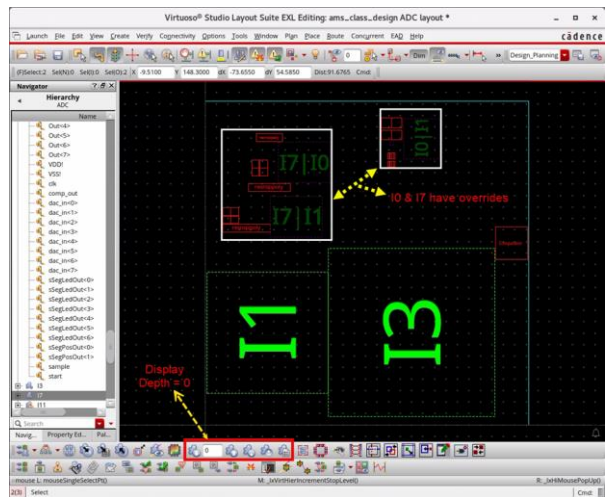
The count in the Global Display Depth field will change to reflect the current level.

You can also type in the required value in the Display Depth field, if needed.

You can use the Decrease Display Depth toolbar button to return to lower level of hierarchy.

Adding Display Overrides for the Virtual Hierarchies

- A global visibility control may not be sufficient during design planning. So, any VH can have a visibility override.
- With one or more virtual hierarchies selected, say, for example, after selecting **I0 & I7**, click the **Increase Display Depth** toolbar button.
 - This changes the display depth of the selected virtual hierarchies, as needed.
 - The count in the *Global Display Depth* field remains **0**, i.e., unchanged.
- The display depth is changed for the selected virtual hierarchies (**I0 & I7**) while the other virtual hierarchies are unchanged.
 - This is due to an override that prevents the global display depth from any changes.



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Adding Display Overrides for the Virtual Hierarchies

Let us see how to Add Display Overrides for the virtual hierarchies.

A global visibility control may not be sufficient during design planning. So, any Virtual Hierarchy can have a visibility override.

With one or more virtual hierarchies selected, say for example, after selecting I0 and I7, click the **Increase Display Depth** toolbar button.

This changes the display depth of the selected virtual hierarchies, as needed.

The count in the Global Display Depth field remains 0, that is, unchanged.

The display depth is changed for the selected virtual hierarchies, that is, I0 and I7 in this case, while the other virtual hierarchies are unchanged.

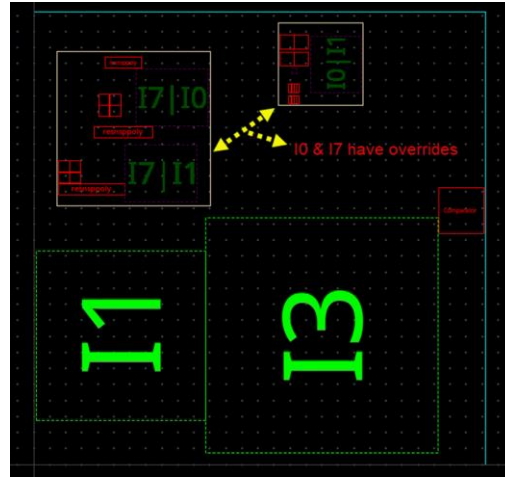
This is due to an override that prevents the global display depth from any changes.

A global display level may not be enough during design planning, and you may wish to see a lower level for some virtual hierarchies that will leave the others opaque.

You can use a display override to do the same.

Removing Display Overrides for the Virtual Hierarchies

Virtual Hierarchies **I0** & **I7** are overridden in the canvas.



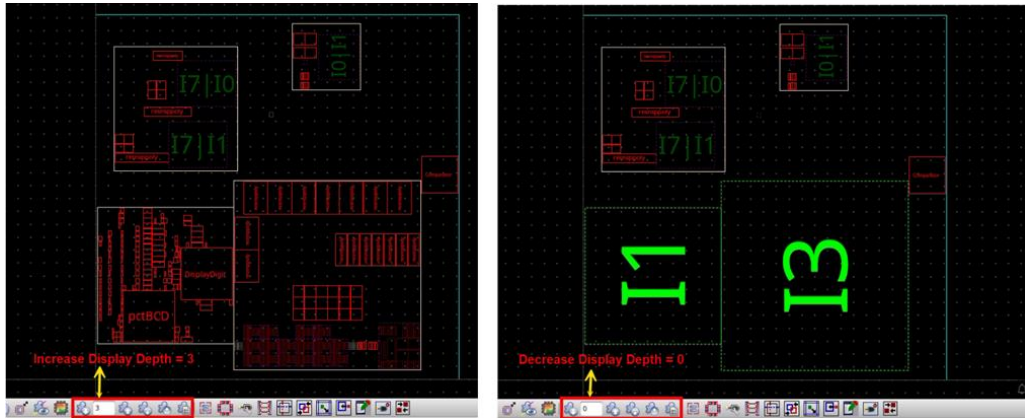
Removing Display Overrides for the Virtual Hierarchies

Let us see how to remove the Display Overrides for the virtual hierarchies.

Virtual Hierarchies **I0** and **I7** are overridden in the canvas as seen here.

Removing Display Overrides for the Virtual Hierarchies (continued)

- With no virtual hierarchy selected, click the **Increase and/or Decrease Display Depth** toolbar buttons.
- This changes the global display depth, as needed.
- The virtual hierarchies with the overrides (**I0 & I7**) are not changed.



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Removing Display Overrides for the Virtual Hierarchies

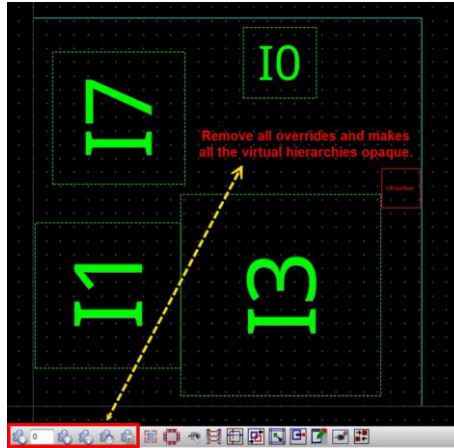
With no virtual hierarchy selected, click the **Increase Display Depth** or **Decrease Display Depth** toolbar button.

This changes the global display depth, as needed.

The virtual hierarchies with the overrides, in this case, I0 and I7 are not changed.

Removing Display Overrides for the Virtual Hierarchies (continued)

- In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar.
 - The overrides are removed, and the *Display Depth* is restored to **0**.



Removing Display Overrides for the Virtual Hierarchies

In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar.

The overrides are removed, and the Display Depth is restored to 0 making all the virtual hierarchies opaque.



Submodule 3-5

Analyzing the Net Connectivity

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Submodule Objectives

In this submodule, you

- Use net connectivity information in the layout EXL canvas
- Select nets in the Navigator assistant
- Use the Analyze Connectivity command to evaluate connections



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Using the Net Connectivity



You can select the nets in the Navigator assistant to get net connectivity to make better design planning decisions.

- Since the layout is flat in layout EXL, you have access to more net connectivity.
- You can use this enhanced net connectivity to make better design planning decisions.



You can use the **Analyze Connectivity** command to get net connectivity to make better design planning decisions.

- Since the layout is flat in layout EXL, you have access to more net connectivity.
- You can use this enhanced net connectivity to make better design planning decisions.



Using the Net Connectivity

You can select the nets in the Navigator assistant to get net connectivity to make better design planning decisions.

Since the layout is flat in layout EXL, you have access to more net connectivity.

You can use this enhanced net connectivity to make better design planning decisions.

You can use the Analyze Connectivity command to get net connectivity to make better design planning decisions.

Since the layout is flat in layout EXL, you have access to more net connectivity.

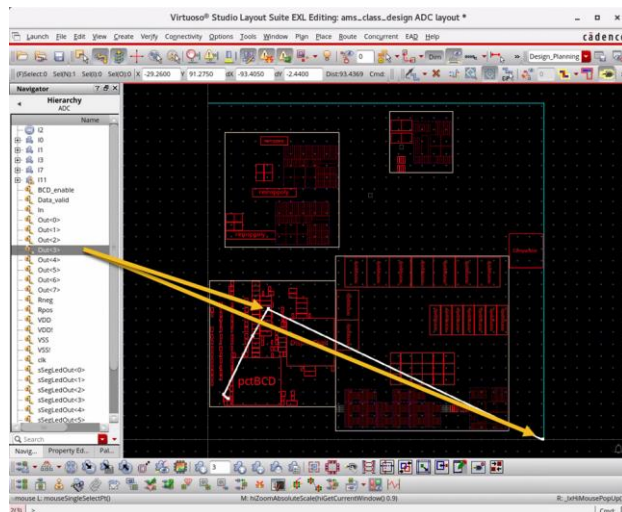
You can use this enhanced net connectivity to make better design planning decisions.

Selecting the Nets in the Navigator Assistant



In the Layout EXL canvas, from the **Navigator** → **DESIGN PLANNING** → **Hierarchy** data set, select the net to get net connectivity to make better design planning decisions.

- Use the net connectivity information to make better design planning decisions.
 - The net **Out<3>** is selected.
 - The net is cross-selected in the canvas.
- You can zoom and observe that the net is connected to a top-level pin **/Out<3>/** and to the virtual hierarchy, **I1**.
 - If **I1** were a real layout, you would only see the connection to its pin, not the instance inside.



Example

With no virtual hierarchy selected, click the **Increase Display Depth** button in the Design Planning toolbar, to say for example **3**, to increase the display levels of the virtual hierarchies in the design.

In the Navigator assistant, in the Design Planning category, click the **Hierarchy** data set.

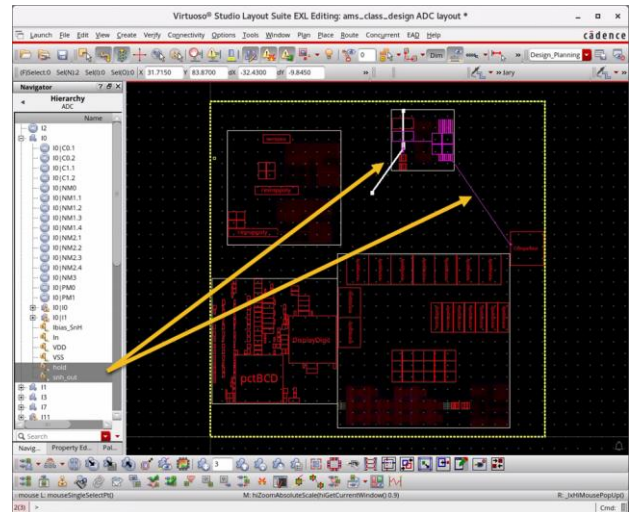
Select the net **Out<3>**.

This view shows the interface nets at each level of the virtual hierarchy.

The net is cross-selected in the layout canvas, as displayed here. You can zoom and observe that the net is connected to a top-level pin **/Out<3>/** and to the virtual hierarchy, **I1**. If **I1** were a real layout, you would only see the connection to its pin, not the instance inside.

Selecting the Nets in the Navigator Assistant (continued)

- Click the (+) button next to **I0** to expand the virtual hierarchy in the *Hierarchy* data set.
 - Press the **Ctrl** key and select any interface net(s), say, for example, **hold** and **Snh_out** to see the connections.
 - The lower-level instances, virtual hierarchy, and the interface nets are listed.
- You can see the connectivity throughout the virtual hierarchy. This extra information will help you make better design planning decisions.



Example (continued)

Click the (+) button next to **I0** to expand the virtual hierarchy in the *Hierarchy* data set.

Press the **Ctrl** key and select any interface net(s), say for example **hold**, and **Snh_out** to see the connections.

The lower-level instances, virtual hierarchy, and the interface nets are listed as shown here.

You can see the connectivity throughout the virtual hierarchy. This extra information will help you make better design planning decisions.

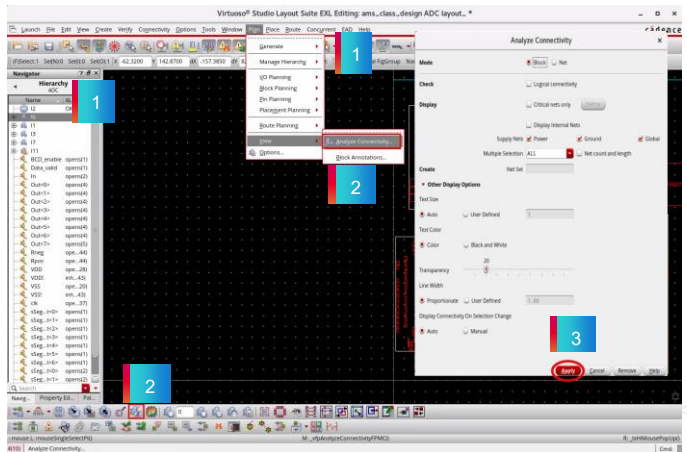
Using the Analyze Connectivity Command



Use the **Analyze Connectivity** command to calculate the number of connections and draw flightlines between and within the virtual hierarchies.

Use this connectivity information to decide upon an appropriate position for each block when planning a design.

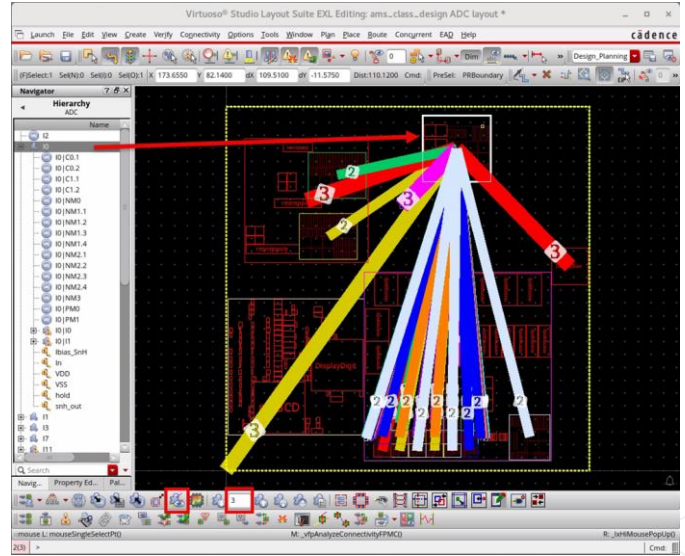
1. Select the virtual hierarchy **IO** in the *Hierarchy* data set in the Navigator assistant.
2. In the layout canvas, click the **Analyze Connectivity** icon in the Design Planning toolbar or choose the **Plan – View – Analyze Connectivity** menu command to launch the Analyze Connectivity form.
3. Keep the default values and click **Apply** to run the *Analyze Connectivity* command.



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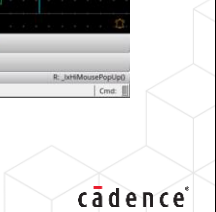
Net Connectivity for the Virtual Hierarchy I0: Depth 3

- All the connections from **I0** are highlighted in the layout.
- Each flightline between the connected pair is overlaid with a number that indicates the total connections along the flightline.
- This information can be used to position the blocks to limit the top-level routing.
- The *Analyze Connectivity* command also displays the contents of a virtual hierarchy.
- Since the virtual hierarchy instance is visible, the connections to the instances are shown.



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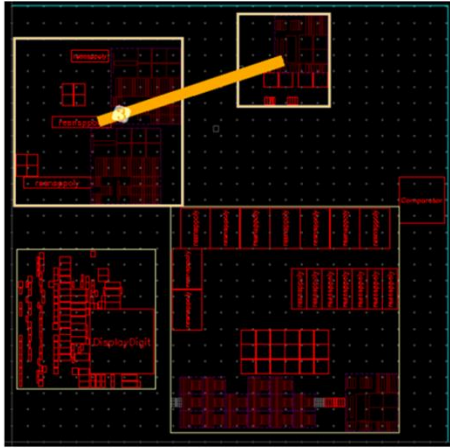
Since the virtual hierarchy is opaque, the instance connectivity is bundled into a single line to the center when running the *Analyze Connectivity* command.



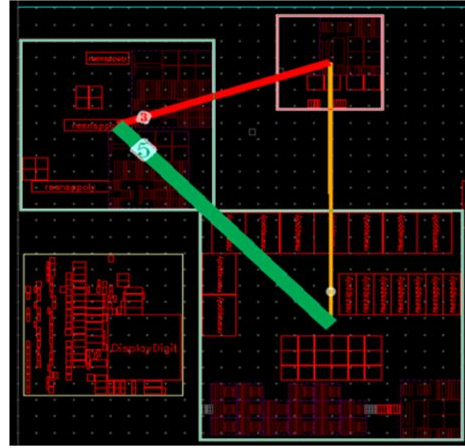
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Net Connectivity Between Virtual Hierarchies

If two or more virtual hierarchies are selected, only the connectivity to the selected virtual hierarchies is displayed.



Connectivity between 2 virtual hierarchies.



Connectivity between 3 virtual hierarchies.

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Quiz: Generate Commands in the Design Planner Toolbar



What are the generate commands available in the Design Planner toolbar?

Select all that apply.

- A. Generate All From Source
- B. Auto Place
- C. Generate Selected From Source
- D. Wire Assistant
- E. All of the Above

Answers: A and C

The Generate All From Source and Generate Selected From Source commands are available in the Design Planner toolbar.



Answers: A and C

The Generate All From Source and Generate Selected From Source commands are available in the Design Planner toolbar.

Quiz: The Design Planning Toolbar



In the Design Planning toolbar, the Generate All From Source command is available under a drop-down menu. This command can be toggled with the Auto Place command.

- A. True
- B. False

Answer: B

False. In the Design Planning toolbar, the Generate All From Source command is available under a drop-down menu. This command can be toggled with the Generate Selected From Source command.



Answer: B

False. In the Design Planning toolbar, the Generate All From Source command is available under a drop-down menu. The command can be toggled with the Generate Selected From Source command.

Quiz: Identifying the Virtual Hierarchy



Virtual hierarchies are displayed with a ----- dashed line to make them stand out from other real hierarchies. What is the color of that line?

Enter the answer in the blank field.

Answer: green

Virtual hierarchies are displayed with a green dashed line to make them stand out from other real hierarchies.



Answer: green

Virtual hierarchies are displayed with a green dashed line to make them stand out from other real hierarchies.

Quiz: Default Display Depth of the Generated Virtual Hierarchies



What is the default display depth of the generated virtual hierarchies?

Select all that apply.

- A. By default, the virtual hierarchies are generated opaque with their contents hidden
- B. By default, the virtual hierarchies are generated with display depth of 32
- C. By default, the virtual hierarchies are generated with display depth of 1
- D. By default, the virtual hierarchies are generated with display depth of 3
- E. None of the Above

Answer: A

By default, the virtual hierarchies are generated opaque with their contents hidden.



Answer: A

By default, the virtual hierarchies are generated opaque with their contents hidden.

Quiz: DESIGN PLANNING Data Sets in the Navigator Assistant



In the Layout EXL canvas, in the Navigator assistant, what are the additional data sets for displaying the generated virtual hierarchies and related components?

Select all that apply.

- A. Virtual Hierarchy
- B. Virtual Hierarchy Clones
- C. Hierarchy Clones
- D. Hierarchy
- E. All of the Above

Answers: A, B, and D

In the Layout EXL canvas, the Navigator assistant supports 3 additional data sets for displaying the generated virtual hierarchies and related components, namely: Virtual Hierarchy, Virtual Hierarchy Clones, and Hierarchy.



Answers: A, B, and D

In the Layout EXL canvas, the Navigator assistant supports 3 additional data sets for displaying the generated virtual hierarchies and related components, namely: Virtual Hierarchy, Virtual Hierarchy Clones, and Hierarchy.

Quiz: Removing Display Overrides for the Virtual Hierarchies



How to remove the display overrides for the virtual hierarchies in the Layout EXL canvas?
Select all that apply.

- A. By clicking the Set Default Display Depth 0 button in the Design Planning toolbar
- B. By clicking the Increase Display Depth button in the Design Planning toolbar
- C. By clicking the Decrease Display Depth button in the Design Planning toolbar
- D. By typing in the value of the required hierarchy in the Design Planning toolbar
- E. None of the Above

Answer: A

In the Layout EXL canvas, when clicking the Set Default Display Depth 0 button in the Design Planning toolbar, the overrides are removed, and the Display Depth is restored to 0.



Answer: A

In the Layout EXL canvas, when clicking the Set Default Display Depth 0 button in the Design Planning toolbar, the overrides are removed, and the Display Depth is restored to 0.

Quiz: Net Connectivity in the Layout EXL Canvas



In the Layout EXL canvas, what are the ways to get the net connectivity to make better design planning decisions?

Select all that apply.

- A. By selecting the nets in the Navigator assistant
- B. By using the Display Connectivity command
- C. By using the Analyze Connections command
- D. By using the Display Connections command
- E. By using the Analyze Connectivity command
- F. All of the above

Answers: A and E

In the Layout EXL canvas, you can get the net connectivity by selecting the nets in the Navigator assistant and by using the Analyze Connectivity command.



Answers: A and E

In the Layout EXL canvas, you can get the net connectivity by selecting the nets in the Navigator assistant and by using the Analyze Connectivity command.

Quiz: Calculating the Number of Connections and Drawing Flightlines



Which command can you use to calculate the number of connections and draw flightlines between and within the virtual hierarchies?

Select all that apply.

- A. Display Connectivity
- B. Analyze Connections
- C. Analyze Connectivity
- D. Display Connections
- E. None of the Above

Answer: C

You can use the Analyze Connectivity command to calculate the number of connections and draw flightlines between and within the virtual hierarchies.



Answer: C

You can use the Analyze Connectivity command to calculate the number of connections and draw flightlines between and within the virtual hierarchies.

Module Summary

By completing this module, you should be able to:

- Generate a virtual hierarchy to organize and represent design data effectively.
- Create and use Soft Block P-cells from CSV files within the Design Planning environment.
- Access and navigate Design Planning datasets using the Navigator assistant.
- Analyze the generated virtual hierarchy to understand design structure and relationships.
- Analyze net connectivity to evaluate interactions and dependencies across design components.
- Perform connectivity analysis for design validation.



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Labs



- Lab 3-1 Generating the Virtual Hierarchy
- Lab 3-2 Setting Up the Soft Block Pcells Using CSV in Design Planning
- Lab 3-3 Accessing the Design Planning Data Sets in the Navigator Assistant
- Lab 3-4 Viewing the Virtual Hierarchy
- Lab 3-5 Analyzing the Net Connectivity



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Module 4

Early Design Planning

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Module Objectives

At the end of this module, you will be able to:

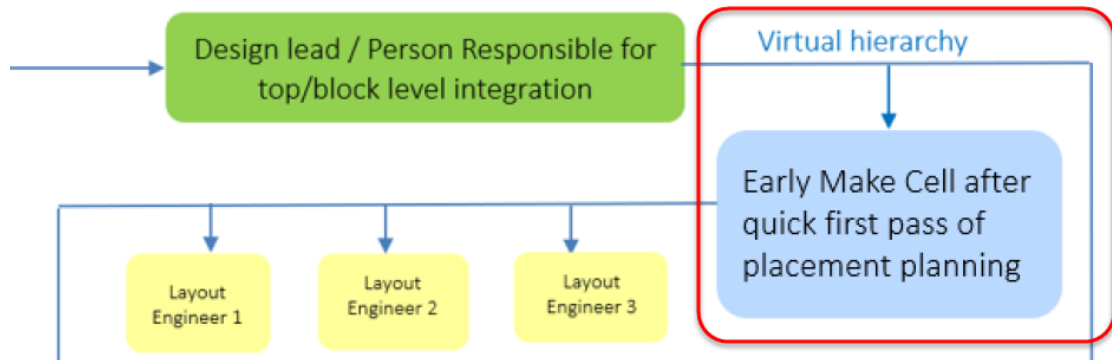
- Manipulate the virtual hierarchy into a suitable design plan
- Apply manual placement and editing in the Virtuoso® Design Planner
- Use the placement status of virtual hierarchies to control placement behavior during design planning
- Adjust the area boundary of the virtual hierarchies
- Analyze connectivity to optimize placement and routing decisions



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Early Design Planning Phase: Manual Placement and Editing

- One of the most used features during design planning is the manual manipulation of the virtual hierarchy.
- After the manual manipulation of the virtual hierarchy, Early Make Cell can be done for further implementation by the assigned Layout Engineer.



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Early Design Planning Phase: Manual Placement and Editing

One of the most used features during design planning is the manual manipulation of the virtual hierarchy.

You can move, stretch, and chop a virtual hierarchy to position it and change the aspect ratio as part of the design planning task.

By default, the analog placer will run to place the contents inside the adjusted area Boundary.

This auto placement can be controlled by using the *Design Planning and Analysis Options* form.

Virtual Hierarchy area boundary *Constant area stretch* option is available.

After the manual manipulation of the virtual hierarchy, early *Make Cell* can be done for further implementation by the assigned layout engineer.

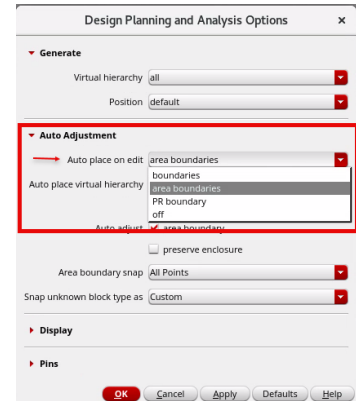
Manual Placement and Editing in the Virtuoso Design Planner



You can move, stretch, and chop the virtual hierarchies to position them and change their aspect ratio for better layout-place-route results.

You can manually place and edit the virtual hierarchies to achieve a tighter plan.

- One of the most used features during design planning is the manual manipulation of virtual hierarchies.
- By default, the analog placer will run to place the contents inside the adjusted area Boundary.
- This auto placement can be controlled by using the *Design Planning and Analysis Options* form.



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Editing a Virtual Hierarchy Using the Context-Sensitive Menu

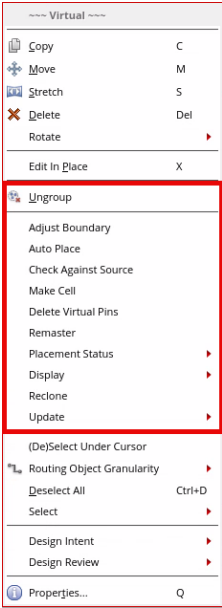


To perform any of the operations supported through the context menu:

From Navigator, **right-click** a virtual hierarchy or **right-click** in the Layout EXL canvas.

- The Design Planner provides quick access to some useful commands through the context-sensitive menu of a virtual hierarchy.

When invoked from the Navigator assistant, the *Virtual* context-sensitive menu displays an additional command: *Show Content*.



Commands supported only for virtual hierarchies



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Level-1 Editing of a Virtual Hierarchy

- A virtual hierarchy can be automatically *generated* when *Generate All From Source* is run with the Virtual Hierarchy option selected.
- If the virtual hierarchy is automatically generated, you can update the contents of the hierarchy at the top level, as required.
- However, for virtual hierarchies that are manually *created*, level-1 editing is restricted to avoid any unexpected edits to the level-1 objects.
- For the contents of a *created* virtual group to be selected and for the virtual group to be edited, the *transparent Group* environment variable must be set to *true*.



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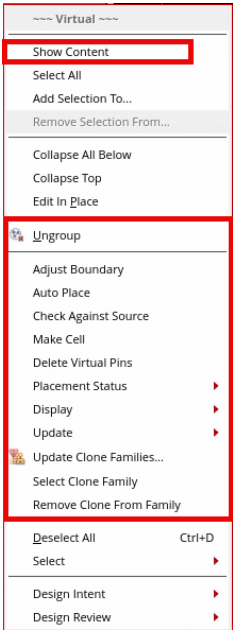
Editing a Virtual Hierarchy Clone Using the Context-Sensitive Menu



To perform any of the operations supported through the context menu:
From Navigator, **right**-click a virtual hierarchy clone or **right**-click in the Layout EXL canvas.

- The Design Planner provides quick access to some useful commands through the context-sensitive menu of a virtual hierarchy clone.

A virtual hierarchy clone supports three additional commands in the context-sensitive menu: *Update Clone Families*, *Select Clone Family*, and *Remove Clone From Family*.



Commands supported only for virtual hierarchy clones



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Editing a Virtual Hierarchy Clone

- Similar to editing a top-level virtual hierarchy, you can perform some basic edits on a virtual hierarchy clone or a clone inside a virtual hierarchy without having to edit it in place.
- These top-level edits are limited to stretching, chopping, and adjusting the area boundary of a virtual hierarchy clone.
- Any changes made to the clone area boundary due to stretching or chopping are automatically reflected across all the synchronous clones.
- This capability of top-level edit support for virtual hierarchy clones ensures ease of working with the clones from the top level.



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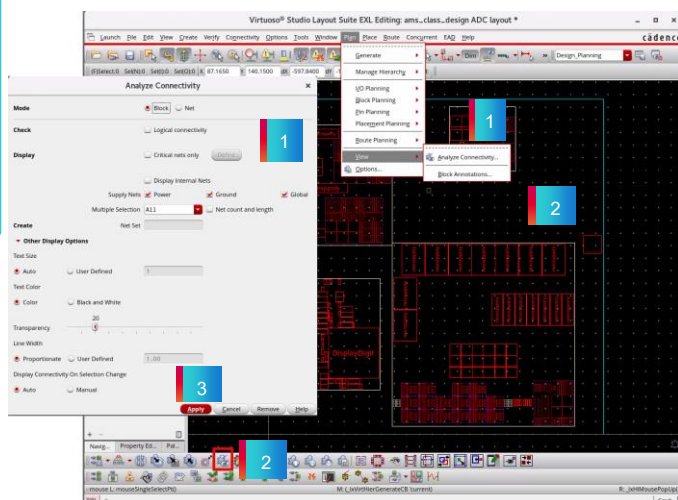
Using the Analyze Connectivity Command



Use the Analyze Connectivity command to calculate the number of connections and draw flightlines between and within the virtual hierarchies — at either the Block or Net level.

Use this connectivity information to decide upon an appropriate position for each block when planning a design

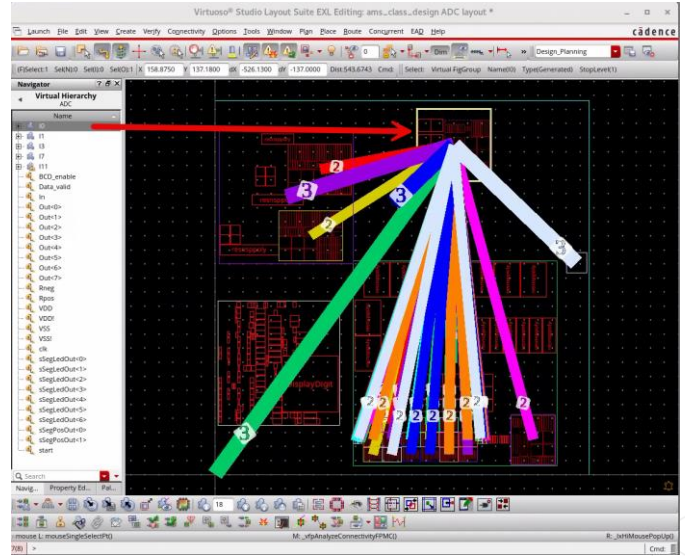
1. In the Navigator assistant, select the virtual hierarchy I0 from the Hierarchy data set
2. In the layout canvas, click the Analyze Connectivity icon in the Design Planning toolbar, or choose **Plan** → **View** → **Analyze Connectivity** to launch the Analyze Connectivity form.
3. In the form, keep Mode = Block (default), confirm the Supply Nets filters (Power, Ground, Global) as needed, and click Apply to run the Analyze Connectivity command.



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Net Connectivity for the Virtual Hierarchy IO

- All the connections from **I0** are highlighted in the layout.
- Each flightline between the connected pair is overlaid with a number that indicates the total connections along the flightline.
- This information can be used to position the blocks to limit the top-level routing.
- The *Analyze Connectivity* command also displays the contents of a virtual hierarchy.
- Since the virtual hierarchy instance is visible, the connections to the instances are shown.



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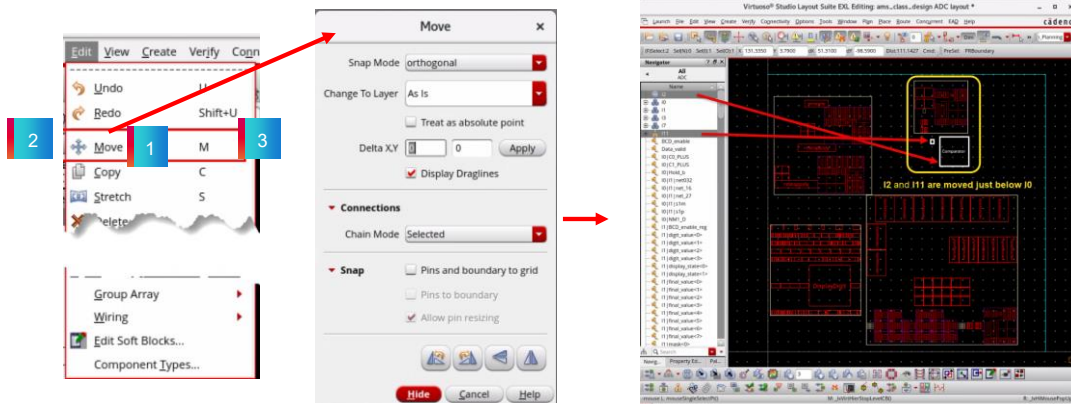
Manual Editing of Hierarchical Objects: Move



In the layout canvas:

1. Choose the **Edit – Move** menu command.
2. Or click the **Move** icon in the Edit toolbar.
3. Or press the bindkey **M** to invoke the Move form.

Use the **Move** command to move the instances **I2** and **I11** to appear just below **I0** to limit the top-level routing.



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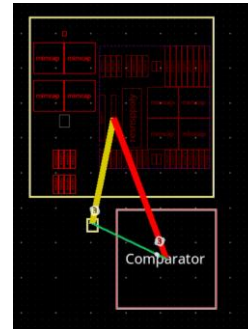
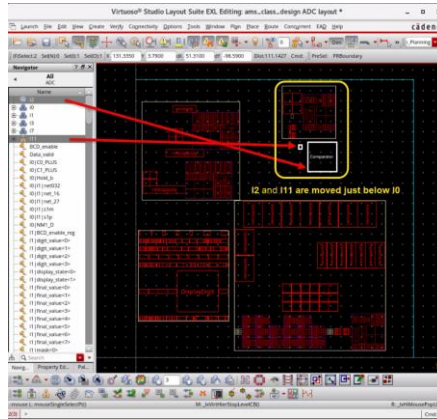
Manual Editing of Hierarchical Objects: Move

In the layout canvas, choose the **Edit – Move** menu command or click the **Move** icon in the Edit toolbar or press the bindkey **M** to invoke the Move form.

Here, you have used the Move command to move the instances **I2** and **I11** to appear just below **I0** to limit the top-level routing.

Manual Editing of Hierarchical Objects: Move Results

Looking at the relative connectivity with *Analyze Connectivity* command, you can see that the top-level connections are now shorter.



Manual Editing of Hierarchical Objects: Move Results

Looking at the relative connectivity with *Analyze Connectivity* command, you can see that the top-level connections are now shorter after moving the instances I2 and I11 just below the instance I0.

Manual Editing of Virtual Hierarchies: Stretch



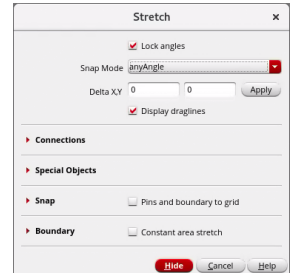
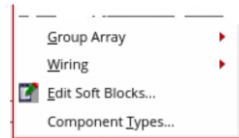
Use the **Stretch** command to reposition the instances and get placement for better routing results.

You can stretch virtual hierarchies to change their aspect ratio and achieve a tighter plan.

To stretch the virtual hierarchy:

1. In the layout canvas, choose the **Edit – Stretch** menu command or click the **Stretch** icon in the Edit toolbar or press the bindkey **S** to invoke the Stretch form.
2. Hover the mouse over **I7** and click at the bottom corner.
3. Move the cursor up and to the right to fill the space to the left of **I0**.

If you want to maintain the area constant, use the *Constant area stretch* option by clicking **F3** when running the Stretch command.



Manual Editing of Virtual Hierarchies: Stretch

You can use the Stretch command to reposition the instances and get placement for better routing results.

You can stretch the virtual hierarchies to change their aspect ratio and achieve a tighter plan.

In the layout canvas, choose the **Edit – Stretch** menu command or click the **Stretch** icon in the Edit toolbar or press the bindkey **S** to invoke the Stretch form.

Hover the mouse over **I7** and click at the bottom corner.

Move the cursor up and to the right to fill the space to the left of **I0**.

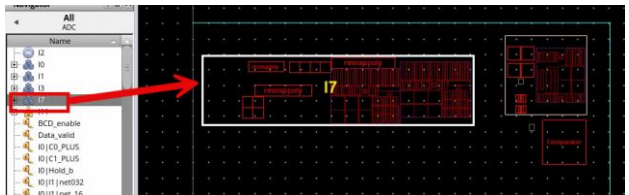
This feature is very useful in early design planning.

The contents of the virtual hierarchy can help you estimate if the boundary is the correct size.

If you want to maintain the area constant, you can use the Constant area stretch option by clicking **F3** when running the Stretch command.

Manual Editing of Virtual Hierarchies: Stretch Results

- Use the **Stretch** command to move I7 to reposition and get the placement, as shown here.
- The contents are automatically repositioned inside the new boundary.
- By default, the analog placer will run to place the contents inside the adjusted area Boundary.
- This auto placement can be controlled by using the *Design Planning and Analysis Options* form.



You can directly edit the contents of a virtual hierarchy, if the virtual hierarchy is not opaque. This avoids having to **Edit In Place** the virtual hierarchy, unless it is a virtual hierarchy clone that you need to edit.

Manual Editing of Virtual Hierarchies: Stretch Results

You can use the Stretch command to move I7 to reposition and get placement as shown here.

The contents are automatically repositioned inside the new boundary.

By default, the analog placer will run to place the contents inside the adjusted area Boundary.

This auto placement can be controlled by using the *Design Planning and Analysis Options* form.

You can directly edit the contents of a virtual hierarchy if the virtual hierarchy is not opaque.

This avoids having to Edit In Place the virtual hierarchy unless it is a virtual hierarchy clone that you need to edit.

Being able to directly edit the virtual hierarchy makes the editing very convenient, if you are working across multiple hierarchy levels.

Manual Editing of Virtual Hierarchy Clones: Stretch



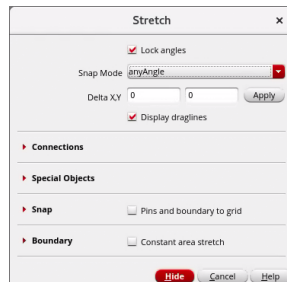
Use the **Stretch** command to reposition the instances and get placement for better routing results.

You can stretch the virtual hierarchy clones to change their aspect ratio and achieve a tighter plan.

To stretch the virtual hierarchy clones:

1. In the layout canvas, choose the **Edit – Stretch** menu command or click the **Stretch** icon in the Edit toolbar or press the bindkey **S** to invoke the Stretch form.
2. Hover the mouse over **I7|I1** and click at the bottom corner.
3. Stretch **I7|I1** downward and click.

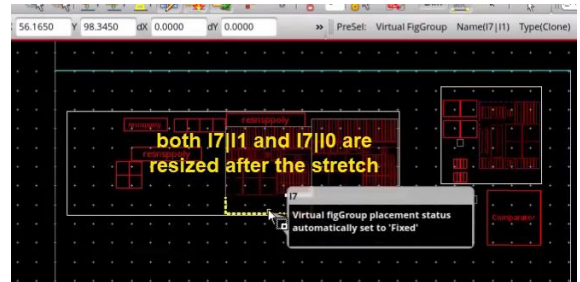
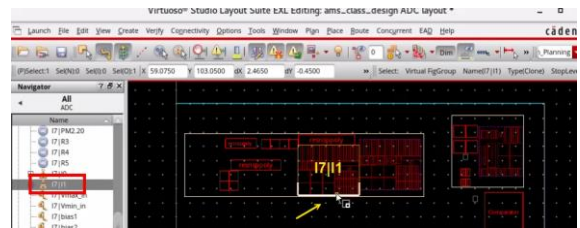
Although the contents of the Virtual Hierarchy Clones cannot be selected without first doing *Edit In Place*, you can stretch the boundary.



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Manual Editing of Virtual Hierarchy Clones: Stretch Results

- Both I7|I1 and I7|I0 are resized and replaced after the stretch.
- This is because they both belong to the same clone family.
- If the *Auto adjust area boundary* option is set to **true** in the *Design Planning and Analysis Options* form, which is the default, the area boundary of I7 is resized to reflect the stretch.
- The info balloon informs you that the placement status is set to “Fixed”.



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Manual Editing of Virtual Hierarchies: Chop



Use the **Chop** command to reposition the instances and get placement for better routing results.

You can chop the virtual hierarchies to create rectilinear virtual hierarchies.

To chop the virtual hierarchy:

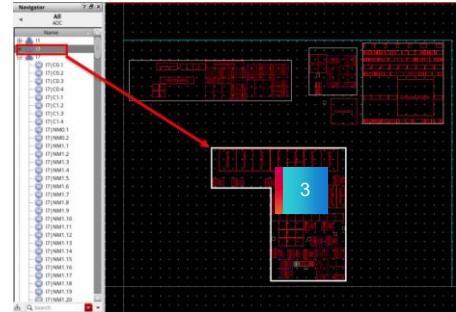
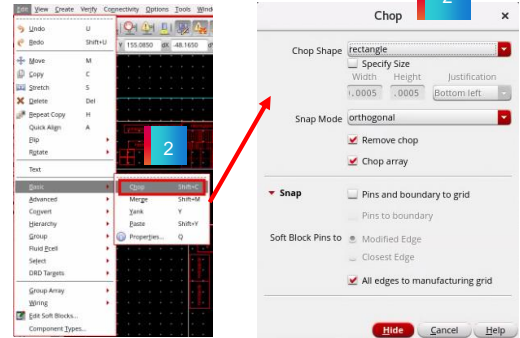
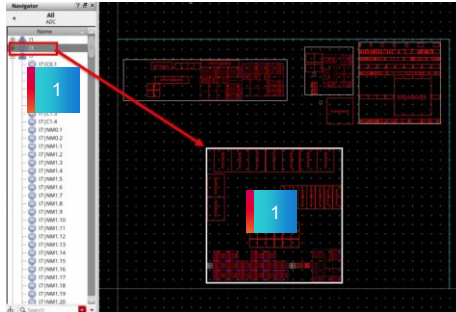
1. Select the virtual hierarchy instance, for example, **I3**.
2. In the layout canvas, choose the **Edit – Basic – Chop** menu command or press the bindkey **Shift+C** to invoke the Chop form.
3. Draw a rectangle over **I3**.



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Manual Editing of Virtual Hierarchies: Chop Results

- Use the **Chop** command to cut away part of an object or cut an object into pieces.
- The rectangle is chopped, and the boundary is redrawn.
- The contents are automatically repositioned inside the new boundary.
- By default, the analog placer will run to place the contents inside the adjusted area Boundary.
- This auto placement can be controlled by using the *Design Planning and Analysis Options* form.



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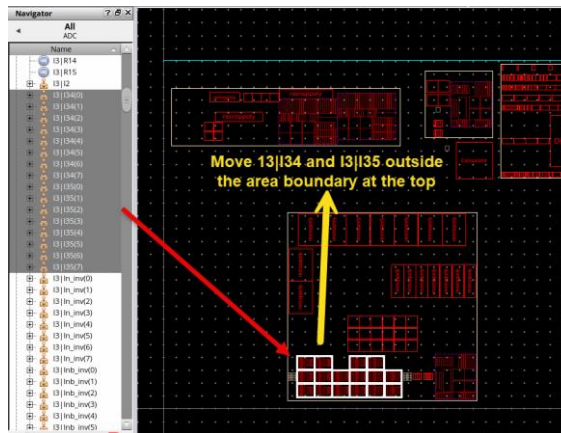
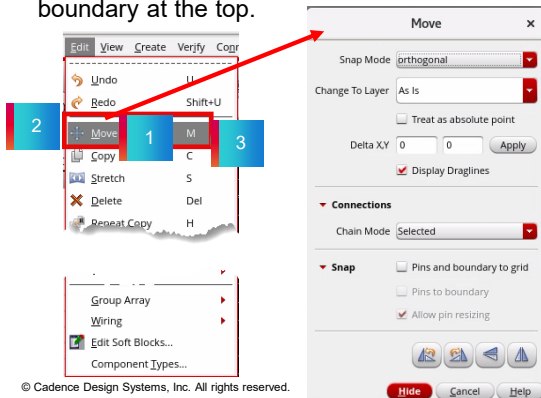
Manual Editing of Virtual Hierarchy Clones: Move



In the layout canvas:

1. Choose the **Edit – Move** menu command.
2. Or click the **Move** icon in the Edit toolbar.
3. Or press the bindkey **M** to invoke the Move form.

- Select all the **I3|I34** and **I3|I35** virtual hierarchy clones and move them outside the area boundary at the top.



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Manual Editing of Virtual Hierarchy Clones: Move

In the layout canvas, choose the **Edit – Move** menu command or click the **Move** icon in the Edit toolbar or press the bindkey **M** to invoke the Move form.

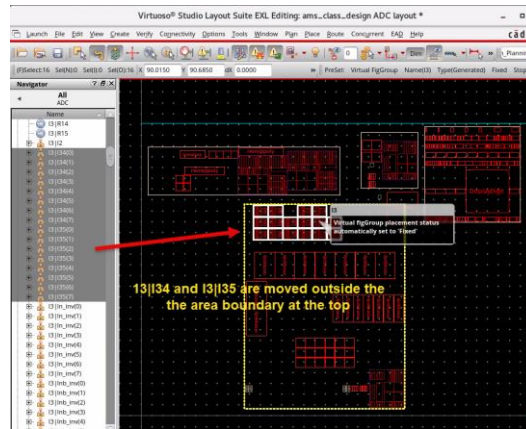
Select all the **I3|I34** and **I3|I35** virtual hierarchy clones and move them outside the area boundary at the top.

The virtual hierarchy clone instances are moved outside the boundary, but they remain inside the virtual hierarchy.

You have only changed the position of the instances, not moved them to the top level.

Manual Editing of Virtual Hierarchy Clones: Move Results

- The VH Clone instances are moved outside the boundary, but they remain inside the virtual hierarchy.
- You have only changed the position of the instances, not moved them to the top level.
- After the instances are placed, the area boundary is resized to accommodate the new placement.
- The info balloon tells you that the placement status has been set to “Fixed”.



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Manual Editing of Virtual Hierarchy Clones: Move Results

The virtual hierarchy clone instances are moved outside the boundary, but they remain inside the virtual hierarchy.

You have only changed the position of the instances, not moved them to the top level.

After the instances are placed, the area boundary is resized to accommodate the new placement.

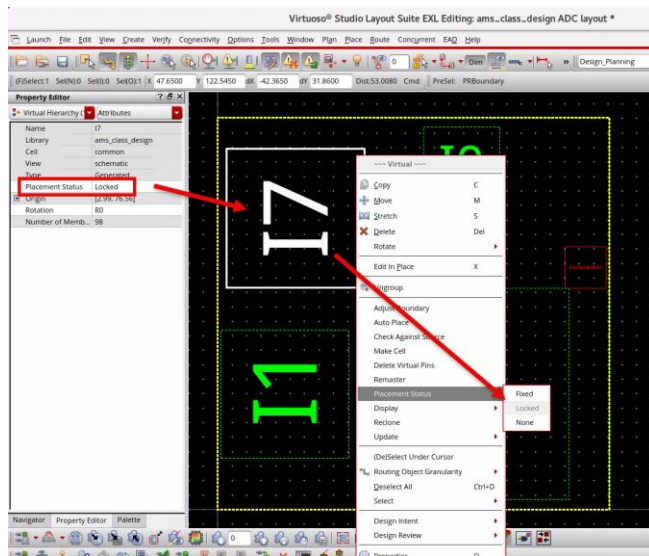
The info balloon tells you that the placement status has been set to “Fixed”.

Viewing the Virtual Hierarchy Placement Status



To know whether a virtual hierarchy has its placement in *Fixed*, *Locked*, or *None* status, you can check its bounding box.

- To know the placement status of a virtual hierarchy, you can now just look at its bounding box to see whether it is a solid or dashed line.
 - If the bounding box of a virtual hierarchy is a solid line, the placement status of the virtual hierarchy is with *Fixed* or *Locked* status.
 - If the bounding box of a virtual hierarchy is a dashed line, the placement status of the virtual hierarchy is *None*.



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Setting the Placement Status of the Virtual Hierarchies



Use the **Placement Status** to set the virtual hierarchies to *Fixed/Locked/None* status. Placement Status as *Fixed/Locked* avoids the contents being moved by automatic placers. Placement Status as *None* makes the contents being moved by automatic placers.

To set the placement status:

1. In the layout canvas, select the virtual hierarchies **I3 & I7** which are in *Fixed/Locked* status.
2. From the context-sensitive menu, **right-click** and select **Placement Status → None**.

Result: The placement status of **I3 & I7** are reset from *Fixed/Locked* to *None*. The area boundaries of **I3 & I7** are returned to dashed lines again.

Virtual hierarchies with *Fixed/Locked* placement status are surrounded by solid lines rather than dashed lines. Selecting *Placement Status → Fixed/Locked* will reset the virtual hierarchies from *None* to *Fixed/Locked*.

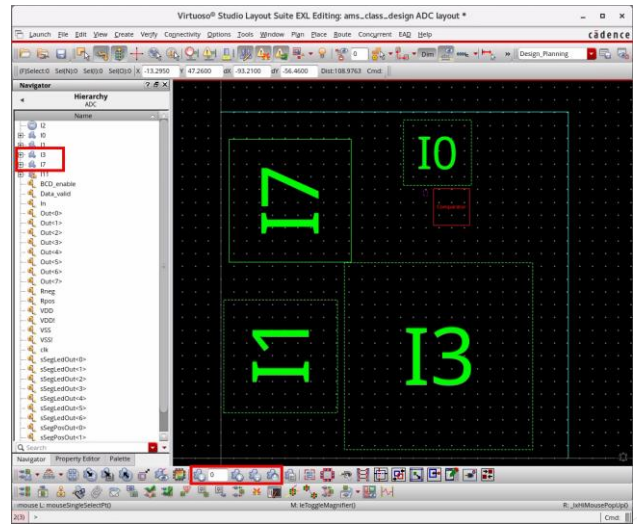


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Setting the Placement Status of the Virtual Hierarchies: Opaque

- In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to return to level 0 making all the virtual hierarchies opaque.
- Virtual hierarchies **I3** & **I7** with *Fixed/Locked* placement status are surrounded by solid lines rather than dashed lines.

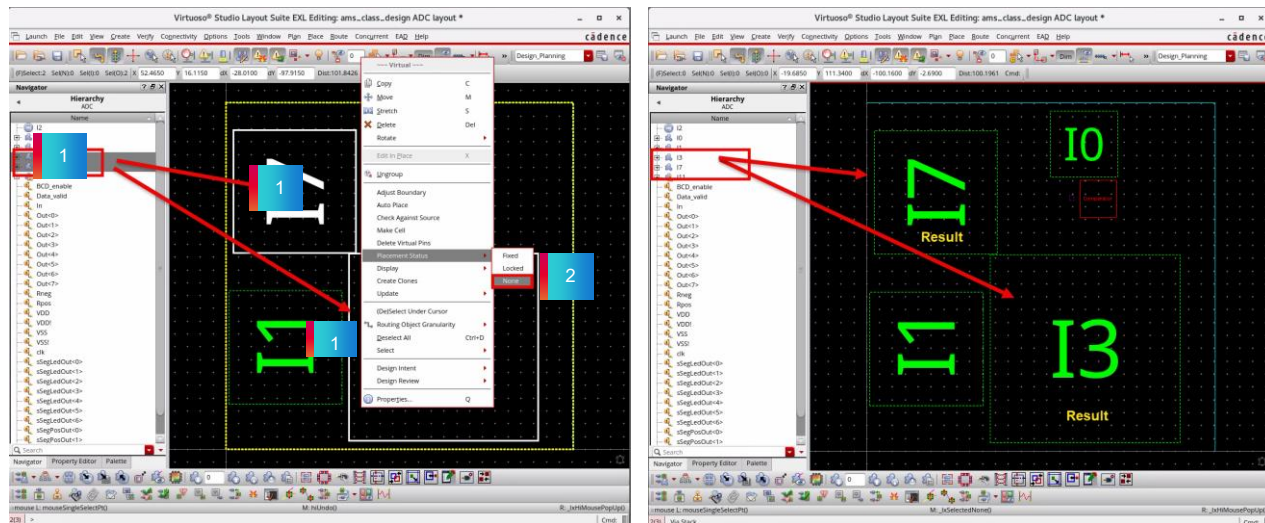
The placement status can be reset if needed.



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Setting the Placement Status of the Virtual Hierarchies: Fixed/Locked/None

Result: After the placement status is reset to **None**, the bounding boxes of I3 & I7 are returned to dashed lines again.



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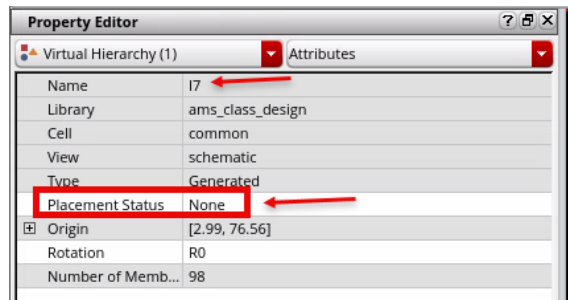
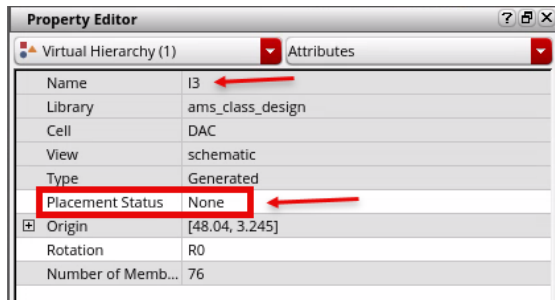
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Setting the Placement Status of the Virtual Hierarchies: Fixed/Locked/None in the Property Editor Assistant

- Use the Property Editor assistant to view the placement status.
- The Property Editor assistant shows that the placement status of **I3** & **I7** are *None* which means they are no more *Fixed/Locked*.

The placement status of **I3** is *None*.

The placement status of **I7** is *None*.



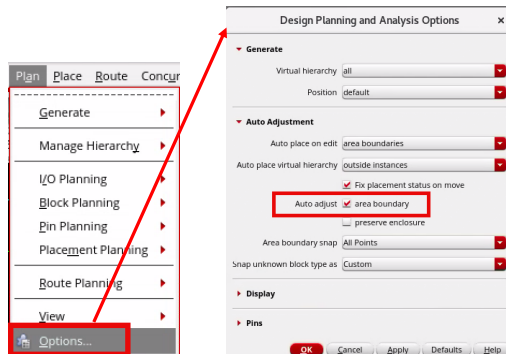
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Automatic Adjustment of an Area Boundary



In the Layout EXL canvas, choose the **Plan – Options** menu command to open the Design Planning and Analysis Options form.

- The Design Planner automatically adjusts the area boundary of a virtual hierarchy when:
 - Moving instances outside a virtual hierarchy
 - Adding instances to a virtual group
- Use the *Design Planning and Analysis Options* form to control the area boundary of the virtual hierarchy.



The resized boundary is always rectangular when automatically adjusted.

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Moving Instances Outside a Virtual Hierarchy



To allow the automatic area boundary adjustment of virtual hierarchies across the hierarchy, in the **Design Planning and Analysis Options** form, select the **Auto adjust area boundary** option.

The Design Planner can automatically resize the area boundary of the virtual hierarchy to accommodate an instance (or instances and figGroups other than row region) that are moved *outside* the boundary.

Design Planning and Analysis Options

Generate

Virtual hierarchy: all

Position: default

Auto Adjustment

Auto place on edit: area boundaries

Auto place virtual hierarchy: outside instances

☒ Fix placement status on move

Auto adjust ☒ area boundary

☐ preserve enclosure

Area boundary snap: All Points

snap unknown block type as: Custom

Display

Pins

OK Cancel Apply Defaults Help



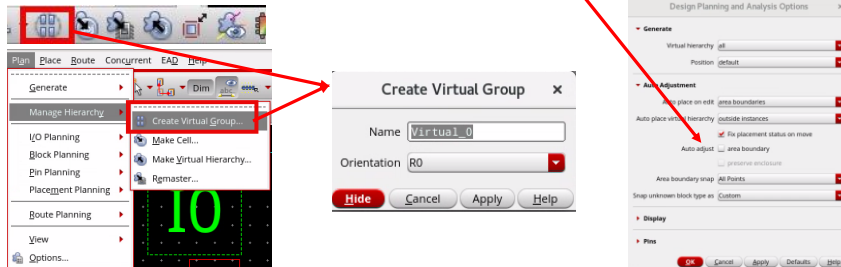
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Adding Instances to a Virtual Group



In the Layout EXL canvas, choose **Plan – Manage Hierarchy – Create Virtual Group** or click the **Create Virtual Group** icon in the Design Planning toolbar to open the Create Virtual Group form.

- Use the *Create Virtual Group* command to add new instances to an existing virtual group created.
 - To accommodate the new instances, the area boundary of the updated virtual group is automatically adjusted.
 - Rectilinear boundaries that are resized are automatically adjusted to form rectangles.
- To prevent the automatic adjustment of the area boundary:
 - Set the *autoAdjustBoundary* environment variable to **nil** (or)
 - Deselect the *Auto adjust area boundary* option in the Design Planning and Analysis Options form.



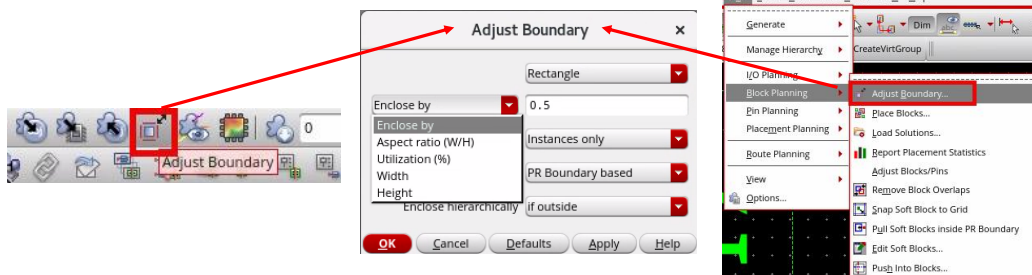
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Adjusting the Area Boundary of the Virtual Hierarchies



In the Layout EXL canvas, click the **Adjust Boundary** icon in the Design Planning toolbar or choose **Plan – Block Planning – Adjust Boundary** menu command to invoke the Adjust Boundary form.

- Use the *Adjust Boundary* command to adjust the boundary of the virtual hierarchy instance to fit around the virtual hierarchy contents.
- In the Design Planning and Analysis Options form, deselect the *Auto adjust area boundary* option to ensure a rectangular boundary is not created for a polygonal virtual hierarchy.

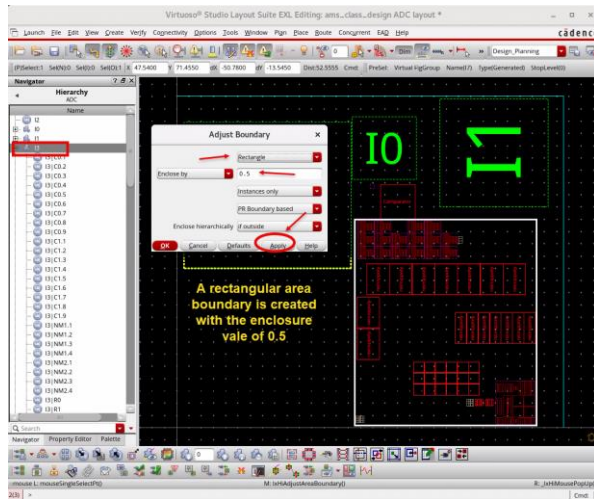


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Creating a Rectangular Area Boundary

- In the Adjust Boundary form, the default option is to create a rectangular boundary, spaced away from the contents by the enclosure value of **0.5**.
- Keep the default values and click **Apply** to adjust the area boundary of **I3**.

A rectangular area boundary is created with an enclosure value of **0.5**.

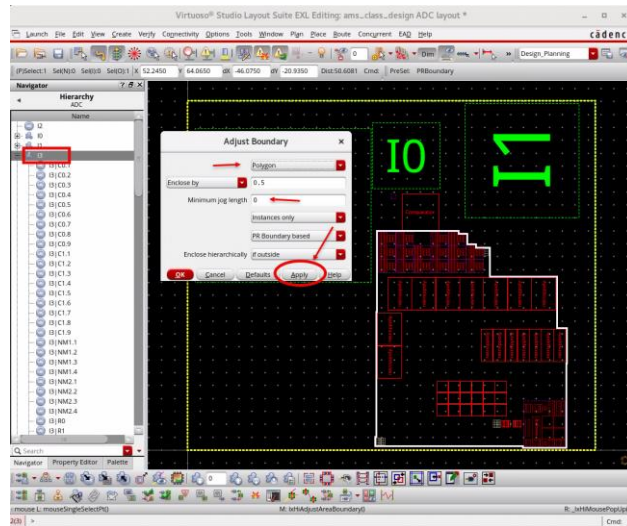


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Creating a Rectilinear Area Boundary with Notches

- In the Adjust Boundary form, change the top field to **Polygon** and click **Apply**.
- The area boundary is adjusted to a polygon surrounding the contents of the virtual hierarchy.

A rectilinear area boundary is created, but it has several notches.

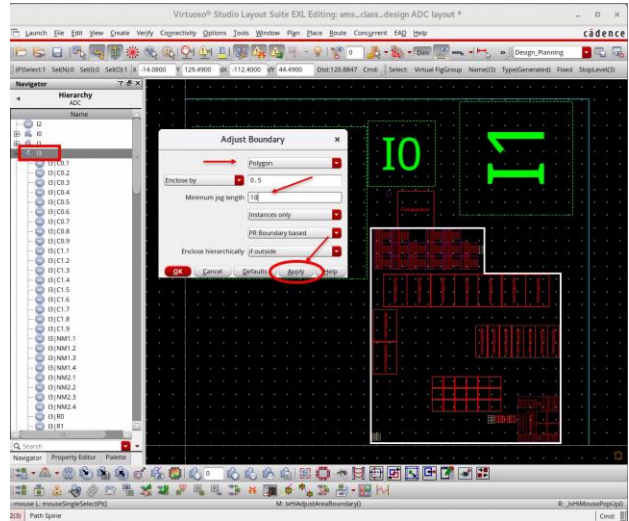


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Creating a Rectilinear Area Boundary without Notches

- In the Adjust Boundary form, set the *Minimum jog length* value to **10** and click **Apply** again.
- A rectilinear area boundary is created without notches.

A rectilinear area boundary is created without notches.



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Quiz: Auto Placement During Manual Editing



When you move/stretch/chop the virtual hierarchies, the analog placer will run to place the contents inside the adjusted area Boundary. You cannot disable this auto placement.

- A. True
- B. False

Answer: B

False. This auto placement can be controlled by using the Design Planning and Analysis Options form.



Answer: B

False. This auto placement can be controlled by using the Design Planning and Analysis Options form.

Quiz: Editing the Contents of a Virtual Hierarchy



You can directly edit the contents of a virtual hierarchy if the virtual hierarchy is not -----.
Enter the answer in the blank field.

Answer: opaque

You can directly edit the contents of a virtual hierarchy if the virtual hierarchy is not opaque.



Answer: opaque

You can directly edit the contents of a virtual hierarchy if the virtual hierarchy is not opaque.

Quiz: Clone Area Boundary



Any changes made to the clone area boundary due to stretching or chopping are automatically reflected across all the synchronous clones.

- A. True
- B. False

Answer: A

True. Any changes made to the clone area boundary due to stretching or chopping are automatically reflected across all the synchronous clones.



Answer: A

True. Any changes made to the clone area boundary due to stretching or chopping are automatically reflected across all the synchronous clones.

Quiz: Virtual Context-Sensitive Menu



When invoked from the Navigator assistant, the Virtual context-sensitive menu displays an additional command, namely ---- -.

Enter the answer in the blank field.

Answer: Show Content

When invoked from the Navigator assistant, the Virtual context-sensitive menu displays an additional command, namely Show Content.



Answer: Show Content

When invoked from the Navigator assistant, the Virtual context-sensitive menu displays an additional command, namely Show Content.

Quiz: Virtual Hierarchy Clone Context-Sensitive Menu



What are the additional commands supported in the context-sensitive menu for virtual hierarchy clones than virtual hierarchies?

Select all that apply.

- A. Update Clone Families
- B. Adjust Boundary
- C. Select Clone Family
- D. Fix placement status
- E. Remove Clone From Family
- F. All of the above

Answers: A, C, and E

A virtual hierarchy clone supports three additional commands in the context-sensitive menu, namely Update Clone Families, Select Clone Family, and Remove Clone From Family.



Answers: A, C, and E

A virtual hierarchy clone supports three additional commands in the context-sensitive menu, namely Update Clone Families, Select Clone Family, and Remove Clone From Family.

Quiz: Auto Adjust Area Boundary Option



If the Auto adjust area boundary option is set to true in the Design Planning and Analysis Options form, which is the default, the area boundary of the virtual hierarchy is resized to reflect the manual edit.

- A. True
- B. False

Answer: A

True. If the Auto adjust area boundary option is set to true in the Design Planning and Analysis Options form, which is the default, the area boundary of the virtual hierarchy is resized to reflect the manual edit.



Answer: A

True. If the Auto adjust area boundary option is set to true in the Design Planning and Analysis Options form, which is the default, the area boundary of the virtual hierarchy is resized to reflect the manual edit.

Quiz: Placement Status of Virtual Hierarchies



To know whether a virtual hierarchy has its placement fixed or not, you can check the following:

Select all that apply.

- A. flightline
- B. net connectivity
- C. bounding box
- D. display overrides
- E. None of the above

Answer: C

To know whether a virtual hierarchy has its placement fixed or not, you can check its bounding box.



Answer: C

To know whether a virtual hierarchy has its placement fixed or not, you can check its bounding box.

Quiz: Placement Status of Virtual Hierarchies: Solid Line



If the bounding box of the virtual hierarchy is a solid line, the placement status of the virtual hierarchy is Fixed/Locked.

- A. True
- B. False

Answer: A

True. If the bounding box of the virtual hierarchy is a solid line, the placement status of the virtual hierarchy is Fixed/Locked.



Answer: A

True. If the bounding box of the virtual hierarchy is a solid line, the placement status of the virtual hierarchy is Fixed/Locked.

Quiz: Placement Status of Virtual Hierarchies: Dashed Line



What is the placement status of the virtual hierarchy if the bounding box of the virtual hierarchy is a dashed line?

Select all that apply.

- A. None
- B. Fixed
- C. Locked
- D. Float
- E. Group
- F. None of the above

Answer: A

If the bounding box of a virtual hierarchy is a dashed line, the placement status of the virtual hierarchy is None.

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Answer: A

If the bounding box of a virtual hierarchy is a dashed line, the placement status of the virtual hierarchy is None.

Quiz: Adjust Boundary Form



You can use the Adjust Boundary form to resize or create the virtual hierarchy area boundary in the following shapes:

Select all that apply.

- A. Circle
- B. Rectangle
- C. Ellipse
- D. Line
- E. Polygon
- F. All of the above

Answers: B and E

You can use the Adjust Boundary form to resize or create the virtual hierarchy area boundary in the Rectangle and Polygon shapes.

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Answers: B and E

You can use the Adjust Boundary form to resize or create the virtual hierarchy area boundary in the Rectangle and Polygon shapes.

Articulate Quiz on Early Design Planning

<https://cadence.sharepoint.com/:u:/s/Assets/EfEHTGV8h3JMphVQeFp4BREBEhCHmAcXjiykEiok9takJg>



Articulate Quiz on Early Design Planning

Module Summary

In this module, you:

- Understood how to **manipulate virtual hierarchies** to create an effective and structured design plan.
- Applied **manual placement and editing techniques** within the Virtuoso Design Planner to refine layout design.
- Used **placement status controls of virtual hierarchies** to influence and manage placement behavior during design planning.
- Adjusted and optimized the **area boundary of virtual hierarchies** to improve layout organization and efficiency.
- Performed **connectivity analysis** to make informed decisions that enhance placement and routing outcomes.



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Lab



Lab 4-1 Manipulating the Virtual Hierarchy

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Module 5

Using the Design Planner in the Team Environment

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Module Objectives

At the end of this module, you will be able to:

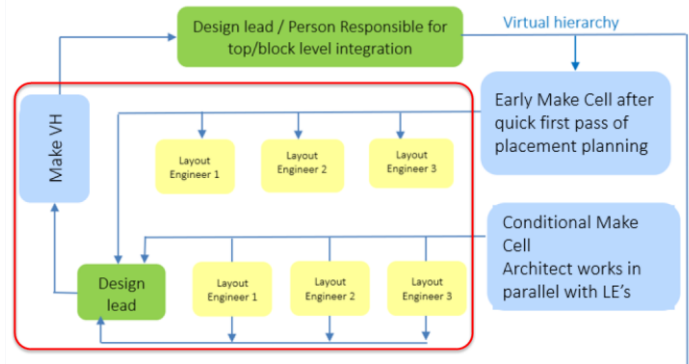
- Create rows and run automatic placement
- Run Congestion Analysis
- Create real layout views for the virtual hierarchies
- Create virtual hierarchies for the layout views



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Using the Design Planner in the Team Environment Phase

- In Using the Early Design Planning Phase, you completed the tasks that the Engineer responsible for the block implementation would follow early in the design planning.
- In this section, Using the Design Planner in the Team Environment, you will perform the steps that should be used to handover a virtual hierarchy to a Layout Engineer for implementation.



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Submodule 5-1

Creating Rows and Running Automatic Placement

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Submodule Objectives

In this submodule, you

- Use the Row Template Manager to view the row templates
- Use the Placement Planning form to create the rows
- Use the Design Planning and Analysis Options form to control the area boundary
- Run the Automatic Placer to place the standard cells



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Adding Display Overrides for the Virtual Hierarchies

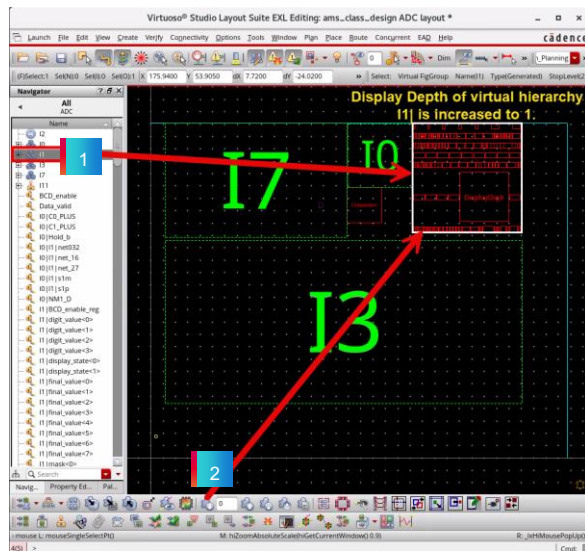


A global visibility control may not be sufficient during design planning.
Virtual hierarchies can have visibility overrides for better planning.

To add a display override for a virtual hierarchy:

1. Firstly, select the virtual hierarchy, for example, **I1**.
2. Then, click the **Increase Display Depth** toolbar button as required.

Result: The Display Depth of **I1** is increased and its contents are displayed.



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What Is the Virtuoso Placer?



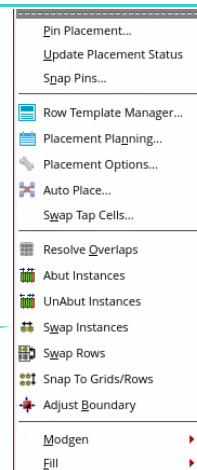
The Virtuoso® Placer is a unified placement solution available in the Virtuoso Layout EXL cockpit in advanced node releases.

It can be used to achieve quick, high-quality, DRC-correct row-based placement of standard cells and devices in analog, digital, and mixed-signal designs.

- The Placer brings together under its umbrella several existing and new tools that utilize the row template infrastructure.
- You can use the row template infrastructure to create rows which define how standard cells and devices are placed.
- You can use the Virtuoso Custom Digital Placer to place the custom digital designs.



In the Layout EXL menu bar, from the **Place** menu bar or from the **Placement** toolbar, access the placer commands.



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The Virtuoso Placer is a unified placement solution available in the Virtuoso Layout EXL cockpit in advanced node releases. It can be used to perform row-based placement of standard cells and devices in analog, digital, and mixed-signal designs.

The Virtuoso Placer is a unified placement solution that can be used to achieve quick, high-quality, DRC-correct placement of standard cells and devices. It supports placement of different types of designs, such as digital designs with only standard cells, analog designs with only devices, and mixed designs that contain a mixture of both. It can also be used to generate connectivity and constraint-driven device placements.

The Placer combines several advanced features. At the core of the Placer is the row template infrastructure that lets you create rows, which, in turn, define how standard cells and devices are placed. Rows, if made compatible with snap patterns, help achieve correct-by-construction placement.

The Placer also brings together under its umbrella several existing and new tools that utilize the row template infrastructure. For example, you can use the Virtuoso Custom Digital Placer to place the custom digital designs and Modgens to construct blocks of analog devices. The new automatic and assisted placement and postprocessing tools can be used to fill the design with dummy devices to meet density DRC rules and to achieve better matching.

The Placer is available in the Virtuoso Layout EXL cockpit in advanced node releases.

What Are Row and Row Attributes?



A **Row** represents the location for placing standard cells, macros, or devices.

The **Row Attributes** are *Row Edge*, *Row Orientation*, *Row Offset*, and *Row Height*.

- In a layout design, you define rows within the PR boundary.
- You can then place the instances within these rows.



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What Are Row and Row Attributes?

A **Row** represents the location for placing standard cells, macros, or devices.

The **Row Attributes** are *Row Edge*, *Row Orientation*, *Row Offset*, and *Row Height*.

In a layout design, you define rows within the PR boundary.

You can then place the instances within these rows.

Row Attributes

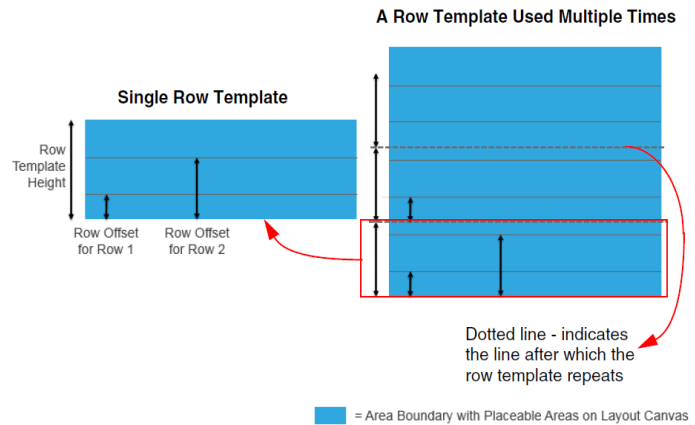
- *Row Edge* is defined by the row offset and row height values.
- *Row Orientation* is the orientation of the row with respect to the x and y axes.
- *Row Offset* refers to the distance from the base line to the reference edge.
- *Row Height* is the distance between the top and bottom edges of the row.

What Is a Row Template?



A **Row Template** is a container that defines one or more rows.

- A *row template* comprises specifications for a set of rows to be generated in the layout canvas.
- It can be repeated multiple times, one on top of the other, based on the required height.
- The graphic depicts how a *row template* can be used multiple times in a design.
 - One instance is zoomed in to view the details.



What Is a Row Template?

A **Row Template** is a container that defines one or more rows.

A *Row Template* comprises specifications for a set of rows to be generated in the layout canvas.

It can be repeated multiple times, one on top of the other, based on the required height.

The graphic depicts how a *Row Template* can be used multiple times in a design.

One instance is zoomed in to view the details.

What Is the Row Template Manager?



The **Row Template Manager** feature provides the capability to define the row templates.

- You can create row templates specific to your design type and then use these templates to generate rows in the layout canvas.
- Rows are created in objects called row regions.
- Row regions reference a row template that determines how rows are created in that region.
- Rows, in turn, contain component type attributes that determine which instances are placed in that row and the manner in which they are placed.
- You can save the row templates and reuse them in multiple designs.
- Row templates can be used to place both analog and digital designs.

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Using the Row Template Manager

Before running the Virtuoso Placer, you must have generated the rows in the layout canvas.

Row generation involves the following steps:

- **Defining one or more row templates:** Use the **Row** tab of the *Placement Planning* form to define the row templates.
- **Defining the area where rows should be generated:** Use the **Create** tab of the *Placement Planning* form to specify the area (*Boundary*) in which the rows must be generated.



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Creating and Editing a Row Template

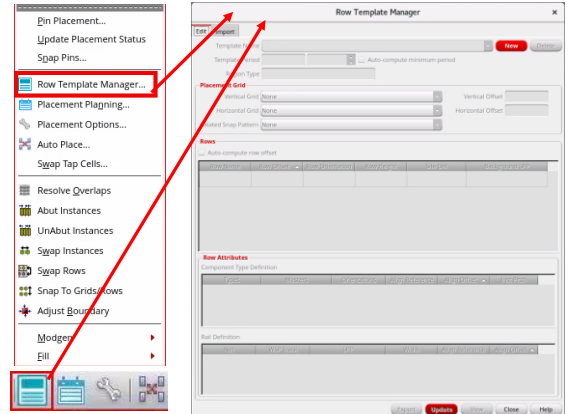


In the layout canvas, choose **Place – Row Template Manager** or click the **Row Template Manager** icon in the Placement toolbar to open the Row Template Manager form.

Use the *Row Template Manager* to create and edit the row templates.

The form has two tabs:

- **Edit:** Use this tab to create new row templates or to edit existing templates. The row templates are stored directly in the layout.
- **Import:** Use this tab to import existing row templates from a different cellview in your design library.



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Using the Row Template Manager to View the Row Templates

RowTemplate1 is set up here.

Row Template Manager

Template Name: RowTemplate1

Template Period: 0

Region Type:

Placement Grid

Vertical Grid: None

Horizontal Grid: None

Related Snap Pattern: None

Auto-compute minimum period: ☒

Vertical Offset: 0

Horizontal Offset: 0

Rows

Auto-compute row offset: ☒

Row Name	Row Offset	Row Orientation	Row Height	Site Def	Background LPP

Row Attributes

Component Type Definition

Types	Masters	Orientations	Align Reference	Align Offset	Inst Pitch

Rail Definition

Net	WSP Track	LPP	Width	Align Reference	Align Offset

Export Create View Close Help

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Deleting a Row Template

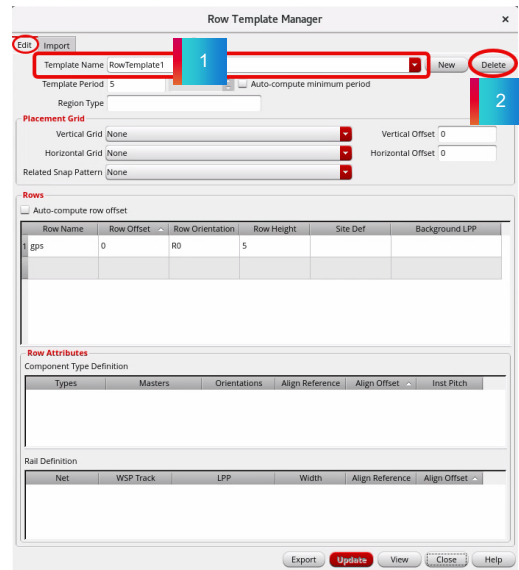


You need to delete the row templates which are not required in your design.

To delete a row template in Row Template Manager:

1. Firstly, with the opened **Edit** tab, choose the *Template Name* from the drop-down list.
2. Then, click **Delete** button.

Result: The selected row template is deleted.



Deleting a Row Template

You need to delete the row templates which are not required in your design.

To delete a row template in Row Template Manager:

- With the *Edit* tab open, choose the *Template Name* from the drop-down list.
- Click the *Delete* button.

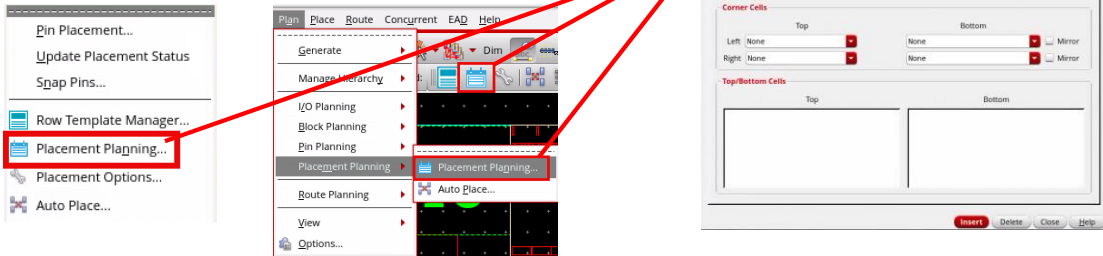
The selected row template is deleted.

Invoking the Placement Planning Form



In the layout canvas, choose **Place – Placement Planning** or **Plan – Placement Planning – Placement Planning** or click the **Placement Planning** icon in the Placement toolbar to open the Placement Planning form.

Use the *Placement Planning* form to define the rows used to accommodate components, define the region for generating the rows, and specify any special components required.



This page does not contain notes.

Using the Placement Planning Form to Create the Rows

After creating the row templates, use the *Placement Planning* form to generate the rows in the layout canvas.

RowTemplate1 is used to create the row and the **rowRegion** is created to correspond to the **I1** boundary.

Placement Planning: ams_class_design ADC layout

Boundary Cell | Tap Cell | **Row** | Create

Row Template: Auto

Vertical Grid: None

Horizontal Grid: None

Use Partial Template: ☒

Vertical Offset: 0

Horizontal Offset: 0

Row Parameters: Number Of Rows: Auto

Rails: Generate Rails: ☐

Insert Delete Close Help

Placement Planning: ams_class_design ADC layout

Boundary Cell | Tap Cell | Row | **Create**

Rows: ☒

Boundary Cells: ☐

Tap Cells: ☐

Region: Boundary grBoundary

Placement Region: ☐

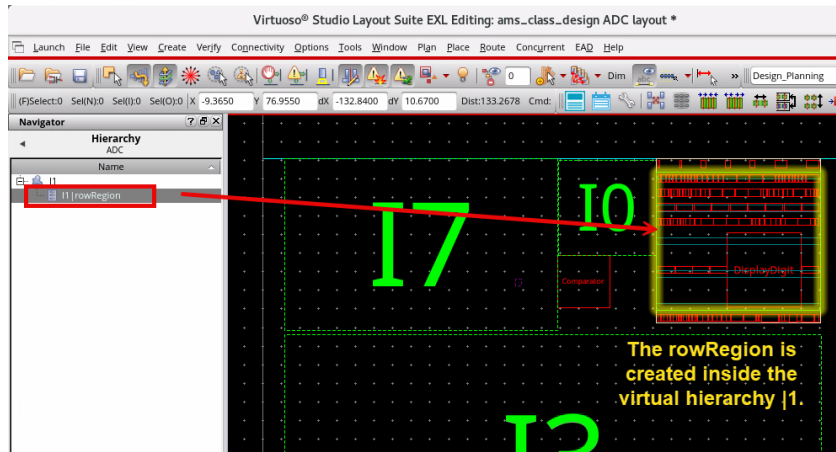
Options:

Insert Delete Close Help

This page does not contain notes.

Viewing the Created Rows in the Virtual Hierarchy I1

The **rowRegion** is created inside the virtual hierarchy I1.



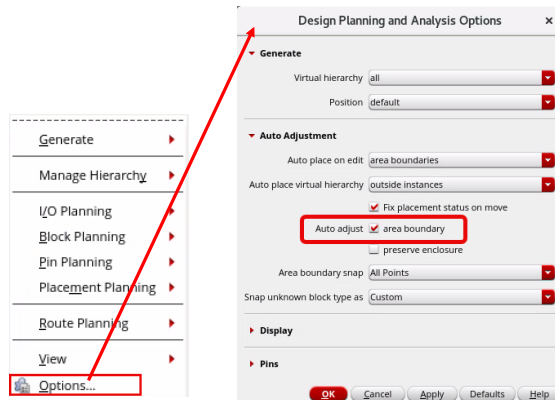
This page does not contain notes.

Using the Design Planning and Analysis Options Form to Control the Area Boundary



In the Layout EXL canvas, choose **Plan – Options** to open the Design Planning and Analysis Options form.

Use the *Design Planning and Analysis Options* form to control the area boundary of the virtual hierarchy.



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Auto Adjusting the Area Boundary Option

In the *Design Planning and Analysis Options* form, deselect the **Auto adjust area boundary** option to ensure a rectangular boundary is not created for a polygonal virtual hierarchy.

Design Planning and Analysis Options

▼ Generate

Virtual hierarchy: all

Position: default

▼ Auto Adjustment

Auto place on edit: area boundaries

Auto place virtual hierarchy: outside instances

☒ Fix placement status on move

Auto adjust: ☒ area boundary

☐ preserve enclosure

Area boundary snap: All Points

Snap unknown block type as: Custom

► Display

► Pins

OK Cancel Apply Defaults Help



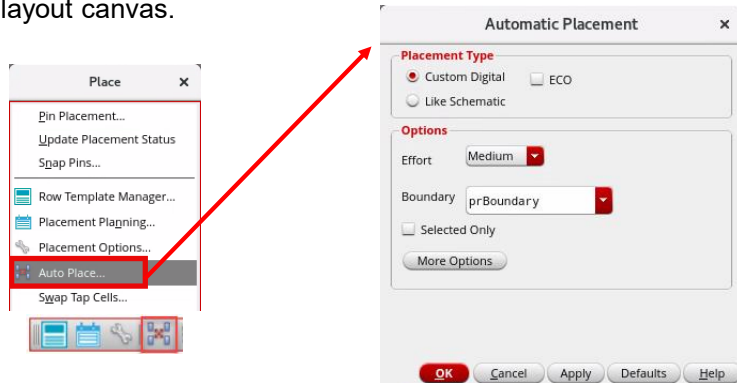
This page does not contain notes.

Invoking the Automatic Placement Form



In the Layout EXL canvas, choose **Place – Auto Place** or click **Auto Place** icon in the Placement toolbar to open the Automatic Placement form.

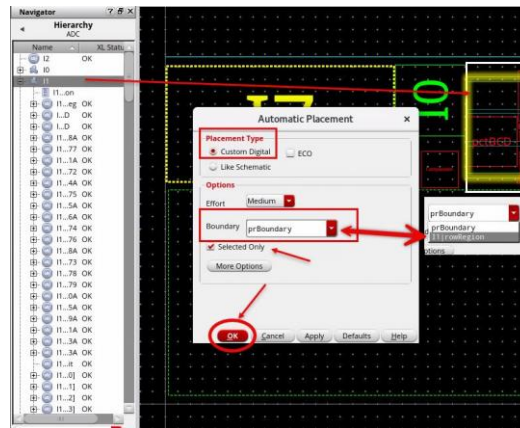
Use the *Automatic Placement* form to place the devices, standard cells, or a mix of both in the rows defined in the layout canvas.



This page does not contain notes.

Running the Automatic Placer

- Use the Automatic Placement form to define the Placement Type as *Custom Digital*, Boundary as *I1|rowRegion*, and enabling the *Selected Only* option.
- The Virtuoso Custom Digital Placer (VCP) runs and places the standard cells contained in the virtual hierarchy **I1** on the rows.



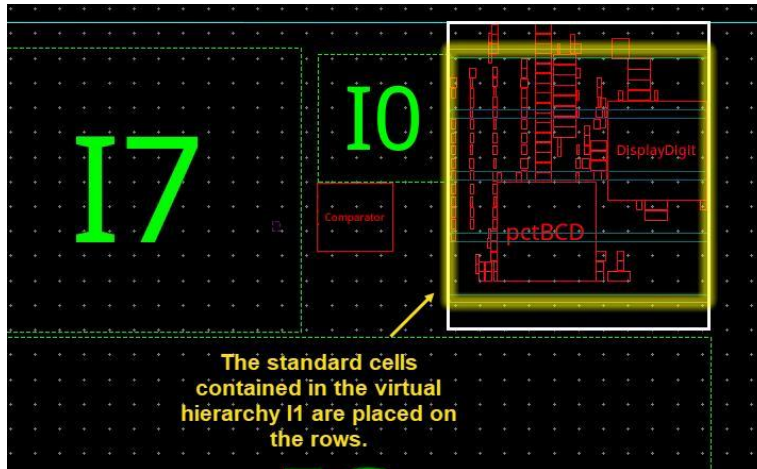
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Result: Running the Automatic Placer

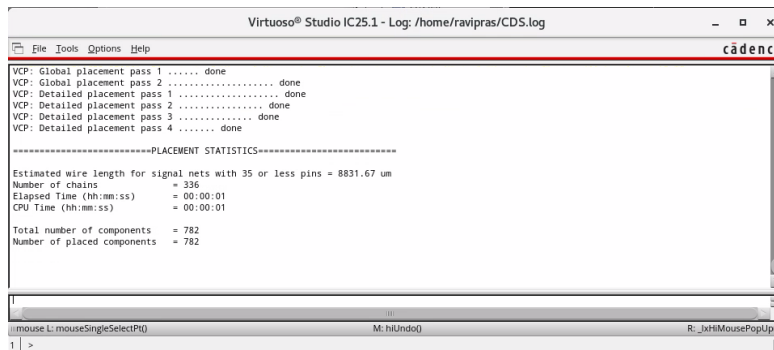
The Virtuoso Custom Digital Placer runs, and the standard cells contained in the virtual hierarchy **I1** are placed on the rows.



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Viewing the Placement Report

- Observe the Placement Statistics report in the CIW window.
- The Placement Statistics report is generated dynamically in the CIW as you run the automatic placer.



The screenshot shows the Virtuoso Studio IC25.1 - Log: /home/ravipras/CDS.log window. The log displays the following text:

```
VCP: Global placement pass 1 ..... done
VCP: Global placement pass 2 ..... done
VCP: Detailed placement pass 1 ..... done
VCP: Detailed placement pass 2 ..... done
VCP: Detailed placement pass 3 ..... done
VCP: Detailed placement pass 4 ..... done

=====PLACEMENT STATISTICS=====
Estimated wire length for signal nets with 35 or less pins = 8831.67 um
Number of chains = 336
Elapsed Time (hh:mm:ss) = 00:00:01
CPU Time (hh:mm:ss) = 00:00:01

Total number of components = 782
Number of placed components = 782
```

The Placement report is available in *CDS.log* file as well.



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Submodule 5-2

Running Congestion Analysis

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Submodule Objectives

In this submodule, you

- Activate the Congestion Analysis assistant in the layout canvas
- Run Global Routing and Congestion Analysis
- Analyze Congestion



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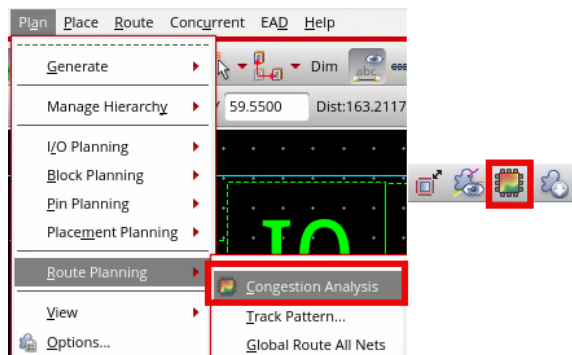
Activating the Congestion Analysis Assistant in the Layout Canvas



In the Layout EXL canvas, choose **Plan – Route Planning – Congestion Analysis** or click the **Congestion Analysis** icon in the Design Planning toolbar to open the Congestion Analysis assistant in the layout window.

Use the *Congestion Analysis* assistant to check whether there are enough resources available for routing.

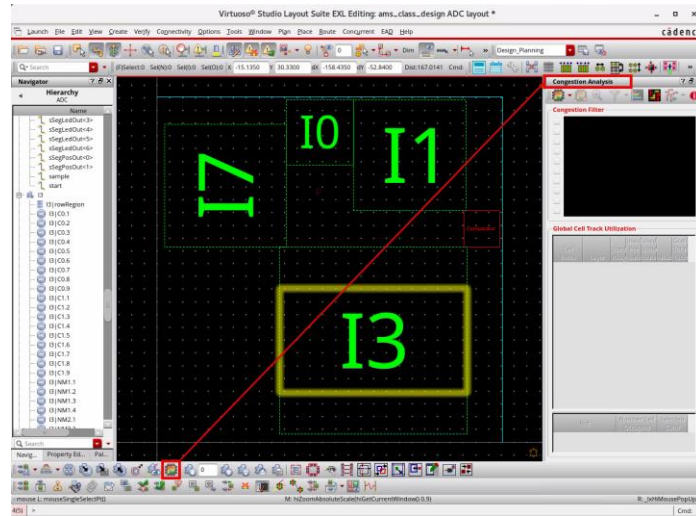
You can invoke the Congestion Analysis assistant from **Window – Assistants – Congestion Analysis** as well.



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Layout Canvas with Congestion Analysis Assistant Enabled

You have the layout window with the Congestion Analysis assistant enabled to check for the availability of routing resources.



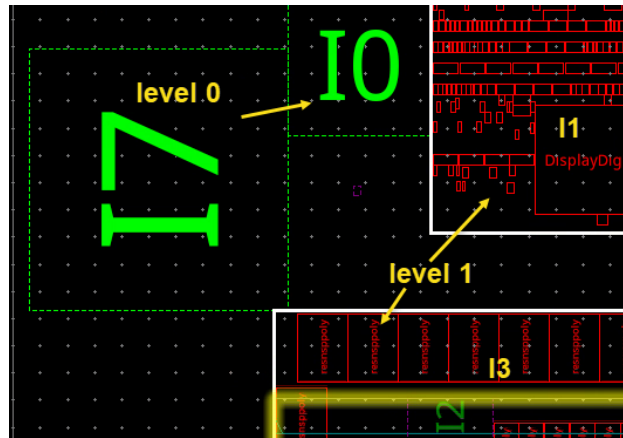
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Setting the Display Depth for the Virtual Hierarchies

- Set the Display Depth for the virtual hierarchies as follows:
 - **Level 1** for the virtual hierarchies **I1** and **I3**.
 - **Level 0** for the other virtual hierarchies.



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Virtual Hierarchy Display Depth Level

- If the contents of a virtual hierarchy have not been placed, it is recommended to set the level to **0**, i.e., opaque.
 - In this case, the virtual hierarchy is considered as a blockage and congestion information is reported only for the interface nets of the virtual hierarchy.
- If the virtual hierarchy is visible, the congestion information for its contents is reported.
 - This is recommended if you have completed some placement within the virtual hierarchy and would like to check the congestion.

The virtual hierarchy display depth level is important when running the congestion analysis.



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What Is Congestion Analysis?



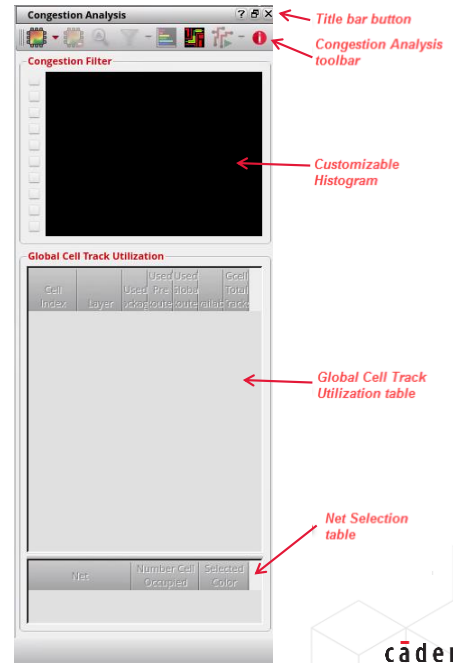
Congestion analysis determines the routing capacity of a design.

You can run congestion analysis both on global routed and detail routed designs.

The congestion in a design can be represented graphically using a heat map, statistically using a histogram, or using the routing metrics table that the global routing engine internally uses to map capacity and plan routing.

The *Congestion Analysis* user interface has five major components:

1. Title Bar Buttons
2. Congestion Analysis Toolbar
3. Customizable Histogram
4. Global Cell Track Utilization Table
5. Net Selection Table



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What Is Congestion Analysis?

Congestion analysis determines the routing capacity of a design.

You can run congestion analysis both on global routed and detail routed designs.

The congestion in a design can be represented graphically using a heat map, statistically using a histogram, or by using the routing metrics table that the global routing engine internally uses to map capacity and plan routing.

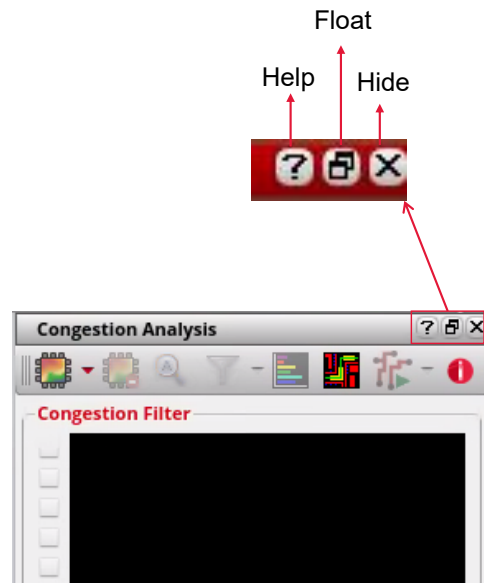
The *Congestion Analysis* user interface has five major components, namely:

- Title Bar Buttons, Congestion Analysis Toolbar, Customizable Histogram, Global Cell Track Utilization Table, and Net Selection Table.

Refer to the *Virtuoso Space-based Router User Guide* for more information on the *Congestion Analysis* user interface.

Title Bar Buttons

- Help
 - Displays information about how to use the *Congestion Analysis* assistant.
- Float/Dock
 - Undocks the *Congestion Analysis* assistant. Click the button again to dock the *Congestion Analysis* assistant.
 - **Note:** You can also drag the *Congestion Analysis* to dock or undock it.
- Hide
 - Hides the *Congestion Analysis* assistant.



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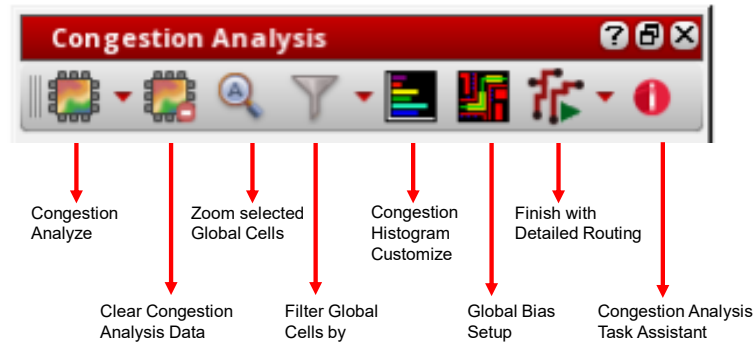


Title Bar Buttons

- Help button displays the information about how to use the *Congestion Analysis* assistant.
- Float/Dock button undocks the assistant. Click the button again to dock the assistant. You can also drag the assistant to dock or undock it.
- Hide button Hides the assistant.

Congestion Analysis Toolbar

The *Congestion Analysis* assistant toolbar lets you access the main functions with a single mouse click.

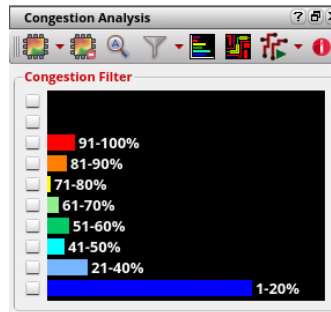


Congestion Analysis Toolbar

You can use the *Congestion Analysis* assistant toolbar to access the main functions with a single mouse click.

Customizable Histogram

- Running congestion analysis displays the congestion data in the histogram.
- The histogram illustrates the congestion results in the design based on the selected filter option.



Customizable Histogram

Running congestion analysis displays the congestion data in the histogram.

The histogram illustrates the congestion results in the design based on the selected filter option.

Global Cell Track Utilization Table

Running Congestion Analysis populates the *Global Cell Track Utilization* table with the capacity and availability information of every Gcell in the design.

Cell Index	Layer	Used Package	Used Pre-route	Used Global-route	Gcell Total Track
(100 95)	Metal1	22	0	3	0 25
(80 90)	Metal2	19	0	1	0 20
(80 91)	Metal2	19	0	1	0 20
(80 92)	Metal2	19	0	1	0 20
(80 93)	Metal2	19	0	1	0 20
(80 94)	Metal2	19	0	1	0 20
(80 95)	Metal2	19	0	1	0 20
(80 96)	Metal2	19	0	1	0 20
(80 97)	Metal2	19	0	1	0 20
(80 98)	Metal2	19	0	1	0 20
(102 96)	Metal2	15	0	5	0 20
(102 97)	Metal2	15	0	5	0 20
(102 98)	Metal2	15	0	5	0 20
(102 102)	Metal2	15	0	5	0 20

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Global Cell Track Utilization Table

Running congestion analysis populates the *Global Cell Track Utilization* table with the capacity and availability information of every Gcell in the design.

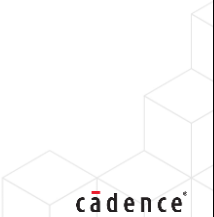
Net Selection Table

- After running Congestion Analysis, when you select a Gcell in the *Global Cell Track Utilization* table, the nets passing through the selected Gcell are displayed in a table below the *Global Cell Track Utilization* table.
- This table is the Net selection table.

The screenshot shows two tables. The top table is titled "Global Cell Track Utilization" and lists various cells and their utilization. The bottom table is the "Net Selection Table" which shows the nets passing through the selected Gcell.

Cell Index	Layer	Used Pre	Used Post	Global	Local	Gcell Total
(100 95)	Metal1	22	0	3	0	25
(80 90)	Metal2	19	0	1	0	20
(80 91)	Metal2	19	0	1	0	20
(80 92)	Metal2	19	0	1	0	20
(80 93)	Metal2	19	0	1	0	20
(80 94)	Metal2	19	0	1	0	20
(80 95)	Metal2	19	0	1	0	20
(80 96)	Metal2	19	0	1	0	20
(80 97)	Metal2	19	0	1	0	20

Net	Number Cell Occupied	Selected Color
I1 final_value<4>	1	Yellow
Out<2>	1	Cyan
VDD!	1	Orange



Net Selection Table

After running congestion analysis, when you select a Gcell in the *Global Cell Track Utilization* table, the nets passing through the selected Gcell are displayed in a table below the *Global Cell Track Utilization* table.

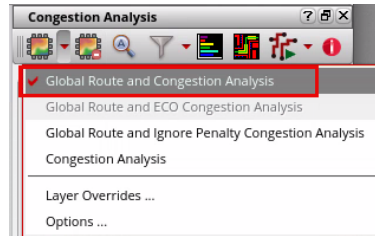
Here, we see the Net selection table.

Running Global Routing and Congestion Analysis



In the Layout EXL window, in the **Congestion Analysis** assistant, click the top-left **Congestion Analyze** icon and select **Global Route and Congestion Analysis**.

To visualize the congestion data, run *Global Route and Congestion Analysis* on the design data.

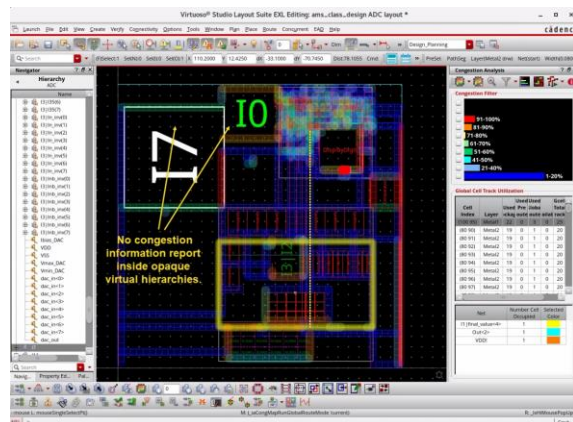


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Result: Running Global Routing and Congestion Analysis

- The Global Route command runs, and the results are displayed in the layout canvas as well as in the *Congestion Analysis* assistant.
- A GCell is a global cell used for routing and congestion analysis.
- Color coding is used to show the level of congestion in each global cell known as Gcell.

Cold colors, such as **Blue** and **Green**, indicate less congestion. Warmer colors, such as **Yellow**, **Orange**, and **Red**, indicate high congestion. Hot colors, such as **Purple** and **White**, indicate over-congestion.



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- ```
Virtuoso® Studio IC25.1 - Log /home/ravi/prj/CDS.log
```
- ```
File Edit Tools Options Help
```
- ```
=====
*
* Begin Route Summary
*
=====
Number of nets in design: 238
Number of nets attempted: 232
Number of nets excluded: 0
Number of terminals excluded: 0
=====
*
* End Route Summary
*
=====
Total Flow Elapsed Time: 5 seconds
Total Flow CPU Time: 5 seconds (user) 1 second (kernel)
REPORTING_STATS : Elapsed Time: 9.0 Sec ; Mem : Current: 2769.21 Mb | End: vca_jammer_modified_locked_routers -net set:4fa0800
=====
*
* End Congestion Map Flow
*
=====

***** Selected elements are ignored when routing with NOT topology style.

End of VSR command ...

Run Pin Post Processing

Deleted the Cover Obstruction

Global Cells(100/95) has 100% left congestion value on layer Metall.
```
- ```
Module L: measureSingleSelect(Pg)          M: LstCompMapRunGlnbaRouteMode Summets)
S: JahnMouse@Plytch
```

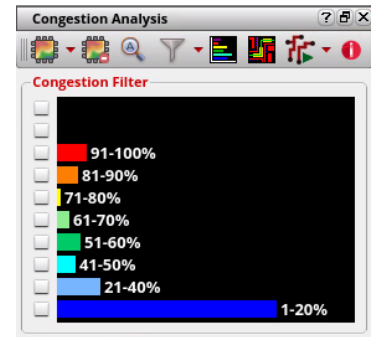
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Using the Congestion Analysis Histogram



Before final routing, a layout engineer reviews the congestion histogram to identify high-congestion GCells across virtual hierarchies, then adjusts block placement and spacing to ensure balanced congestion and sufficient routing resources.

- Analyze congestion early and refine placement in iterations
- Use histogram filtering to target the most congested GCells first
- Maintain adequate spacing between hierarchies for routing
- Keep congestion balanced inside and across virtual hierarchies
- Re-check congestion after each major placement change



- The congestion analysis histogram shows the breakdown of the congestion across the GCells.
- The results are displayed in ranges that can be filtered, allowing you to review only the areas of interest.



Using the Congestion Analysis Histogram

Use the congestion information for two purposes:

- To ensure that there are enough routing resources between the virtual hierarchies for the top-level routing.
- The virtual hierarchies themselves are not under or over congested.

The congestion analysis histogram shows the breakdown of the congestion across the GCells.

The results are displayed in ranges that can be filtered, allowing you to review only the areas of interest.

Analyzing the Congestion

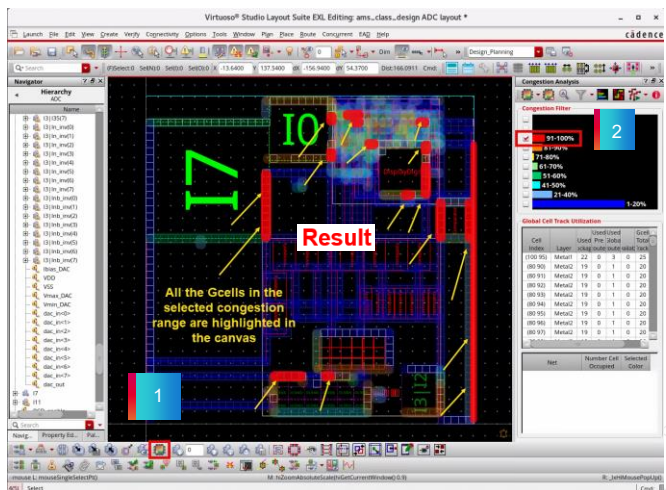


Use the heat map, the histogram, and the global cell track utilization table to analyze the congestion, quickly debug hot spots, and find nets that pass through congested regions of the design.

To analyze the congestion after running Global Routing and Congestion Analysis:

1. Make sure that the *Congestion Analysis* assistant is active.
2. In the *Congestion Analysis* assistant, click the **91-100%** checkbox in the *Filter by Histogram* pane.

Result: All the GCells in the selected congestion range are highlighted in the canvas.



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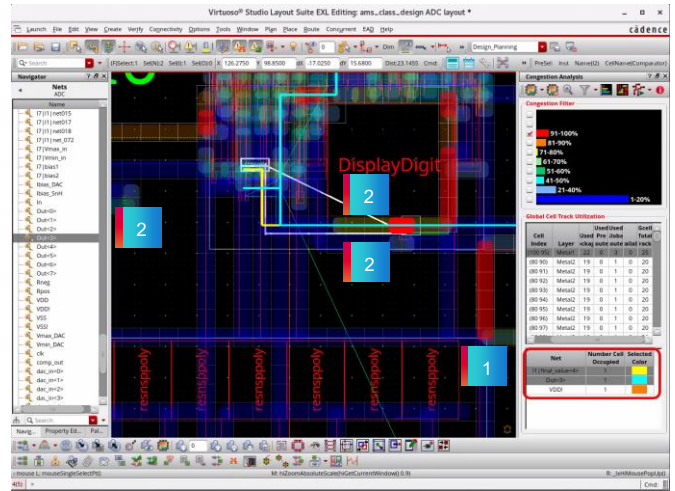
1. Select the top entry in the *Global Cell Track Utilization* table.
2. The GCell is highlighted in the layout canvas and the breakdown of the congestion is shown in the utilization table.

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Using the Net Selection Table

1. Select the nets shown in the bottom pane of the *Congestion Analysis* assistant.
2. The selected nets are displayed in the layout canvas and in the Navigator assistant.

This congestion information can be used to refine the design plan.



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Running the Global Router in the Design Planner

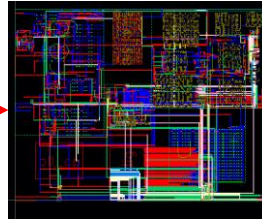
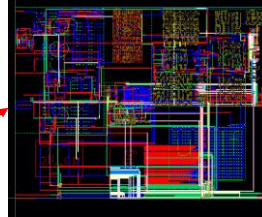
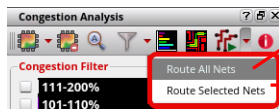


Run the Global Router to get a measure of congestion in the design.

- Routing is an important part of the flow in the Design Planner, as in the *Make Cell* command, pins will be created where routes cross the boundary.



Available from the *Congestion Analysis* toolbar (in pull-down menu)



- Early in the Design Planning flow, the placement of the virtual hierarchies will not be fixed, but the user may want to do a global route to get a measure of congestion.

Although these results look very similar, when *Selected Nets* is used, Croute is called to spread apart the global nets to avoid shorts, when pins are created in *Make Cell*.



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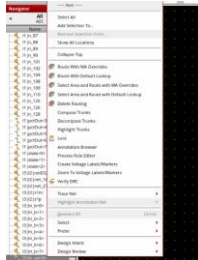
Running the Detailed Router in the Design Planner



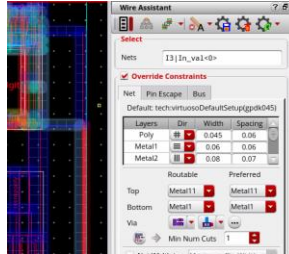
Run the Detailed Router to route the selected set of nets or the entire design.

- Routing is an important part of the flow in the Design Planner, as in the Make Cell command, pins will be created where routes cross the boundary.

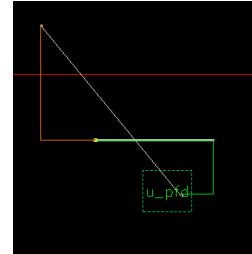
- Later in the Design Planning flow, as the positioning of the Virtual Hierarchies becomes more fixed, then the user will want to move onto detailed route for more critical nets.



Select the interface net(s) in the Virtual Hierarchy view in the Layout Navigator and use the RMB to route



... or use the Wire Assistant



... or manually route using the Create Wire command

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Submodule 5-3

Creating Real Layout Views for the Virtual Hierarchies

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Submodule Objectives

In this submodule, you

- Run *Make Cell* with the *Create pins: Congestion aware* option
- Run *Make Cell* with the *Create pins: Below boundary* option
- Run *Make Cell* with the *Create pins: On boundary* option
- Run *Remaster* to handle layouts created outside of the virtual hierarchy



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What Is Make Cell?



Make Cell will create a real cellview from a virtual hierarchy.

When the design planning is complete, you can create a real cell to pass on for implementation.

- If the design is opened in Layout EXL and no virtual hierarchy is pre-selected, the design must contain a virtual hierarchy for the Make Cell form to display the options related to the virtual hierarchies.
- If the design is opened in Layout EXL and the *Make Cell* command is invoked post selection, the selected figGroup must be a virtual hierarchy figGroup.

The screenshot shows the 'MakeCell' dialog box with the following settings:

- Library: **ams_class_design**
- Cell: (empty)
- View: **Layout**
- Type: **none**
- ☒ Replace selected figures
- Origin: **LowerLeft**
 - X.Y: 59.455, 76.485
 - Set by cursor
- Hierarchy:
 - ☒ Transparent instance
 - ☒ Preserve connectivity
 - ☒ Create pins
 - ☐ Pins below boundary
- P&R Boundary: **Rectangle**
- Cluster: **none**
 - Type: **Inclusive**

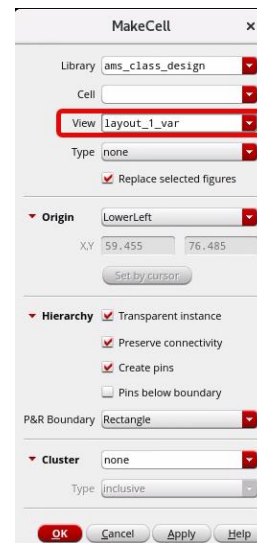
Buttons at the bottom: **OK**, **Cancel**, **Apply**, **Help**.



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Make Cell: Layout Variant View

- In the Make Cell form, **View** specifies the name of the view to be created for the virtual hierarchy blocks.
- By default, *Make Cell* creates a layout variant view.
- You can edit the fields if you want the new layout variant to be created in a different name.



The screenshot shows the 'MakeCell' dialog box with the following settings:

- Library: ams_class_design
- Cell: (empty)
- View: layout_1_var (highlighted with a red rectangle)
- Type: none
- ☒ Replace selected figures
- Origin: LowerLeft
- X,Y: 59.455, 76.485
-
- Hierarchy:
 - ☒ Transparent instance
 - ☒ Preserve connectivity
 - ☒ Create pins
 - ☐ Pins below boundary
- P&R Boundary: Rectangle
- Cluster: none
- Type: inclusive
- Buttons: OK, Cancel, Apply, Help

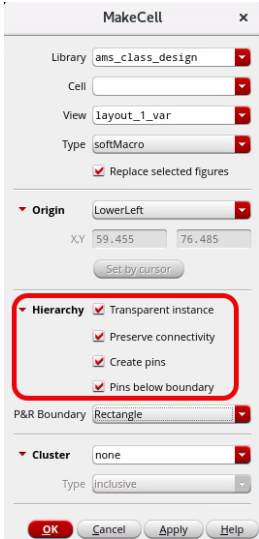


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Make Cell: Hierarchy Options

Make Cell converts the selected virtual hierarchy into a soft macro cell, with the Hierarchy section controlling the conversion.

- Transparent instance — Keeps the new cell visible and editable from the top level.
- Preserve connectivity — Maintains all existing net connections during conversion.
- Create pins — Automatically generates interface pins on the boundary based on connected nets.
- Pins below boundary — Places the generated pins just inside the boundary instead of on the edge.



The screenshot shows the 'MakeCell' dialog box with the following settings:

- Library: ams_class_design
- Cell: (empty)
- View: Layout_1_var
- Type: softMacro
- ☒ Replace selected figures
- Origin: LowerLeft
- XY: 59.455, 76.485
- Set by cursor
- Hierarchy** (highlighted with a red box):
 - ☒ Transparent instance
 - ☒ Preserve connectivity
 - ☒ Create pins
 - ☒ Pins below boundary
- P&R Boundary: Rectangle
- Cluster: none
- Type: inclusive
- Buttons: OK, Cancel, Apply, Help



This page does not contain notes.

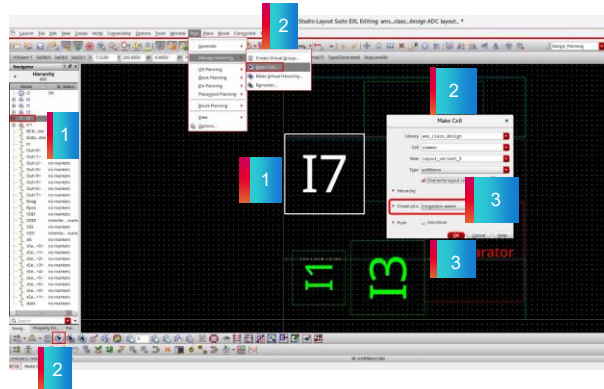
Running *Make Cell* with the *Create pins: Congestion aware* Option



Use **Make Cell** to create a cellview using the selected top-level virtual hierarchy with the *Create pins: Congestion aware* option.

1. In the layout canvas, select the virtual hierarchy **I7**.
2. Click the **Make Cell** icon in the Design Planning toolbar or choose the **Plan – Manage Hierarchy – Make Cell** menu command to open the Make Cell form.
3. Select **Congestion aware** in the **Create pins** field and click **OK** to run the Make Cell command.

Result: Running the *Make Cell* command on the selected virtual hierarchy **I7** creates a layout view for **I7** using the selected options. The area boundary is updated to a PR Boundary.



Reducing the virtual hierarchy display depth to 0 makes the virtual hierarchy opaque, allowing the *Create pins: Congestion aware* option in the *Make Cell* form to create pins anywhere on the boundary. This is recommended for the virtual hierarchies that are not yet placed.

This page does not contain notes.

-
- The screenshot shows the Virtuoso® Studio IC25.1 interface with a log window titled "Log: /home/ravipras/CDS.log". The log contains the following text:
- ```

* *
* Begin Route Summary *
* *

Number of nets in design: 227
Number of nets attempted: 225
Number of nets excluded: 0
Number of terminals excluded: 0

* *
* End Route Summary *
* *

Total Flow Elapsed Time: 1 second
Total Flow CPU Time: 1 second (user) 0 seconds (kernel)
REPORTING_STATS | Elapsed Time: 0.0 Secs | Mem: "Current: 2673.52" Mb | Cmd: vca_unmark_modified_locked_routes -set sel:51062010

* *
* End Congestion Map Data Clearing *
* *

INFO (VPA-10823): The pins, when placed on the boundary of the block, would be assigned the layer as per its preferred routing direction
defined in the technology database. The layer assigned could be different from the layer provided by the user in the 'Create Pins on
Boundary' option in 'Make Cell' form
Loading lverf.ctt
INFO (L-1908): Cellview ans_class_designDisplay@igt:layout has been made editable.
```
- The bottom status bar shows the file path "/mouse L: mouseSingsSelect(Pti)", the current cell "M: jrhMakeCell()", and the right-click context menu item "R: jrhMousePopUp()".

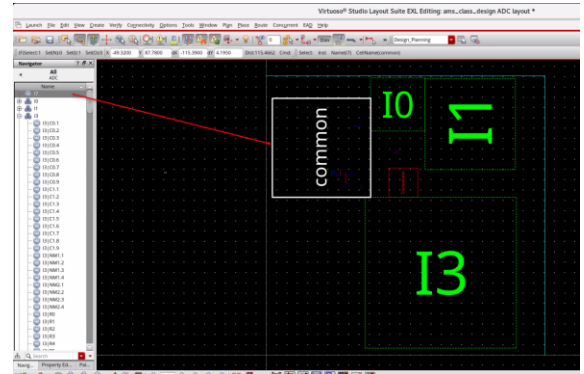
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## Viewing the Made Cell

- In the Navigator assistant, in the *Hierarchy* view, select **I7**.

### Observation

- You can notice that **I7** is no longer a virtual hierarchy.
- The icon in the Navigator assistant has changed to reflect that the virtual hierarchy is now an instance of a layout.
- The XL Status is **OK**, showing that the created instance is fully XL compliant.
- In the layout canvas, the instance is now colored **red**.



The layout instance **I7** is now changed from  to  icon as shown in the Navigator assistant.

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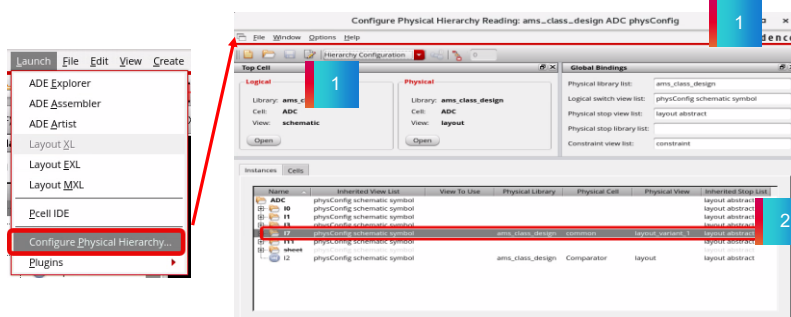
## Reviewing the *physConfig* View of the Made Cell



In the layout canvas, choose **Launch – Configure Physical Hierarchy** to open the Configure Physical Hierarchy (CPH) form.

1. In the Configure Physical Hierarchy form, in the *Hierarchy Configuration* cyclic section, review the *Instances* tab.
2. The *physConfig* view has been updated so that the newly created layout view automatically replaces the virtual hierarchy.

**Result:** A physical binding is created to match the options set in the Make Cell form.



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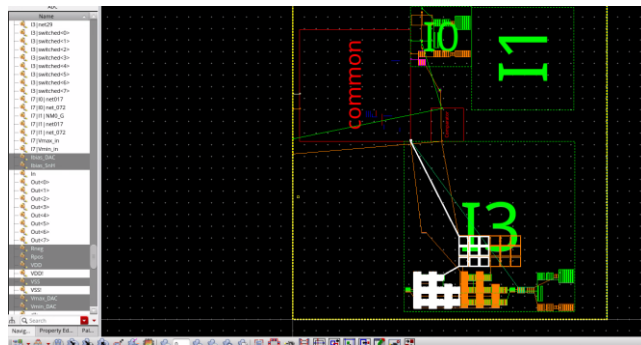
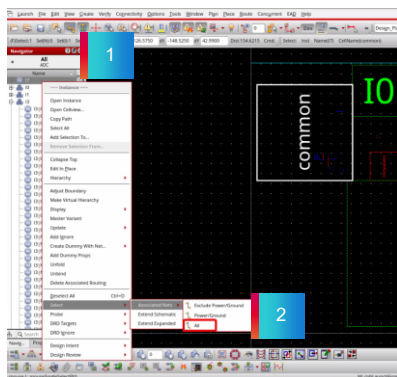
cadence

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## Viewing the Pins Created from the *Make Cell* with the *Create pins: Congestion aware* Option

1. In the layout canvas, select the newly created layout instance I7.
2. Right-click the selected instance I7 and choose **Select – Associated Nets – All** from the context-sensitive menu.

In this mode, pins are created on the boundary, attempting to minimize the top-level net length.



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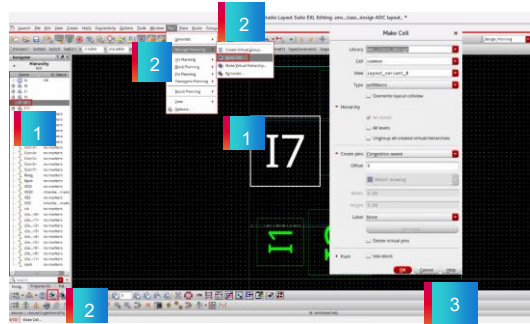
## Running *Make Cell* with the *Create pins: Below boundary* Option



Use **Make Cell** to create a cellview using the selected top-level virtual hierarchy with the *Create pins: Below boundary* option.

1. In the layout canvas, select the virtual hierarchy **I7**.
2. Click the **Make Cell** icon in the Design Planning toolbar or choose the **Plan – Manage Hierarchy – Make Cell** menu command to open the Make Cell form.
3. Set the options as shown here and click **OK** to run the Make Cell command.

**Result:** Running the Make Cell command on the selected virtual hierarchy **I7** creates a layout view for **I7** using the selected options. The area boundary is updated to a PR Boundary.



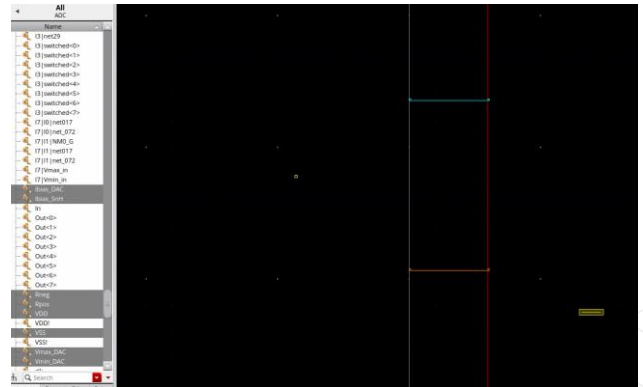
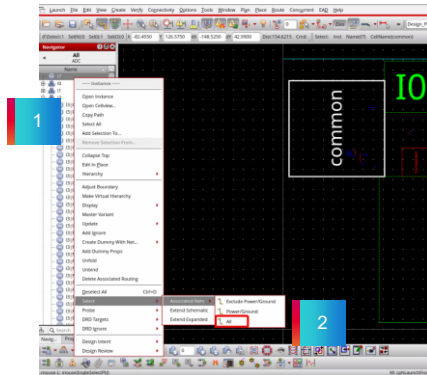
The *Create pins: Below boundary* mode does not run global routing and allows you to specify the pin layer and size.

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### Viewing the Pins Created from the *Make Cell* with the *Create pins: Below boundary* Option

1. In the layout canvas, select the newly created layout instance **I7**.
2. **Right-click** the selected instance **I7** and choose **Select – Associated Nets – All** from the context-sensitive menu.

In this mode, the pins are created at the bottom corner, so further pin optimization may be required.



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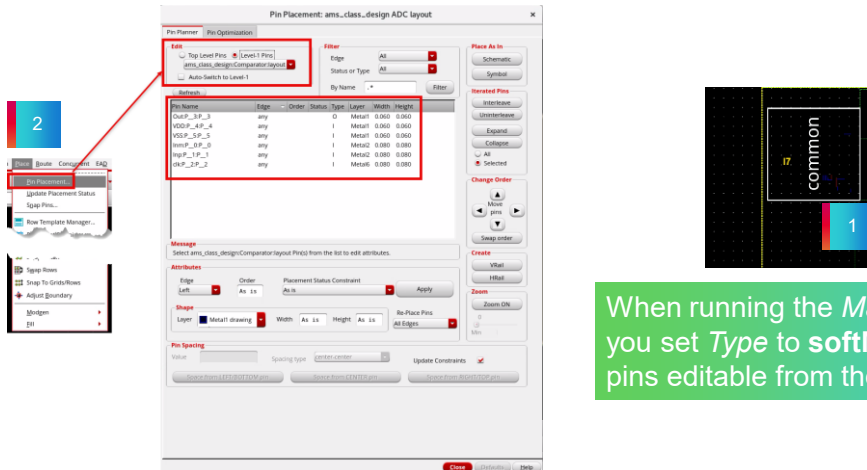
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## Pin Optimization Using the Pin Placement Form

1. In the layout window, select the layout instance I7.
2. Choose **Place – Pin Placement** to open the Pin Placement form.

**Result:** The pin positions can be further edited after creation with the *Pin Planner*/*Pin Optimization* commands.

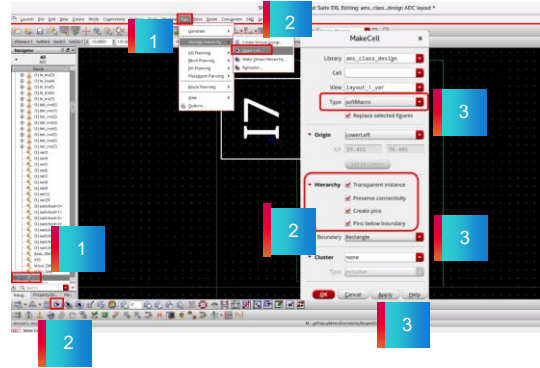


When running the *Make Cell* command, you set *Type* to **softMacro** to make the pins editable from the top level.

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Use **Make Cell** to create a cellview using the selected top-level virtual hierarchy with the *Create pins: On boundary* option.

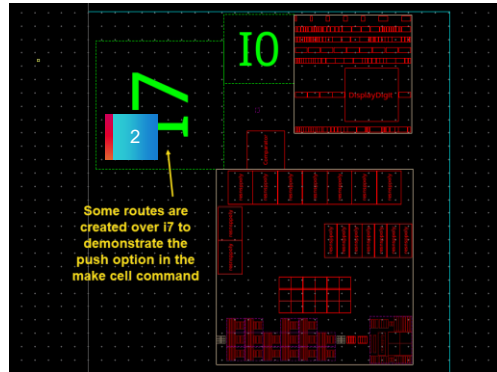
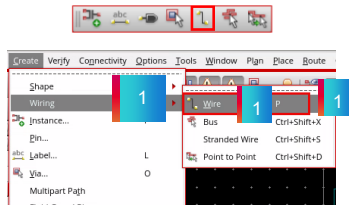
- Result:** Running the Make Cell command on the selected virtual hierarchy **17** creates a layout view for **17** using the selected options. The area boundary is updated to a PR Boundary.



The *Create pins: On boundary* mode does not run global routing and allows you to specify the pin layer and size.

## Creating Routes to Demonstrate the Push Option in Make Cell

1. In the layout canvas, choose the **Create – Wiring – Wire** menu command or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to create some routes over **I7**.
2. Use the Create Wire command to create some routes over **I7** to demonstrate the Push option in the Make Cell command.



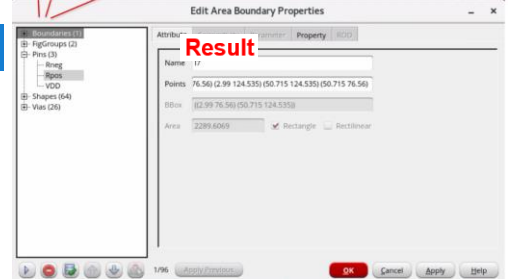
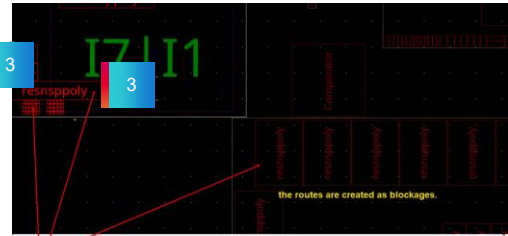
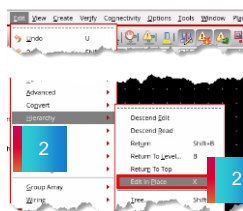
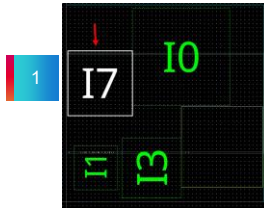
**Note:** Before running Make Cell, create these routes to observe the Push option in the Make Cell command.



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## Viewing the Routes Created Which Are Pushed as Blockages

1. In the layout canvas, select the layout instance **I7**.
2. Choose the **Edit – Hierarchy – Edit In Place** menu command or press the bindkey **X** to edit in place of the instance **I7**.
3. Select the routes created in the layout cell to examine their properties.
4. Choose the **Edit – Basic – Properties** menu command or click the **Edit Properties** icon in the Edit toolbar or press the bindkey **Q** to query the routes in the layout canvas.



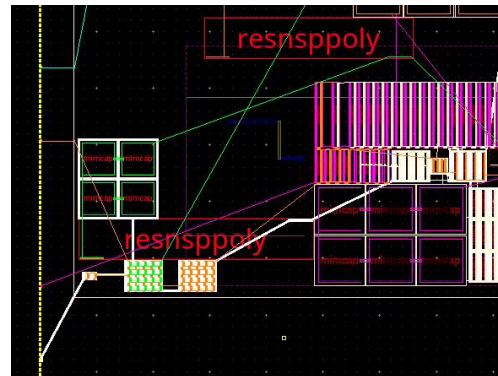
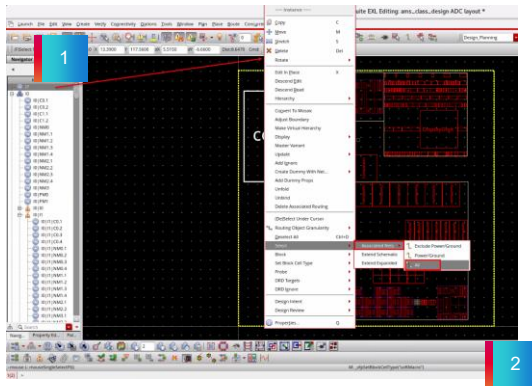
**Result:** You can see these are blockages.

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## Viewing the Pins Created from the *Make Cell* with the *Create pins: On boundary* Option

1. In the layout canvas, select the newly created layout instance **I7**.
2. **Right-click** the selected instance **I7** and choose **Select – Associated Nets – All** from the context-sensitive menu.

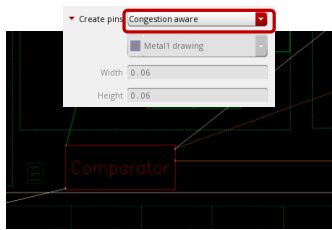
When the *On boundary* option is used, the pins are created on the boundary, as expected.



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## Design Planning: Make Cell: Flexible Pin Creation



### Congestion aware: Default Mode

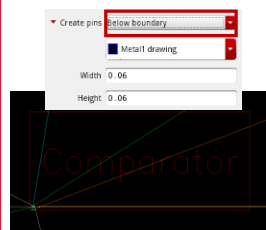
- Pin position, layer, and size determined by global router

### Advantages

- Congestion aware pin creation
- Pin targets elongated on opaque neighbors

### Disadvantages

- Runtime may be longer



### Below boundary

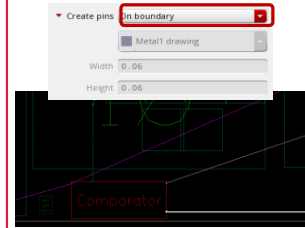
- Pins positioned bottom left
- Pin layer and size are set in options

### Advantages

- Pin location easy to identify

### Disadvantages

- Created pins will need to be optimized after creation



### On boundary

- Pin position determined by flightlines
- Pin layer and size are set in options

### Advantages

- Quick method to create pins on boundary of the made cell

### Disadvantages

- Not aware of congestion
- Pin positions affected by targets that may not be placed



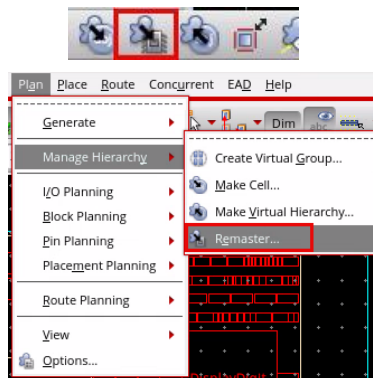
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## Running Remaster to Handle Layouts Created Outside of the Virtual Hierarchy



In the layout canvas, click the **Remaster** button in the Design Planning toolbar or choose **Plan – Manage Hierarchy – Remaster** to open the Remaster form.

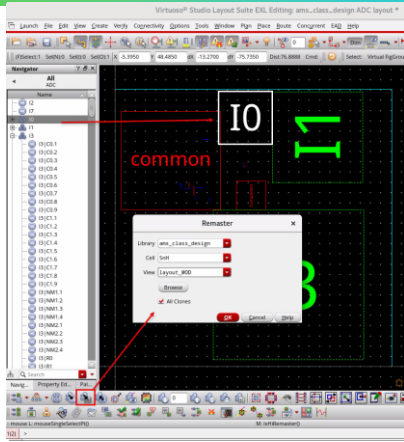
- There may be cases where you create a virtual hierarchy for a schematic, and the layout has been implemented outside of the Design Planning environment.
- You can replace a virtual hierarchy with an existing layout using the *Remaster* command.
- You can use the *Remaster* form to replace the selected virtual hierarchy with the selected layout master that exists on the disk.



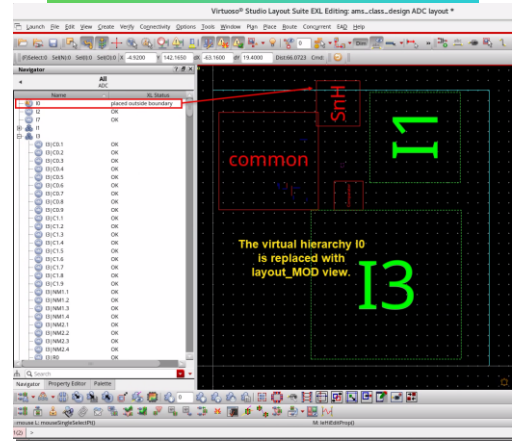
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## Remastering the Virtual Hierarchy I0

After selecting the virtual hierarchy **I0**, in the Remaster form, in the **View** pull-down, select **layout\_MOD** and click **OK**.



The virtual hierarchy **I0** is replaced with the **layout\_MOD** view.



In the Navigator assistant, the XL Status shows as *placed outside boundary*.

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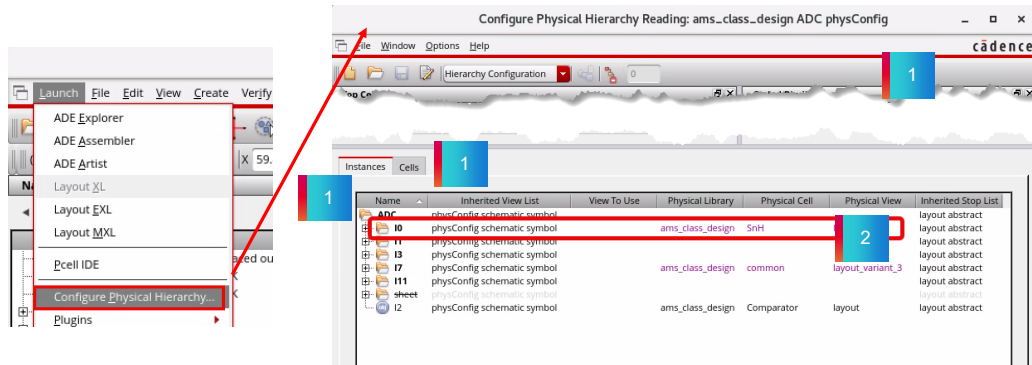


## Reviewing the *physConfig* View of the Remastered Layout



In the layout canvas, choose **Launch – Configure Physical Hierarchy** to open the Configure Physical Hierarchy (CPH) form.

1. In the CPH form, in the *Hierarchy Configuration* cyclic section, review the *Instances* tab.
2. The *physConfig* view has been updated so that after remastering is done, and as for the *Make Cell* command, the physical bindings in Configure Physical hierarchy are handled automatically.



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## **Submodule 5-4**

### **Creating Virtual Hierarchies for the Layout Views**

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## Submodule Objectives

In this submodule, you

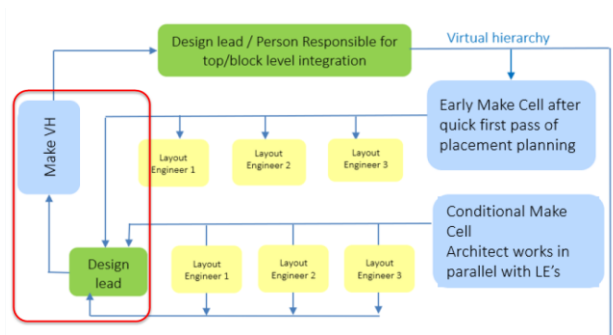
- Create a virtual hierarchy using the Make Virtual Hierarchy command
- Run the Make Virtual Hierarchy command



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## Creating a Virtual Hierarchy Using the Make Virtual Hierarchy Command

- You will bring a layout view back into the virtual hierarchy using the *Make Virtual Hierarchy* command.
- In a real design scenario, this task is likely to be performed by the Layout Engineer responsible for implementing the top level.
- To do so, the Layout Engineer will bring a layout implemented by another Engineer back into the virtual hierarchy to make some changes to the embedded cellview in the context of the top level.



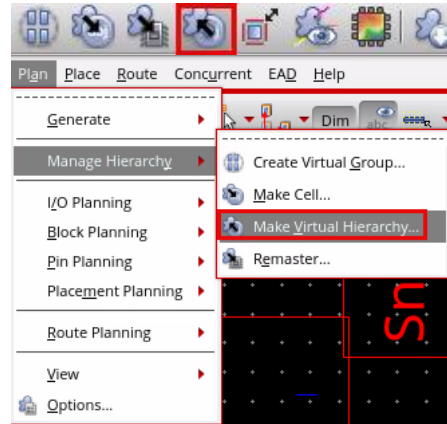
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## Running the Make Virtual Hierarchy Command



In the layout canvas, click the **Make Virtual Hierarchy** button in the Design Planning toolbar or choose the **Plan – Manage Hierarchy – Make Virtual Hierarchy** menu command to open the Make Virtual Hierarchy form.

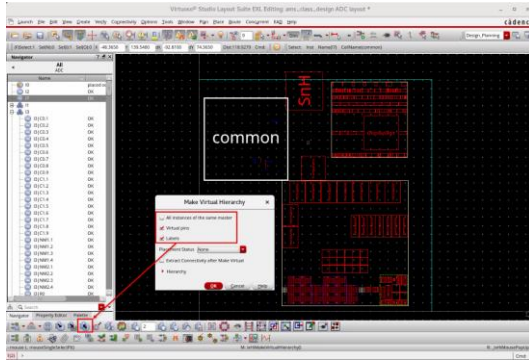
- Use the *Make Virtual Hierarchy* form to integrate a layout hierarchy that was realized outside the design.
- Use this command to create virtual hierarchies from real cellviews.
- You can make a real cell to virtual again to make some edits, such as DRC fixes, in the context of the top level.



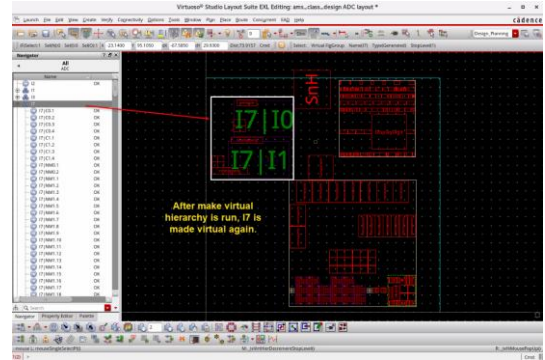
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## Making the Virtual Hierarchy for the Layout I7

Select the layout instance I7. In the *Make Virtual Hierarchy* form, set the options as shown here and click **OK**.



After *Make Virtual Hierarchy* is run, I7 is made virtual again.



The virtual hierarchy I7 is represented using an amoeba-shaped icon as seen in the Navigator assistant.

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## Reviewing the *physConfig* View of the Made Virtual Hierarchy I7



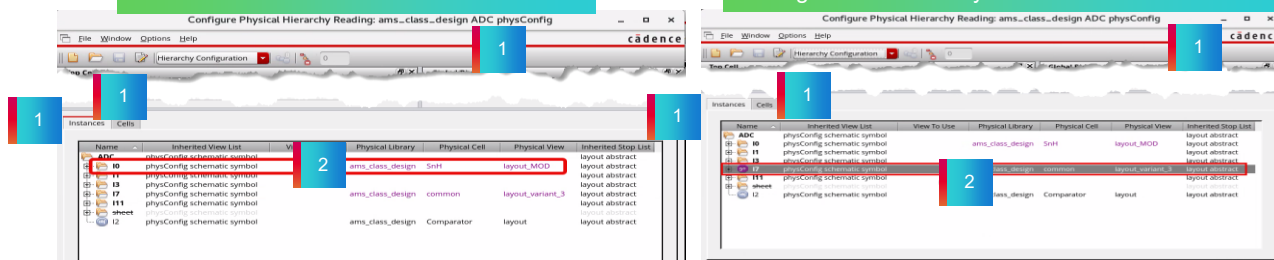
In the layout canvas, choose **Launch – Configure Physical Hierarchy** to open the Configure Physical Hierarchy (CPH) form.

1. In the CPH form, in the *Hierarchy Configuration* cyclic section, review the *Instances* tab.
2. The *Make Virtual Hierarchy* command makes a virtual hierarchy from the layout view I7.

**Result:** It also automatically updates the *physConfig* view so that the virtual hierarchy is used instead of the layout.

Before *Make Virtual Hierarchy* is run, layout instance I7 has the *layout\_variant\_3* view as seen in the CPH form.

After *Make Virtual Hierarchy* is run, a force descend has been added for I7 so that the *layout\_variant\_3* is no longer used in the layout.



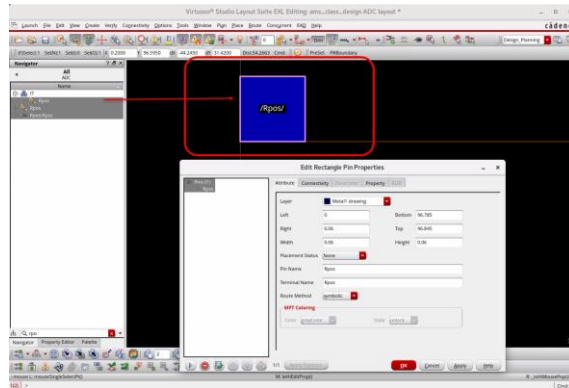
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## Reviewing the Properties of the Virtual Pins of Virtual Hierarchy I7

Zoom in to the edge of I7 to locate any of the virtual pins and review the properties.



The virtual pin is not a real pin but a shape with connectivity.



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## Quiz: Defining the Rows



What is a container that defines one or more rows?

A row ----- is a container that defines one or more rows.

Enter the answer in the blank field.

Answer: template

A row template is a container that defines one or more rows.



Answer: template

A row template is a container that defines one or more rows.

## Quiz: Defining the Row Templates



Which feature provides the capability to define the row templates?

Select all that apply.

- A. Row Template Organizer
- B. Row Template Provider
- C. Row Template Manager
- D. Row Template Attributes
- E. None of the above

Answer: C

The Row Template Manager feature provides the capability to define the row templates.



Answer: C

The Row Template Manager feature provides the capability to define row templates.

## Quiz: Controlling the Area Boundary of the Virtual Hierarchies



Which form can you use to control the area boundary of the virtual hierarchy?

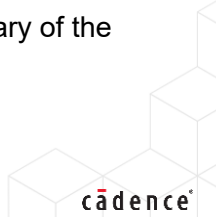
Select all that apply.

- A. Adjust Area Boundary
- B. Control Area Boundary
- C. Control Virtual Hierarchy
- D. Design Planning and Analysis Options
- E. None of the above

Answer: D

You can use the Design Planning and Analysis Options form to control the area boundary of the virtual hierarchy.

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Answer: D

You can use the Design Planning and Analysis Options form to control the area boundary of the virtual hierarchy.

## Quiz: Placement Statistics Report of the Auto Placer



Where can you observe the Placement Statistics report of the automatic placer run?  
Select all that apply.

- A. Library Manager
- B. CIW
- C. Auto Placer
- D. CDS.log
- E. All of the above

Answers: B and D

You can observe the Placement Statistics report of the automatic placer run in the CIW and in the CDS.log file.



Answers: B and D

You can observe the Placement Statistics report of the automatic placer run in the CIW and in the CDS.log file.

## Quiz: The Assistant for Checking the Resources for Routing



Which assistant can you use to check whether there are enough resources available for routing?

Select all that apply.

- A. Annotation Browser
- B. Congestion Analysis
- C. Constraint Manager
- D. Wire Assistant
- E. None of the above

Answer: B

You can use the Congestion Analysis assistant to check whether there are enough resources available for routing.

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Answer: B

You can use the Congestion Analysis assistant to check whether there are enough resources available for routing.

## Quiz: Creating Cellview from the Virtual Hierarchy



Which form can you use to create a cellview using the selected top-level virtual hierarchy?  
Select all that apply.

- A. Make Cell
- B. Make Virtual Hierarchy
- C. Make Layout Cell
- D. Make Layout Virtual Hierarchy
- E. None of the above

Answer: A

You can use the Make Cell form to create a cellview using the selected top-level virtual hierarchy.



Answer: A

You can use the Make Cell form to create a cellview using the selected top-level virtual hierarchy.

## Quiz: The *Create pins* Options in the Make Cell Form



What are the *Create pins* options in the Make Cell form?

Select all that apply.

- A. Congestion aware
- B. On boundary
- C. Below boundary
- D. Top boundary
- E. All of the above

Answers: A, B, and C

The *Create pins* options in the Make Cell form are *Congestion aware*, *On boundary*, and *Below boundary*.

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Answers: A, B, and C

The *Create pins* options in the Make Cell form are *Congestion aware*, *On boundary*, and *Below boundary*.

## Quiz: Reviewing the *physConfig* View of the Layout Cell



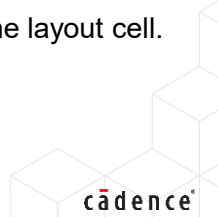
Which form can you use to review the *physConfig* view of the layout cell?

Select all that apply.

- A. Configure Layout Hierarchy
- B. Configure Virtual Hierarchy
- C. Configure Physical Hierarchy
- D. Configure physConfig Hierarchy
- E. None of the above

Answer: C

You can use the Configure Physical Hierarchy form to review the physConfig view of the layout cell.



Answer: C

You can use the Configure Physical Hierarchy form to review the physConfig view of the layout cell.



## Quiz: Replacing the Virtual Hierarchies



Which command can you use to replace the selected virtual hierarchy with the selected layout master that exists on the disk?

Select all that apply.

- A. Make Cell
- B. Remaster
- C. Make Virtual Hierarchy
- D. Remastering Cell
- E. None of the above

Answer: B

You can replace a virtual hierarchy with an existing layout using the Remaster command.

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Answer: B

You can replace a virtual hierarchy with an existing layout using the Remaster command.

## Quiz: Creating the Virtual Hierarchies from Real Cellviews



Which command can you use to create virtual hierarchies from real cellviews?

Select all that apply.

- A. Make Cell
- B. Make Layout Cell
- C. Make Layout Virtual Hierarchy
- D. Make Virtual Hierarchy
- E. None of the above

Answer: D

You can use the Make Virtual Hierarchy command to create virtual hierarchies from real cellviews.



Answer: D

You can use the Make Virtual Hierarchy command to create virtual hierarchies from real cellviews.

## Module Summary

In this module, you learned to:

- Create floorplan rows and run automatic placement to organize instances efficiently across the design.
- Analyze routing congestion to identify problem areas and optimize the overall layout quality.
- Generate real layout views that accurately represent the virtual hierarchies in your design.
- Build virtual hierarchies from existing layout views to streamline hierarchical management.



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## Labs



- Lab 5-1 Creating Rows and Running Automatic Placement
- Lab 5-2 Running Congestion Analysis
- Lab 5-3 Creating Real Layout Views for the Virtual Hierarchies
- Lab 5-4 Creating Virtual Hierarchies for the Layout Views



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## Module 6

### Course Conclusions

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## Summary

In this course, you learned to

- Set up and analyze the Design Planning Environment
- Use the virtual hierarchy in Design Planning
- Analyze Early Design Planning
- Use the Design Planner in the Team Environment



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## References

Refer to these links for more details on the topics discussed in this course.

- [Virtuoso Design Planning and Analysis User Guide Product Version IC25.1](#)
- [Virtuoso Placer User Guide Product Version IC25.1](#)



- [Virtuoso Design Planning and Analysis User Guide Product Version IC25.1](#)
- [Virtuoso Placer User Guide Product Version IC25.1](#)



## Module 7

### Next Steps

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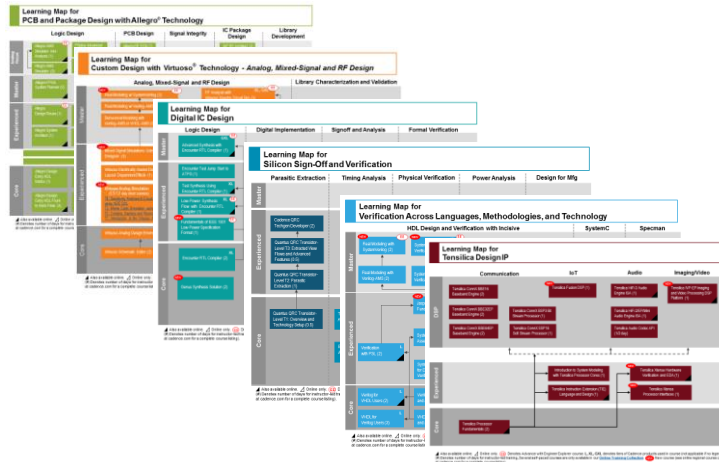
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Click [here](#) to see all our courses in each technology area and the recommended order in which to take them.



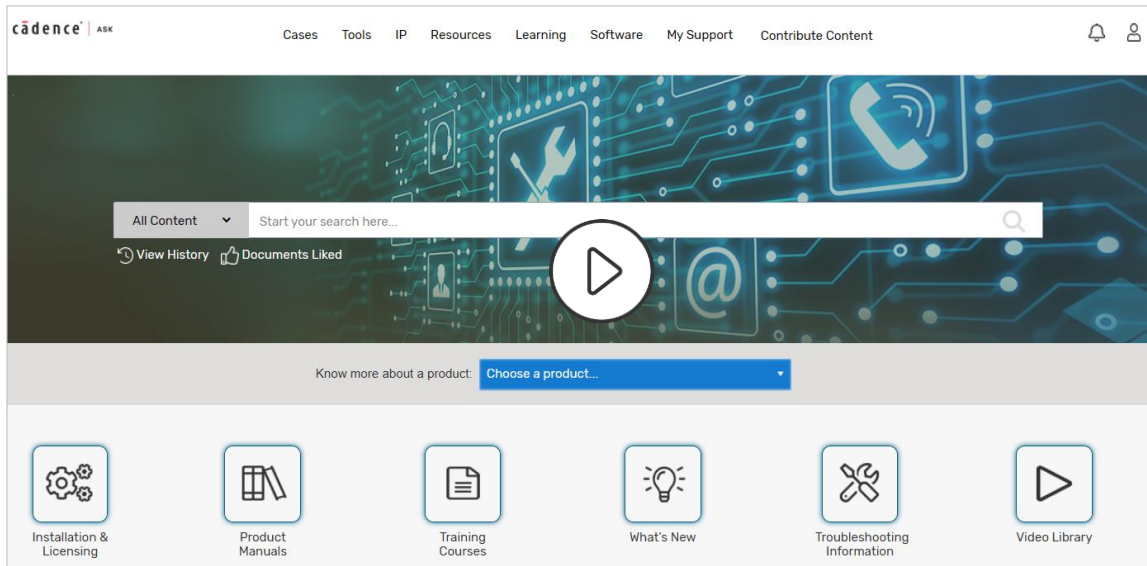
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[http://www.cadence.com/Training/Pages/learning\\_maps.aspx](http://www.cadence.com/Training/Pages/learning_maps.aspx)

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## Wrap Up

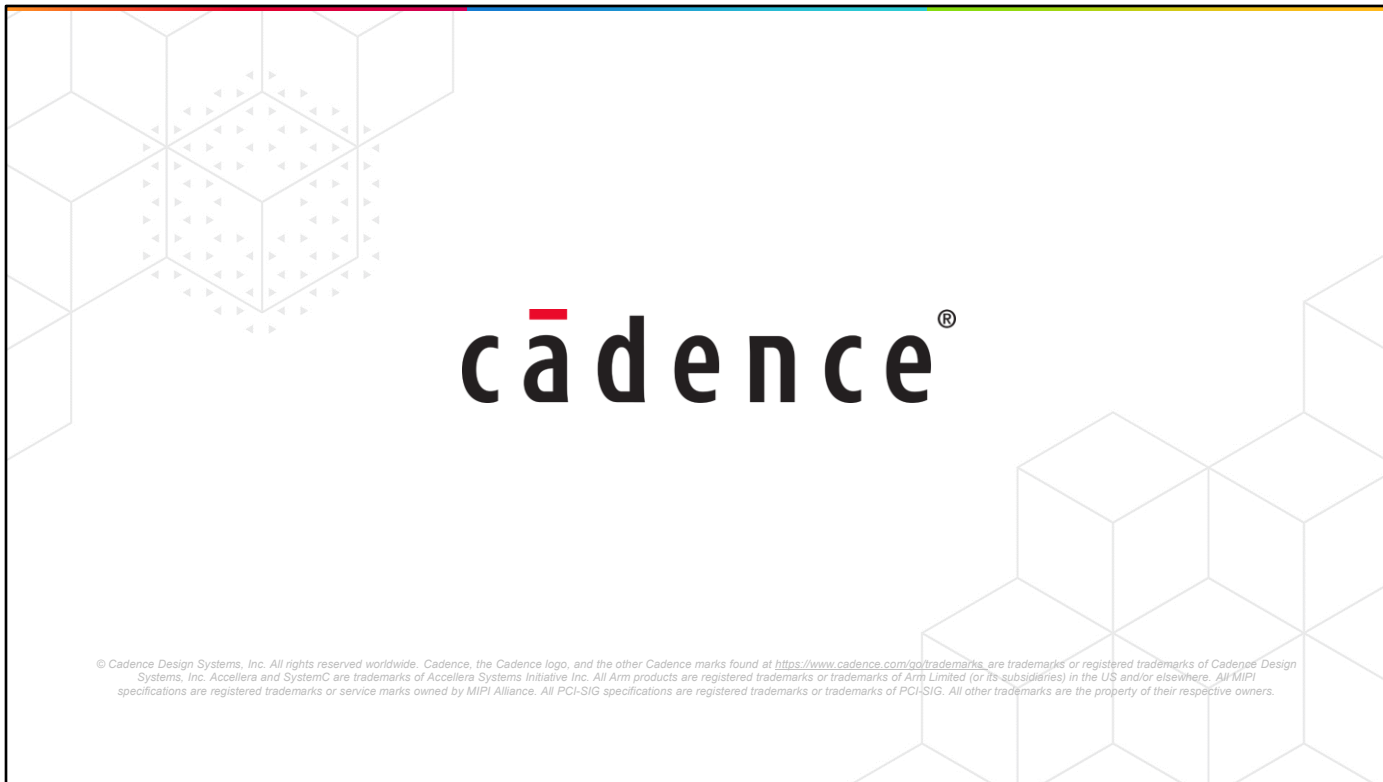
- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion
- Complete the Course's Digital Badge exam to earn your Badge

# Thank you!

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